Calibration of the LIBOR Market Model

Hugo Wei

MRGR Rates, JPMorganChase

CentraleSupélec

Apprenticeship report

August 2026

Acknowledgements

I would like to sincerely thank the MRGR Rates team at JPMorganChase for their help and support throughout this apprenticeship. I am especially grateful to my supervisors, Matteo Basei, Ming Gao and Patrick Roome, for the time they dedicated to guiding me, for their patience, and for the many insights they shared along the way. I am also grateful to Yury Kravchenko for his assistance with the implementation in the MRGR Benchmarking library and for the patient, detailed feedback he provided to improve my coding quality.

I am very thankful to Marc Rozensztajn for his mentorship and for taking the time to share his experience with me. I am equally thankful to Youssef Randjiou for giving me the opportunity to join this team and for his invaluable advice and encouragement. I would also like to thank Wayan Crain for his day-to-day support and for teaching me how to navigate the tooling infrastructure used by the team.

Finally, I would like to express my gratitude to my academic tutor, Laurence Carassus, for her guidance and constructive feedback during the preparation of this report.

Contents

Acknowledgements 1

Abstract 4

Context 5

1 Introduction 6

2 Literature Review 8
- 2.1 Origins of the LIBOR Market Model . . . 8
- 2.2 Calibration Strategies: Local vs. Global . . . 9
- 2.3 Volatility and Correlation Specifications . . . 10
- 2.4 Optimisation Methods for Calibration . . . 11
- 2.5 Monte Carlo Simulation and American Monte Carlo . . . 11
- 2.6 Spread Option Pricing and Correlation Calibration . . . 12
- 2.7 Factor Reduction and Principal Component Analysis . . . 13
- 2.8 Local and Stochastic Volatility Extensions . . . 13
- 2.9 The Forward Market Model and the Transition to Risk-Free Rates . . . 14
- 2.10 Further Extensions and Related Developments . . . 14

3 The LIBOR Market Model 15
- 3.1 Setup and Notation . . . 15
- 3.2 Change of Numéraire and Drift Correction . . . 15
  - 3.2.1 Spot Measure . . . 16
  - 3.2.2 Terminal Measure . . . 16
- 3.3 Volatility . . . 16
- 3.4 Correlation Structure . . . 17
- 3.5 Factor Reduction via Principal Component Analysis . . . 18
  - 3.5.1 Factor Loading Construction . . . 20

4 Calibration Methodology 22
- 4.1 Selection of Calibration Targets . . . 22
  - 4.1.1 Global vs. Local Calibration . . . 22
  - 4.1.2 Calibration Basket . . . 22
  - 4.1.3 Grid-Based Volatility . . . 22

4.1.4 Correlation Estimation from Historical Data . . . 23
4.2 Objective Function . . . 27
4.3 Regularisation . . . 27
4.3.1 Selection of First-Order Expiry Weight . . . 27
4.3.2 Selection of Second-Order Regularisation Weights . . . 28
4.4 Approximate Model-Implied Volatilities . . . 29
4.5 Levenberg–Marquardt Algorithm . . . 30

5 Results **32**
5.1 Market Data . . . 32
5.2 Unregularised Calibration (Baseline) . . . 34
5.2.1 Validation of the Swaption Volatility Approximation . . . 34
5.2.2 Calibration Fit: Approximate Model vs. Market . . . 35
5.2.3 Calibrated Volatility Grid . . . 35
5.3 Regularised Calibration . . . 37
5.4 Computational Performance . . . 38
5.5 Sensitivity to Starting Point . . . 39

6 Extension of the Calibration to Spread Options **42**
6.1 Purpose and Set-up . . . 42
6.2 Cost function and approximation of spread options . . . 42
6.2.1 Cost function . . . 42
6.2.2 Approximation of Model-Implied Normal Volatility of Spread Options . . . 43
6.3 Results . . . 44
6.3.1 Fitting results . . . 44
6.3.2 Impact on correlation . . . 47

7 Extension to Displaced Log-Normal Specification **49**
7.1 Purpose . . . 49
7.2 Definitions and dynamics . . . 49
7.3 Calibration . . . 50
7.3.1 Cost function . . . 50
7.3.2 Market-implied swaption skew . . . 50
7.3.3 Enhancement of swap rate approximation . . . 51
7.3.4 Spread option approximations from the literature . . . 53
7.3.5 Internally developed approximation . . . 55
7.4 Results . . . 61

8 Conclusion **68**

A Derivative of a swap rate with respect to a LIBOR forward rate **70**

B Markovian projection of the swap rate onto a displaced log-normal process **72**

Abstract

This report discusses the specification and calibration of the LIBOR Market Model (LMM). We first consider the deterministic normal-volatility case, then extend the calibration to CMS spread options, and finally generalise the model to a displaced log-normal specification. We describe the model framework, the specification of volatility and correlation structures, the formulation of the calibration problem, and the numerical methods employed. The goal of this project is to provide a flexible and reliable implementation of the LMM in the team's internal C++ library, enabling benchmarking with production models.

Context

This research is carried out as part of the apprenticeship program within the Model Risk Governance and Review (MRGR) department at JPMorganChase.

The MRGR team is tasked with performing independent model reviews and governance activities to identify, measure, and mitigate model risk within the firm. Their goal is to ensure that models are suitable for their intended purpose, appropriately utilised within the approved business context, and that users are informed of the model limitations and their potential impact on business decisions. Specifically, the MRGR - Commercial & Investment Bank Trading - Rates team provides independent oversight through the validation and review of models related to rates derivatives.

In managing model risk, MRGR collaborates closely with Market Risk, Model Developers, Trading, and Finance.

Chapter 1

Introduction

The LIBOR Market Model (LMM), introduced independently by Brace, Gatarek, and Musiela [5] and Jamshidian [20], is a widely adopted framework for modelling the joint dynamics of forward interest rates. Unlike short-rate models (e.g. Hull–White [18], Black–Karasinski [4]) or the Heath–Jarrow–Morton framework [13], the LMM directly models market-observable forward LIBOR rates, and naturally recovers market-standard pricing formulas for caps and swaptions.

Given a tenor structure $0 = T_0 < T_1 < \cdots < T_N$, the forward LIBOR rate $L_n(t)$ is defined as the simply-compounded forward rate prevailing at time t for the accrual period $[T_n, T_{n+1}]$:

$$L_n(t) = \frac{1}{T_{n+1} - T_n}\left(\frac{P(t,\, T_n)}{P(t,\, T_{n+1})} - 1\right), \tag{1.1}$$

where $P(t, T)$ denotes the time-t price of a zero-coupon bond maturing at T. The T_{n+1}-forward measure $\mathbb{Q}^{n+1}$ is the probability measure under which the numéraire is $P(t,\, T_{n+1})$; under $\mathbb{Q}^{n+1}$ the forward rate $L_n(t)$ is a martingale.

In the normal (Bachelier) specification of the LMM with deterministic volatility, the dynamics take the form

$$\mathrm{d}L_n(t) = \lambda_n(t)^\top \,\mathrm{d}W^{n+1}(t), \tag{1.2}$$

where $\lambda_n(t) \in \mathbb{R}^m$ is a deterministic instantaneous volatility vector and $W^{n+1}(t)$ is an m-dimensional Brownian motion under $\mathbb{Q}^{n+1}$.

The central challenge in deploying the LMM in practice is *calibration*: the determination of the model's volatility and correlation parameters such that the model reproduces observed market prices of liquid instruments. This report addresses the following questions:

(i) What calibration targets should be selected (restriction to a specific subset of contracts vs. calibrating on all liquid contracts)?

(ii) What functional forms for volatility and correlation are appropriate?

(iii) How should the calibration problem be formulated and solved efficiently?

A core activity of MRGR Rates is the benchmarking of production models against independently developed alternatives, enabling the review team to assess pricing accuracy, hedge stability, and sensitivity to modelling assumptions.

In this context, the LMM is ideally suited to this role. As a multi-factor model that directly evolves the market-observable forward rates, it provides a natural reference frame against which the behaviour of production models can be assessed. Its multi-factor structure captures the decorrelation between forward rates at different tenors—a feature that is critical for pricing correlation-sensitive instruments such as spread options and Bermudans.

The goal of my apprenticeship project is to implement the LMM within MRGR internal library in C++. The library is designed to support independent model review activities by providing self-contained implementations of benchmark models that do not depend on production pricing infrastructure.

The LMM calibration module we developed accepts as inputs a market data snapshot (discount curve, swaption and caplet volatility surfaces) and user-defined configuration parameters (correlation structure, PCA dimension, regularisation weights, Monte Carlo path count, regression bases for American Monte Carlo), and returns a fully calibrated model object capable of Monte Carlo simulation and derivative pricing. An interactive Excel interface exposes the calibration and pricing functionality to reviewers who require rapid scenario analysis without direct interaction with the C++ codebase.

Results at this stage are encouraging. For the baseline 12×12 grid of calibration swaptions, the methodology achieves a root-mean-square error (RMSE) of 1.11 bps in swaption basis-point volatility space, computed in 948 ms. Extending the calibration to ATM CMS spread options in the normal specification reproduces the market spread volatilities to a RMSE of 0.24 bps (10Y-2Y) and 0.80 bps (30Y-2Y), without degrading the swaption fit (e.g. RMSE 0.54 bps for the 10Y-2Y calibration), in about 17 and 28 s respectively.

In an effort to calibrate the dynamics of LIBOR rates to the skew of swaptions, we then considered a displaced log-normal specification. In the displaced log-normal specification, the dynamics are given by:

$$\mathrm{d}L_n(t) = \Big(b_n(t)L_n(t) + (1 - b_n(t))L_n(0)\Big)\lambda_n(t)^\top \,\mathrm{d}W^{n+1}(t), \tag{1.3}$$

where $b_n(t)$ is a deterministic displacement function.

The displaced log-normal extension calibrates to the swaption smile (RMSE 0.52–0.76 bps across the three strike offsets) and to the ATM CMS spreads (RMSE 0.30 and 1.40 bps), in about 96 s. Extension of the specification of the model to stochastic volatility in an effort to fit the wings of the swaption smile is currently work in progress.

The remainder of this report is organised as follows. Chapter 2 surveys the relevant literature. Chapter 3 presents the model framework. Chapter 4 details the calibration methodology, and Chapter 5 presents the calibration results for the baseline deterministic normal-volatility case. Chapter 6 then extends the calibration to CMS spread options, and Chapter 7 to a displaced log-normal specification that captures the swaption skew. Chapter 8 concludes.

Chapter 2

Literature Review

2.1 Origins of the LIBOR Market Model

The modelling of interest rate term structures has evolved through several generations of increasingly flexible frameworks.

Among the pioneering approaches, we cite the short-rate models of Vasicek [47] and Cox, Ingersoll, and Ross (CIR) [8]. Although analytically tractable, these early frameworks are single-factor models—Vasicek with constant (Gaussian) volatility, CIR with a state-dependent square-root diffusion—that cannot reproduce the market-observed volatility smile across strikes or the decorrelation between forward rates at different tenors, limitations that become acute when pricing caps and swaptions against quoted Black implied volatilities. Extensions such as the Hull–White model [18] and the Black–Karasinski model [4] introduced time-dependent parameters that permit exact calibration to the initial discount curve and greater flexibility in matching the term structure of at-the-money volatility. However, in their standard single-factor formulations these models still rely on a single source of randomness and therefore fail to capture the rich correlation structure observed across tenors in the market.

The Heath–Jarrow–Morton (HJM) framework [13] addressed several of these shortcomings by modelling the dynamics of the entire instantaneous forward rate curve. Under the risk-neutral measure, the requirement that discounted bond prices be martingales fully determines the drift of each instantaneous forward rate in terms of its volatility function, with Girsanov's theorem providing the machinery for translating between equivalent measures. In the general case, state-dependent volatility specifications render the forward rate dynamics non-Markovian, and the infinite-dimensional state space necessitates parametric restriction of the volatility structure for any practical implementation. The Gaussian sub-class, in which the volatility functions are deterministic, restores analytical tractability and yields closed-form prices for bonds, caps, and swaptions. The binding limitation of Gaussian HJM is its inability to generate a volatility smile: implied volatility is flat across strikes for each expiry, a restriction incompatible with the pronounced skew and curvature observed in cap and swaption markets. Non-Gaussian parametric sub-classes of HJM—such as those employing CEV or stochastic-volatility specifications[1]— can

[1]Non-Gaussian parametric sub-classes of HJM that can generate smile dynamics include specifications with CEV-type instantaneous volatility [43] and stochastic-volatility extensions in the spirit of [16]; see also [7] for a systematic treatment.

generate smile dynamics, but their state variables remain unobservable instantaneous forward rates, meaning that an indirect calibration mapping from model parameters to traded instrument prices is required.

The LIBOR Market Model resolved these tensions not by departing from the HJM no-arbitrage structure—indeed, the LMM is itself a sub-class of HJM obtained by specifying the volatility functional such that discrete forward LIBOR rates follow prescribed diffusions—but by recasting the state variables as the market-observable forward rates themselves.

The theoretical foundations were laid independently by Brace, Gatarek, and Musiela [5], who proved the existence of a consistent family of forward measures and derived the drift corrections required to express all rates under a common numéraire, and by Jamshidian [20], who connected the forward rate dynamics to the swap market model and showed how swaptions can be priced under the swap measure. Musiela and Rutkowski [34] provided the measure-theoretic foundations, including the formal construction of the forward measure family and the regularity conditions under which the discrete forward rate dynamics are well-defined.

Taken together, these works established the LMM as a canonical framework that directly models the quantities quoted in interest rate markets: each forward rate prices its corresponding caplet at the observed Black volatility by construction, and the model's state variables are market-observable rates rather than latent instantaneous short rates, eliminating a layer of abstraction that complicates calibration and hedging in earlier frameworks.

2.2 Calibration Strategies: Local vs. Global

The practical deployment of the LMM requires specifying the instantaneous volatility structure of each forward rate and the correlation structure governing their joint dynamics—a combined parameter space that can be large. The market provides constraints in the form of caplet implied volatilities, which determine the magnitude of each rate's volatility; a matrix of swaption implied volatilities, which depend jointly on the volatilities and correlations; and, where liquid quotes are available, spread options and yield-curve-spread caps, which provide direct constraints on the inter-rate correlation structure that are not fully captured by the swaption grid.[2] How one maps these market constraints to model parameters defines the calibration strategy, and two broad paradigms have emerged.

Product-specific (or *local*) calibration restricts the calibration basket to a minimal subset of instruments directly relevant to the trade being priced—for instance, co-terminal swaptions when valuing a Bermudan swaption [37]. This approach is computationally inexpensive and typically achieves exact repricing of the selected instruments, but it produces parameter sets that vary from product to product, leading to inconsistent valuations and hedges across a portfolio of exotic trades.

Global calibration fits the model simultaneously to a broad grid of caplets and swaptions spanning all liquid expiries and tenors.

Rebonato [39] demonstrated that global calibration is tractable by combining an analytical

[2]Swaption implied volatilities constrain a weighted combination of volatilities and correlations, making it difficult to disentangle the two. Spread options, whose payoffs depend on the difference between two rates, isolate the correlation component and thus provide complementary calibration information.

swaption volatility approximation with parsimonious parameterisations of the volatility term structure (the *abcd* form) and the correlation matrix (exponential-decay or angles-based specifications), together with a cascade architecture that fixes caplet-implied norms before optimising correlation parameters against the swaption grid.

An alternative strand of the literature dispenses with parametric functional forms entirely, specifying volatilities on a discrete grid with regularisation penalties to enforce smoothness (Pedersen [36]; Sidenius [45]; Andersen and Piterbarg [2]). The details of this specification are discussed in Section 2.3.

Piterbarg [37] further showed that callable LIBOR exotics—whose exercise boundaries depend on overlapping sets of forward rates—require consistent global calibration to ensure coherent portfolio-level hedging and risk management.

2.3 Volatility and Correlation Specifications

The choice of volatility structure is central to the trade-off between model flexibility and calibration stability. Rebonato [38] proposed time-homogeneous parametric volatilities of the form $\sigma(T_n - t) = (a + b(T_n - t))\, e^{-c(T_n - t)} + d$, where $T_n - t$ is the time remaining until the n-th forward rate's reset date, a governs the initial level at reset, b controls the slope of the volatility hump, c determines the rate of exponential decay, and d sets a positive long-end floor—a four-parameter family that captures the empirical hump shape of forward rate volatility. While parsimonious and conducive to stable calibration, the homogeneity assumption—that all forward rates share the same volatility shape as a function of time-to-reset—limits the ability to match the full caplet and swaption surface when the market exhibits non-homogeneous volatility patterns across tenors, a limitation that becomes binding under global calibration.

An alternative, the grid-based non-parametric approach mentioned in Section 2.2, is now widely regarded as the industry standard for global calibration. Under this specification (Pedersen [36]; Sidenius [45]; Andersen and Piterbarg [2]), the instantaneous volatility is defined at the nodes of a rectangular grid over expiries and times-to-reset, with inter-node values obtained by linear interpolation. This provides as many degrees of freedom as there are grid nodes, enabling near-exact calibration to the full caplet and swaption volatility surface, though it necessitates regularisation penalties to enforce smoothness and prevent overfitting to noise in market quotes. The grid construction and calibration procedure adopted in this work are detailed in Chapter 4.

The correlation structure among forward rates plays a decisive role in the pricing of multi-rate exotic instruments such as CMS spread options and Bermudan swaptions. Empirical studies consistently show that correlations between forward rates decay approximately exponentially with the distance between their reset dates [7, 39], with the rate of decorrelation and the asymptotic correlation level both depending on the absolute maturity of the rates involved [2].

Schoenmakers and Coffey [44] developed a systematic framework for generating parametric correlation structures that satisfy positive semi-definiteness constraints and reproduce empirical decorrelation patterns. Rebonato and White [42] showed that different correlation assumptions produce materially different swaption implied volatilities even when caplet volatilities are held fixed, demonstrating that the swaption market contains information that can identify the inter-rate dependence structure beyond what is recoverable from caplets alone. These two results

together establish that the correlation parameterisation must be both mathematically valid and sufficiently flexible to capture the dependence information embedded in swaption prices. The time-stationary parametric form proposed by Andersen and Piterbarg [2] and adopted in this work (the precise functional form and its calibration are presented in Chapter 4) meets both requirements: it combines an exponential decorrelation kernel with a tenor-dependent asymptotic floor and a maturity-dependent speed of mean reversion, thereby capturing the principal empirical features of LIBOR rate correlations within a six-parameter structure.

2.4 Optimisation Methods for Calibration

The calibration of the LMM is naturally formulated as a nonlinear least-squares problem, in which the objective is to minimise the sum of squared differences between model-implied and market-implied volatilities across the calibration basket. This structure invites the use of the Levenberg–Marquardt (LM) algorithm [32], which exploits the decomposition of the objective into individual residuals by computing the full Jacobian matrix of the residual vector with respect to the model parameters. Each LM iteration thus leverages $M \times P$ pieces of sensitivity information—where M is the number of residuals and P the number of parameters—to construct a curvature-aware update step, interpolating between Gauss–Newton and gradient descent via an adaptive damping parameter.

Andersen and Piterbarg [2] employ LM in conjunction with approximate closed-form swaption volatility formulas, replacing the expensive Monte Carlo evaluation of model-implied volatilities during optimisation with analytical approximations that are accurate to within approximately one basis point. This combination yields a calibration procedure that is both fast and robust.

General-purpose quasi-Newton methods such as L-BFGS, while effective for generic smooth optimisation, observe only one aggregated gradient vector per iteration and build a low-rank Hessian approximation from a limited history of past gradients [23]; they are therefore less efficient for problems exhibiting the specific least-squares structure of model calibration, where methods such as Levenberg–Marquardt exploit the full Jacobian of the residual vector to construct a superior curvature approximation [27, 35].[3]

A growing body of recent work has also investigated the use of neural networks to accelerate calibration. Horvath et al. [17] trained a neural network as a surrogate pricer for the rough Bergomi model, achieving calibration in under 10 milliseconds, while Brigo et al. [6] proposed direct parameter regression from market observables; both strategies trade lengthy offline training for rapid online calibration but are susceptible to accuracy degradation in market regimes far from the training distribution.

2.5 Monte Carlo Simulation and American Monte Carlo

Once the LMM has been calibrated, pricing of exotic path-dependent products proceeds via Monte Carlo simulation of the joint forward rate dynamics.

[3] More precisely, L-BFGS guarantees only superlinear convergence, whereas LM achieves local quadratic convergence near the solution and superlinear convergence otherwise.

The comprehensive treatment by Glasserman [10] remains the standard reference for variance reduction techniques—including antithetic variates and control variates—discretisation schemes, and convergence analysis in the context of financial Monte Carlo methods.

The discretisation of the LMM dynamics requires care, since the drift depends on the forward rates themselves; naive Euler schemes can introduce significant bias. Hunter, Jäckel, and Joshi [19] proposed the predictor-corrector scheme, in which the forward rates are first advanced using the drift evaluated at the current state (predictor step), and the drift is then re-evaluated at the predicted state and averaged with the original drift to produce the final update (corrector step), yielding substantial improvements in accuracy without appreciably increasing computational cost.

For products with early exercise features—most notably Bermudan swaptions—the least-squares regression approach of Longstaff and Schwartz [24], commonly known as American Monte Carlo (AMC), provides a tractable method for approximating the continuation value at each exercise date by regressing discounted future cash flows on a set of basis functions of the current state variables. Andersen [1] applied this methodology specifically to Bermudan swaptions under the LMM, demonstrating that with suitably chosen exercise strategies and regression bases, Monte Carlo methods can achieve accuracy comparable to lattice-based approaches while retaining the flexibility to handle high-dimensional factor structures.

This technique has become the workhorse for pricing callable and cancellable structures under the LMM.

2.6 Spread Option Pricing and Correlation Calibration

Yield curve spread options—options on the difference between two swap rates of different tenors—provide direct market-implied information about the correlation between forward rates and are therefore natural calibration targets for the LMM correlation surface. The classical pricing framework for spread options originates from Margrabe [26], who derived a closed-form formula for the option to exchange one asset for another. For non-zero strikes, Kirk [21] proposed a widely-used approximation that reduces the spread option to a modified Black–Scholes problem. Within the LMM, the model-implied spread option volatility depends on the individual swap rate volatilities and on their correlation, providing a direct channel through which market spread option prices constrain the correlation parameters. Andersen and Piterbarg [2] provided approximate formulas for spread option pricing under the LMM that are consistent with the swaption volatility approximations used elsewhere in the calibration, enabling the simultaneous calibration to swaptions, caps, and spread options within a unified optimisation framework.

While the spread option literature typically frames the payoff in terms of swap rates, a closely related and particularly liquid class of instruments are *yield curve spread caps* (YCS caps), which consist of a strip of caplets on the spread between two forward rates of different tenors (e.g. the 10Y–2Y spread or the 30Y–2Y spread). Each caplet pays $\max(L_{\text{long}}(T_i) - L_{\text{short}}(T_i) - K,\, 0)$ at the observation date T_i, where L_{long} and L_{short} are the long-tenor and short-tenor forward rates respectively and K is the strike. Because YCS caps are traded in significant volume and their prices are highly sensitive to the term structure of inter-rate correlation, they constitute a natural calibration target for the LMM correlation surface. The strip structure provides correlation

information at multiple time horizons simultaneously, making it a richer calibration target than a single spread option maturity. The extension of the calibration framework developed in this report to include YCS caps as calibration targets is the subject of ongoing work.

2.7 Factor Reduction and Principal Component Analysis

A practical concern in any LMM implementation is factor reduction. The instantaneous correlation matrix of N forward rates is of rank N, requiring N independent Brownian drivers in the simulation. However, empirical studies dating back to Litterman and Scheinkman [22] have shown that the term structure of interest rates is predominantly driven by three factors—level, slope, and curvature—which together explain over 95% of the total variance. This motivates replacing the full correlation matrix by its best rank-m approximation obtained via eigendecomposition.

Rebonato [39] demonstrated that retaining three to five principal components captures over 95% of the total variance in typical interest rate systems, dramatically reducing computational cost while preserving accurate pricing of correlation-sensitive products. The approximation error introduced by factor truncation is concentrated in the fine-grained decorrelation structure between adjacent forward rates, which is immaterial for vanilla swaption pricing but can affect the pricing of spread options and other correlation-sensitive instruments [2].

The factor reduction strategy adopted in this work is described in Section 3.5.

2.8 Local and Stochastic Volatility Extensions

The deterministic-volatility LMM adopted in this work cannot reproduce the volatility smile observed across strikes for caps and swaptions—a limitation that has motivated two principal classes of extension.

The first is the *local volatility* approach, in which the instantaneous volatility of each forward rate is made a deterministic function of the rate level, typically through a constant-elasticity-of-variance (CEV) specification or a displaced diffusion [2]. This generates strike-dependent implied volatility (skew) at modest additional computational cost, but produces smile dynamics that flatten unrealistically over time—the well-known sticky-strike behaviour. The second is the *stochastic volatility* approach, in which one or more additional stochastic factors drive the volatility process, yielding both skew and a realistic evolution of the smile surface. Andersen and Piterbarg [2] provide a comprehensive treatment of both local and stochastic volatility extensions within the LMM framework.

Among stochastic volatility formulations, the unification of the LMM with the SABR model of Hagan et al. [12] has received particular attention: Rebonato [40] proposed a time-homogeneous SABR-consistent extension, Henry-Labordère [15] unified the BGM and SABR frameworks via stochastic volatility models on Riemannian manifolds, and Mercurio and Morini [31] provided a practical calibration approach that fits SABR parameters per rate before embedding them into the LMM correlation structure. These extensions lie outside the scope of the present work, which restricts attention to deterministic volatility, but they represent natural directions for enriching the model's ability to capture smile dynamics.

2.9 The Forward Market Model and the Transition to Risk-Free Rates

The global transition away from LIBOR towards risk-free rates (RFRs) such as SOFR, ESTR, and SONIA has prompted a fundamental rethinking of the market model paradigm. Unlike LIBOR, which is a forward-looking term rate determined at the beginning of the accrual period, RFR-based rates are typically *backward-looking*, computed as compounded averages of overnight rates observed over the accrual period. This structural difference renders the classical LMM inapplicable without modification, since the modelled quantities are no longer known at the start of each period.

Lyashenko and Mercurio [25] addressed this challenge by proposing the *Forward Market Model* (FMM), a generalisation of the LMM that accommodates both forward-looking and backward-looking rate definitions within a unified framework. The FMM models forward rates that converge to the realised compounded overnight rate at maturity, and it recovers the classical LMM as a special case when the underlying rate is forward-looking. Mercurio [30] further developed the framework for SOFR-based derivatives, providing closed-form approximations for caps and swaptions under the new rate conventions.

2.10 Further Extensions and Related Developments

The 2007–2008 financial crisis revealed that persistent basis spreads between LIBOR rates of different tenors could no longer be ignored, giving rise to the *multicurve* framework formalised by Henrard [14] and Mercurio [29], in which separate curves are maintained for discounting and for projecting forward rates of each tenor. Moreni and Pallavicini [33] extended the LMM to this setting with additional basis spread dynamics. The present work operates within a single-curve framework; the multicurve extension is noted here as an important modelling consideration for production deployment, since neglecting basis spreads can introduce systematic pricing errors for instruments spanning multiple tenor frequencies.

Chapter 3

The LIBOR Market Model

3.1 Setup and Notation

We now formalise the model introduced in Chapter 2 and establish the notation used throughout the remainder of this report.

Consider a tenor structure $0 = T_0 < T_1 < \cdots < T_N$ with accrual fractions $\tau_n = T_{n+1} - T_n$. The simply-compounded forward rate for the accrual period $[T_n, T_{n+1}]$ is

$$L_n(t) = \frac{1}{\tau_n}\left(\frac{P(t,T_n)}{P(t,T_{n+1})} - 1\right), \tag{3.1}$$

where $P(t,T)$ is the time-t price of a zero-coupon bond maturing at T.

Under the T_{n+1}-forward measure $\mathbb{Q}^{n+1}$, whose numéraire is $P(t,T_{n+1})$, the rate $L_n(t)$ is a martingale. This suggests modelling the diffusion of $L_n(t)$ as

$$\mathrm{d}L_n(t) = \sigma_n(t)^\top \,\mathrm{d}W^{n+1}(t), \tag{3.2}$$

for some stochastic process $\sigma_n(t) \in \mathbb{R}^m$ where $W^{n+1}(t)$ is an m-dimensional Brownian motion under $\mathbb{Q}^{n+1}$.[1]

We refer to [2] for a comprehensive treatment of more general volatility specifications, including local and stochastic volatility extensions. In this project, we consider the LMM with deterministic volatility:

$$\mathrm{d}L_n(t) = \lambda_n(t)^\top \,\mathrm{d}W^{n+1}(t), \tag{3.3}$$

where $\lambda_n(t) \in \mathbb{R}^m$ is a deterministic vector.

3.2 Change of Numéraire and Drift Correction

The dynamics (3.3) express each forward rate $L_n(t)$ as a martingale under its own forward measure $\mathbb{Q}^{n+1}$. To simulate the joint evolution of all forward rates, it is convenient to re-express the dynamics under a single common measure. Two standard choices exist: the *terminal measure* $\mathbb{Q}^N$, whose numéraire is the longest-dated zero-coupon bond $P(t,T_N)$, and the *spot measure* $\mathbb{Q}^{\text{spot}}$, whose numéraire is the discrete rolling money-market account. We adopt the

[1] The log-normal (Black) counterpart replaces (3.2) with $\mathrm{d}L_n(t) = L_n(t)\,\sigma_n(t)^\top \,\mathrm{d}W^{n+1}(t)$.

spot measure throughout this work; the terminal measure dynamics are recorded below for completeness, as they appear frequently in the literature and provide a useful cross-check.

3.2.1 Spot Measure

Under the spot measure $\mathbb{Q}^{\text{spot}}$ (associated with the discrete rolling money-market account; see Andersen and Piterbarg [2], Ch. 14), the drift of $L_n(t)$ acquires a correction that couples it to all live forward rates with indices between the next reset date and n. Applying Girsanov's theorem to change from $\mathbb{Q}^{n+1}$ to $\mathbb{Q}^{\text{spot}}$ yields, in the normal specification,

$$\mathrm{d}L_n(t) = \sum_{k=q(t)}^{n} \frac{\tau_k \, \lambda_n(t)^\top \lambda_k(t)}{1 + \tau_k \, L_k(t)} \, \mathrm{d}t + \lambda_n(t)^\top \, \mathrm{d}W^{\text{spot}}(t), \tag{3.4}$$

where $q(t) = \min\{k : T_k > t\}$ is the index of the next reset date after t and $W^{\text{spot}}(t)$ is an m-dimensional Brownian motion under $\mathbb{Q}^{\text{spot}}$.

The spot measure is preferred over the terminal measure for simulation because the denominator $1 + \tau_k \, L_k(t)$ in the drift involves only *short-dated* forward rates (indices $k \leq n$), whose magnitudes remain moderate along typical simulation paths. Under the terminal measure, by contrast, the analogous sum runs over *long-dated* forward rates (indices $k > n$), and the accumulated drift corrections can become numerically unstable for long-dated tenor structures—a well-known pathology noted by Glasserman [10].

3.2.2 Terminal Measure

Under the terminal measure $\mathbb{Q}^N$, the Brownian motions $W^{n+1}(t)$ of the individual forward measures are related to the terminal Brownian motion $W^N(t)$ by the Girsanov drift

$$\mathrm{d}W^{n+1}(t) = -\sum_{j=n+1}^{N-1} \frac{\tau_j \, \lambda_j(t)}{1 + \tau_j \, L_j(t)} \, \mathrm{d}t + \mathrm{d}W^N(t), \tag{3.5}$$

where the summand is an m-dimensional vector for each j. Substituting (3.5) into (3.3) gives the dynamics of the n-th forward rate under $\mathbb{Q}^N$:

$$\mathrm{d}L_n(t) = -\sum_{j=n+1}^{N-1} \frac{\tau_j \, \lambda_n(t)^\top \lambda_j(t)}{1 + \tau_j \, L_j(t)} \, \mathrm{d}t + \lambda_n(t)^\top \, \mathrm{d}W^N(t). \tag{3.6}$$

The derivations of (3.4) and (3.6) are standard applications of Girsanov's theorem to the family of forward measures; detailed proofs can be found in Andersen and Piterbarg [2], Ch. 14, and in Musiela and Rutkowski [34].

3.3 Volatility

At any given time t, we need to determine $p(t)$ vectors $\lambda_n(t)$, $n = q(t), \ldots, N-1$, that produce the desired correlations between LIBOR rates and match the prices of calibration instruments.

The correlation between increments $\mathrm{d}L_j(t)$ and $\mathrm{d}L_k(t)$ is given by

$$\mathrm{Corr}(\mathrm{d}L_j(t), \mathrm{d}L_k(t)) = \frac{\lambda_j(t)^\top \lambda_k(t)}{\|\lambda_j(t)\| \, \|\lambda_k(t)\|}. \tag{3.7}$$

The model assumes that the correlation between $\mathrm{d}L_j(t)$ and $\mathrm{d}L_k(t)$ depends only on time differences $T_j - T_{q(t)}$ and $T_k - T_{q(t)}$, but not on the absolute values of time points. This keeps the correlation matrix constant as t progresses from $T_{q(t)-1}$ to $T_{q(t)}$. Further, it assumes that correlations do not depend on the level of L.

Let us designate the $p(t) \times p(t)$ correlation matrix as $\mathbf{R}_{q(t)}$:

$$\begin{aligned} \left(\mathbf{R}_{q(t)}\right)_{i,j} &= \mathrm{Corr}(\mathrm{d}L_{q(t)+i-1}(t), \mathrm{d}L_{q(t)+j-1}(t)) \\ &= \rho\left(T_{q(t)+i-1} - T_{q(t)}, T_{q(t)+j-1} - T_{q(t)}\right), \quad i,j = 1,\ldots,p(t), \end{aligned} \tag{3.8}$$

where $\rho(t_1, t_2)$ is a parametric function that defines the pairwise correlations. If we approximate $\mathbf{R}_{q(t)}$ as the product

$$\mathbf{R}_{q(t)} \approx \mathbf{D}_{q(t)} \mathbf{D}_{q(t)}^\top, \tag{3.9}$$

where $\mathbf{D}_{q(t)}$ is a $p(t) \times m$ matrix obtained by principal components analysis (PCA), then vectors $\lambda_n(t)$ can be expressed as products of their norms $\|\lambda_n(t)\|$ by rows of $\mathbf{D}_{q(t)}$:

$$\lambda_n(t)^\top = \|\lambda_n(t)\| \cdot \left(\mathbf{D}_{q(t)}\right)_{n-q(t)+1,*} \tag{3.10}$$

If $p(t) < m$, the rightmost $m - p(t)$ columns of $\mathbf{D}_{q(t)}$ are filled with zeros.

The calculation of volatility then reduces to two independent tasks:

1. Specify $\rho(t_1, t_2)$ and calculate $\mathbf{D}_n$ for all $n = 1, \ldots, N-1$;
2. Calibrate $\|\lambda_n(t)\|$ for $0 < t < T_{N-1}$ and $q(t) \le n \le N-1$.

3.4 Correlation Structure

We adopt the time-stationary parametric form proposed by Andersen and Piterbarg [2]:

$$\rho(\tau_i, \tau_j) = \rho_\infty(\tau_{\min}) + \left(1 - \rho_\infty(\tau_{\min})\right) e^{-a(\tau_{\min})\,|\tau_i - \tau_j|}, \tag{3.11}$$

$$a(\tau) = a_\infty + (a_0 - a_\infty)\, e^{-\kappa\,\tau}, \tag{3.12}$$

$$\rho_\infty(\tau) = b_\infty + (b_0 - b_\infty)\, e^{-\alpha\,\tau}, \tag{3.13}$$

where $\tau_i = T_i - t$ and $\tau_j = T_j - t$ are the residual maturities of the i-th and j-th forward rates, $\tau_{\min} = \min(\tau_i, \tau_j)$, and the six parameters $\{a_0,\, a_\infty,\, \kappa,\, b_0,\, b_\infty,\, \alpha\}$ are subject to the constraints

$$a_0 \ge a_\infty > 0, \qquad 0 \le b_0 \le b_\infty \le 1, \qquad \kappa > 0, \qquad \alpha > 0. \tag{3.14}$$

The structure encodes three empirically motivated features of forward rate correlations. First, the exponential kernel $e^{-a\,|\tau_i - \tau_j|}$ produces decorrelation that increases with the distance

between reset dates, consistent with the observation that forward rates spanning nearby accrual periods co-move more closely than those spanning distant periods. Second, the maturity-dependent asymptotic floor $\rho_\infty(\tau_{\min})$ allows long-dated forward rates to remain highly correlated even at large tenor separations (the "parallel shift" component of the yield curve), while permitting short-dated rates to decorrelate more completely. Third, the maturity-dependent decorrelation speed $a(\tau_{\min})$ captures the empirical observation that short-dated rates decorrelate faster with tenor distance than long-dated rates.

The estimation of these six parameters from historical data is described in Section 4.1.4.

Remark 3.4.1 (Positive semi-definiteness)**.** The parametric form (3.11) does not guarantee positive semi-definiteness (PSD) for arbitrary parameter values, since the dependence of $a(\cdot)$ and $\rho_\infty(\cdot)$ on $\tau_{\min}$ breaks the stationarity conditions that ensure PSD for simple exponential kernels [44]. The PSD property must therefore be verified numerically for each set of fitted parameters by confirming that all eigenvalues of the correlation matrix evaluated at the simulation tenors are non-negative. The verification for the parameters used in this work is reported in Section 3.5.

3.5 Factor Reduction via Principal Component Analysis

To determine the methodology for factor reduction, we retrieve the empirical correlation matrix from realised correlations of daily absolute changes in EUR forward rates over the two-year window from 20 April 2024 to 20 April 2026. The forward rates $L_n(t)$ for $n = 1, \ldots, 121$ are 6-month simple forward rates extracted from the EUR PAR discount curve, spanning accrual periods $[0.5\text{Y}, 1.0\text{Y}]$ through $[60.5\text{Y}, 61.0\text{Y}]$ in 6-month steps[2].

The empirical correlation matrix $\hat{C} \in \mathbb{R}^{121\times121}$ is computed as the sample Pearson correlation of the daily absolute changes $\Delta L_n(t) = L_n(t) - L_n(t-1)$:

$$\hat{C}_{n,k} = \frac{\sum_{t=1}^{T}(\Delta L_n(t) - \overline{\Delta L_n})(\Delta L_k(t) - \overline{\Delta L_k})}{\left[\sum_{t=1}^{T}(\Delta L_n(t) - \overline{\Delta L_n})^2 \sum_{t=1}^{T}(\Delta L_k(t) - \overline{\Delta L_k})^2\right]^{1/2}}, \tag{3.15}$$

where $T \approx 520$ is the number of business days in the observation window and $\overline{\Delta L_n}$ denotes the sample mean of the daily changes in the n-th forward rate.

We then solve for the six parameters of the correlation function (3.11) (see detailed methodology in Subsection 4.1.4) and evaluate the resulting parametric correlation matrix at the same 121 forward rate tenors. The eigenvalue spectrum of this matrix is reported in Table 3.1 and Figure 3.1.

The parametric correlation matrix $\rho(\tau_i, \tau_j)$ (3.11) evaluated at the $N_{\text{correl}} = 121$ forward rate tenors of the LMM simulation yields a 121×121 positive semi-definite matrix C. To reduce the

[2]The correlation estimation grid (121 forward rates, 6M to 60Y6M in 6-month steps) extends well beyond the swaption calibration grid (12 tenors, 6M to 30Y) to accommodate the forward rates required for pricing yield curve spread caps in future work. The most liquid YCS caps in the EUR market are the 10Y–2Y and 30Y–2Y spreads. A 30Y–2Y YCS cap strip with observation dates extending to 30Y requires the 30-year forward rate at the final observation date, corresponding to an accrual period ending at approximately 60Y. The correlation estimation grid is therefore chosen to span the full range of forward rates that would enter the Monte Carlo simulation when pricing these instruments.

dimensionality of the Monte Carlo simulation, we decompose C via its eigendecomposition:

$$C = V\Gamma V^\top, \qquad \Gamma = \mathrm{diag}(\gamma_1, \ldots, \gamma_{N_{\mathrm{correl}}}), \quad \gamma_1 \geq \cdots \geq \gamma_{N_{\mathrm{correl}}} \geq 0, \tag{3.16}$$

where $V = [v_1 \mid \cdots \mid v_{N_{\mathrm{correl}}}]$ is the orthogonal matrix of eigenvectors. The factor loading matrix is constructed by retaining the leading m eigenvectors:

$$A = V_m \Gamma_m^{1/2} \in \mathbb{R}^{N_{\mathrm{correl}} \times m}, \qquad V_m = [v_1 \mid \cdots \mid v_m], \quad \Gamma_m = \mathrm{diag}(\gamma_1, \ldots, \gamma_m), \tag{3.17}$$

so that $AA^\top \approx C$. The volatility vector of the n-th forward rate is then $\lambda_n(t) = \|\lambda_n(t)\| \cdot a_n$, where a_n is the n-th row of A normalised to unit length, and the Monte Carlo simulation requires only m independent Brownian motions rather than N_{correl}.

Table 3.1 reports the eigenvalue spectrum of the 121×121 parametric correlation matrix evaluated at the simulation tenors (6-month forward rates from 6M to 60Y6M) with the fitted parameters of Table 4.1.

Factor	Eigenvalue	Cumulative variance (%)
1	95.486	78.9
2	9.669	86.9
3	4.838	90.9
4	2.813	93.2
5	1.658	94.6
6	1.115	95.5
7	0.791	96.2
8	0.592	96.7
9	0.462	97.0
10	0.369	97.4
Factors 11–121 (total)		100.0

Table 3.1: Eigenvalue spectrum of the 121×121 parametric correlation matrix (3.11) evaluated at the simulation tenors (6M to 60Y6M in 6-month steps) with the fitted parameters of Table 4.1. All eigenvalues are strictly positive (minimum $= 4.29 \times 10^{-3}$), confirming PSD.

As we can see on Figure 3.1, the first three factors capture 90.9% of the total variance and correspond to the classical level, slope, and curvature modes of the yield curve. We retain $m = 10$ factors, capturing 97.4% of the total variance. The remaining 2.6% resides in higher-order factors that represent fine-grained decorrelation patterns between adjacent forward rates. While this residual variance is immaterial for vanilla swaption pricing (which depends on integrated variance dominated by the leading factors), it can affect the pricing of spread options, where the payoff depends on the difference between two rates and is therefore sensitive to the precise correlation structure. The choice of $m = 10$ reflects a balance between computational cost and the accuracy requirements of correlation-sensitive instruments.

All 121 eigenvalues of the parametric correlation matrix are strictly positive, with the smallest eigenvalue equal to 4.29×10^{-3}. The PSD property was therefore verified numerically for the fitted parameters and must be rechecked whenever the correlation parameters are re-estimated.

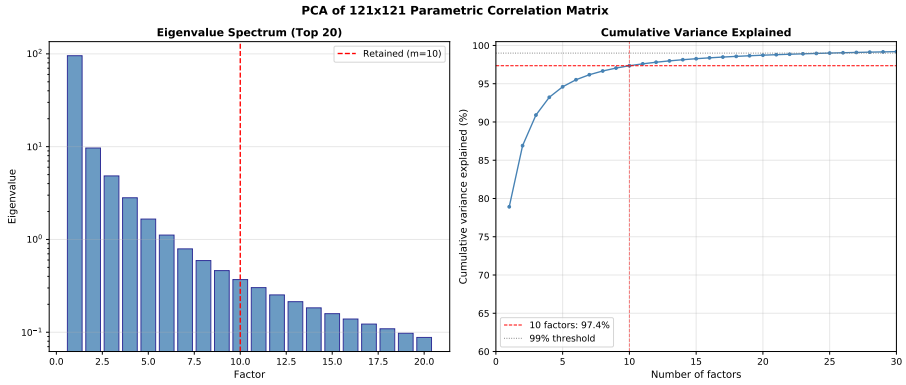


Figure 3.1: Eigenvalue spectrum of the 121×121 parametric correlation matrix. The first three factors (level, slope, curvature) capture 90.9% of the total variance; 10 factors capture 97.4%. The rapid decay justifies the low-dimensional factor approximation used in the Monte Carlo simulation.

3.5.1 Factor Loading Construction

Given the m leading eigenvalues and eigenvectors of the parametric correlation matrix, the factor loading matrix $A \in \mathbb{R}^{N_{\text{correl}} \times m}$ must satisfy two properties: (i) $AA^\top \approx C$ (the reduced-rank correlation approximation), and (ii) $(AA^\top)_{nn} = 1$ for all n (each forward rate's total variance is preserved). Property (ii) is not automatically satisfied by the truncated eigendecomposition $A_0 = V_m \Gamma_m^{1/2}$, since discarding the trailing eigenvectors reduces the diagonal elements below unity.

We implement the following two approaches. The results reported in this work use the rescaled PCA approach as it offers faster simulation and as the difference between the two methods is negligible for our purposes.

Rescaled PCA ("Poorman's PCA"). Following Andersen and Piterbarg [2, Ch. 14], the truncated loading matrix $A_0 = V_m \Gamma_m^{1/2}$ is rescaled row-wise:

$$A = S^{-1} A_0, \qquad S = \operatorname{diag}\left(\sqrt{(A_0 A_0^\top)_{11}}, \ldots, \sqrt{(A_0 A_0^\top)_{N_{\text{correl}} N_{\text{correl}}}} \right), \tag{3.18}$$

ensuring $(AA^\top)_{nn} = 1$ for all n. This is computationally inexpensive but distorts the off-diagonal correlations: the rescaling systematically inflates correlations between rates whose truncated variances were most deficient.

Optimised PCA. Following Rebonato and Jäckel [41], we seek a diagonal perturbation $\mu \in \mathbb{R}^{N_{\text{correl}}}$ such that the truncated eigendecomposition of $C + \operatorname{diag}(\mu)$ yields a loading matrix D with $\operatorname{diag}(DD^\top) = \mathbf{1}$ exactly. The perturbation is found by minimising

$$\min_{\mu \in \mathbb{R}^{N_{\text{correl}}}} \sum_{n=1}^{N_{\text{correl}}} \left[(DD^\top)_{nn} - 1 \right]^2, \tag{3.19}$$

where $D = V_m(\mu) \Gamma_m^{1/2}(\mu)$ and $\Gamma_m(\mu), V_m(\mu)$ are the m leading eigenvalues and eigenvectors of $C + \operatorname{diag}(\mu)$. The optimisation is performed via Levenberg–Marquardt. For large matrices

($N_{\text{correl}} > \max(20, 4m)$), the eigendecomposition at each iteration is accelerated using the Lanczos algorithm with full reorthogonalization, reducing the cost from $O(N_{\text{correl}}^3)$ to $O(N_{\text{correl}}^2 m)$ per iteration.

Chapter 4

Calibration Methodology

4.1 Selection of Calibration Targets

4.1.1 Global vs. Local Calibration

Local calibration fits the model to a minimal subset of instruments relevant to a specific trade (e.g. co-terminal swaptions for a Bermudan swaption), whereas global calibration fits simultaneously to a broad grid of caplets and swaptions spanning all liquid expiries and tenors. Throughout this project, we adopt the global approach for the following reasons:

1. **Consistency**: a single calibrated model prices all products in the portfolio without ambiguity.
2. **Avoidance of selection bias**: selecting instruments for local calibration is non-trivial and can introduce uncontrolled bias.
3. **Computational feasibility**: with efficient algorithms (Section 5.4), the runtime is acceptable for benchmarking.
4. **Subsumes local calibration**: the global framework can be restricted to any sub-basket, enabling comparison with local strategies.

4.1.2 Calibration Basket

The calibration basket consists of an $N_t \times N_x$ grid of at-the-money swaptions with expiries $\{t_1, \ldots, t_{N_t}\}$ and tenors $\{x_1, \ldots, x_{N_x}\}$.

4.1.3 Grid-Based Volatility

Following Andersen and Piterbarg [2], we express $\|\lambda_n(t)\|$ as a function of two variables:

$$\|\lambda_n(t)\| = G(t, T_n - t).$$

The function $G(t, x)$ is then interpolated on a discrete grid $\{n_i\} \times \{p_j\}$, for expiry indices $i = 1, \ldots, N_t$ and tenor indices $j = 1, \ldots, N_x$. This piecewise-linear representation provides $N_t \times N_x$ degrees of freedom, sufficient to match the entire swaption and caplet surface.

Between grid nodes, the instantaneous volatility norm is recovered by bilinear interpolation in both the expiry and time-to-reset dimensions. For each forward rate n and time step k ($1 \le n \le k \le N-1$), we have

$$\|\lambda_{n,k}\| = w_{++}\, G_{i,j} + w_{+-}\, G_{i,j-1} + w_{-+}\, G_{i-1,j} + w_{--}\, G_{i-1,j-1}, \tag{4.1}$$

where, denoting $\tau_{n,k} = T_k - T_{n-1}$, the grid cell indices are

$$i = \min\{a : t_a \ge T_{n-1}\}, \qquad j = \min\{b : x_b \ge \tau_{n,k}\}, \tag{4.2}$$

and the bilinear weights are

$$w_{++} = \frac{(T_{n-1} - t_{i-1})\,(\tau_{n,k} - x_{j-1})}{(t_i - t_{i-1})\,(x_j - x_{j-1})}, \qquad w_{+-} = \frac{(T_{n-1} - t_{i-1})\,(x_j - \tau_{n,k})}{(t_i - t_{i-1})\,(x_j - x_{j-1})}, \tag{4.3}$$

$$w_{-+} = \frac{(t_i - T_{n-1})\,(\tau_{n,k} - x_{j-1})}{(t_i - t_{i-1})\,(x_j - x_{j-1})}, \qquad w_{--} = \frac{(t_i - T_{n-1})\,(x_j - \tau_{n,k})}{(t_i - t_{i-1})\,(x_j - x_{j-1})}. \tag{4.4}$$

The four weights are non-negative and sum to unity by construction, ensuring that the interpolated volatility lies within the convex hull of the surrounding grid values [2].

4.1.4 Correlation Estimation from Historical Data

The parametric form (3.11) is fitted to $\hat{C}$ (see Section 3.5) by minimising the sum of squared errors over the $\binom{121}{2} = 7{,}260$ unique off-diagonal pairs:

$$\theta^* = \arg\min_{\theta \in \Theta} \sum_{n<k} \left(\hat{C}_{n,k} - \rho_\theta(\tau_n, \tau_k)\right)^2, \tag{4.5}$$

subject to the constraints $a_0 \ge a_\infty > 0$, $0 \le b_0 \le b_\infty \le 1$, $\kappa > 0$, and $\alpha > 0$. The constraints are enforced via reparametrisation: $a_0 = a_\infty + \delta_a$ with $\delta_a \ge 0$, and $b_0 = r \cdot b_\infty$ with $r \in [0, 1]$, eliminating the need for constrained optimisation.

The optimisation is performed using differential evolution (DE) [46] with multiple restarts across three strategies (`best1bin`, `best2bin`, `randtobest1bin`) and three random seeds, for a total of nine independent runs with population size $20 \times 6 = 120$ and a maximum of 1,500 iterations per run. All nine runs converge to the same objective value (sum of squared errors (SSE) = 44.18), providing strong evidence that the global minimum has been found. The fitted parameters are reported in Table 4.1 and the corresponding parametric correlation matrix is reported in Table 4.2 and Figure 4.1. A comparison with empirical correlation surface is shown in Figure 4.2 and Figure 4.3, and the residual surface is shown in Figure 4.4.

Parameter	Symbol	Value	Constraint
Short-maturity decorrelation speed	a_0	0.1686	$a_0 \geq a_\infty$
Long-maturity decorrelation speed	a_∞	0.0688	$a_\infty > 0$
Decorrelation speed mean-reversion	κ	0.3350	$\kappa > 0$
Short-maturity asymptotic correlation	b_0	0.0968	$0 \leq b_0 \leq b_\infty$
Long-maturity asymptotic correlation	b_∞	0.7506	$b_\infty \leq 1$
Asymptotic correlation mean-reversion	α	0.2640	$\alpha > 0$

Table 4.1: Fitted correlation parameters for the parametric structure (3.11). Estimation is by differential evolution from realised correlations of daily absolute changes in EUR 6-month forward rates over the period 20 April 2024 to 20 April 2026 (approximately 520 business days, 121 forward rates spanning 6M to 60Y6M in 6-month steps, 7,260 unique correlation pairs). The parametric fit achieves an RMSE of 0.079 and a maximum absolute error of 0.324 against the empirical correlations.

	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	1.00	0.94	0.83	0.74	0.66	0.59	0.48	0.37	0.32	0.27	0.22	0.19
1Y	0.94	1.00	0.90	0.82	0.74	0.68	0.57	0.46	0.41	0.35	0.30	0.26
2Y	0.83	0.90	1.00	0.93	0.86	0.81	0.71	0.61	0.56	0.50	0.44	0.39
3Y	0.74	0.82	0.93	1.00	0.95	0.90	0.81	0.72	0.67	0.61	0.55	0.49
4Y	0.66	0.74	0.86	0.95	1.00	0.96	0.88	0.79	0.75	0.69	0.63	0.56
5Y	0.59	0.68	0.81	0.90	0.96	1.00	0.93	0.85	0.81	0.75	0.69	0.62
7Y	0.48	0.57	0.71	0.81	0.88	0.93	1.00	0.93	0.89	0.84	0.77	0.71
10Y	0.37	0.46	0.61	0.72	0.79	0.85	0.93	1.00	0.96	0.91	0.85	0.77
12Y	0.32	0.41	0.56	0.67	0.75	0.81	0.89	0.96	1.00	0.95	0.88	0.80
15Y	0.27	0.35	0.50	0.61	0.69	0.75	0.84	0.91	0.95	1.00	0.92	0.83
20Y	0.22	0.30	0.44	0.55	0.63	0.69	0.77	0.85	0.88	0.92	1.00	0.87
30Y	0.19	0.26	0.39	0.49	0.56	0.62	0.71	0.77	0.80	0.83	0.87	1.00

Table 4.2: Fitted parametric correlation surface $\rho(\tau_i, \tau_j)$ as a function of forward rate residual maturities τ_i and τ_j, evaluated with the parameters of Table 4.1. The surface exhibits the characteristic exponential decorrelation with tenor distance and the maturity-dependent asymptotic floor.

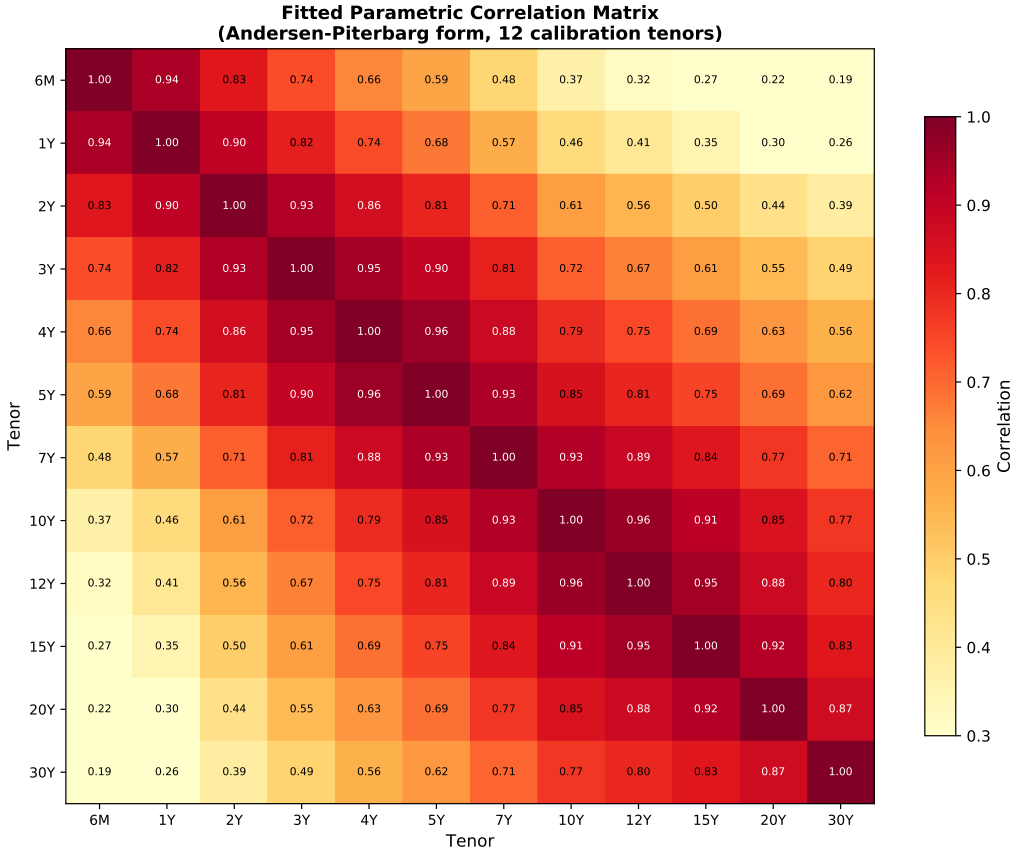


Figure 4.1: Heatmap of Table 4.2.

Interpretation of the fitted parameters The fitted value $b_\infty = 0.751$ implies that forward rates with long residual maturities are asymptotically correlated at approximately 75%, even for widely separated tenors. The low value $b_0 = 0.097$ indicates that short-dated forward rates exhibit much weaker long-range correlation, consistent with the empirical observation that the short end of the curve is driven by idiosyncratic monetary policy expectations while the long end is dominated by common macroeconomic factors. The decorrelation speed $a_0 = 0.169$ for short-maturity rates is approximately 2.4 times the long-maturity speed $a_\infty = 0.069$, reflecting the faster decorrelation of short-dated rates with tenor distance. This is consistent with Figure 4.1.

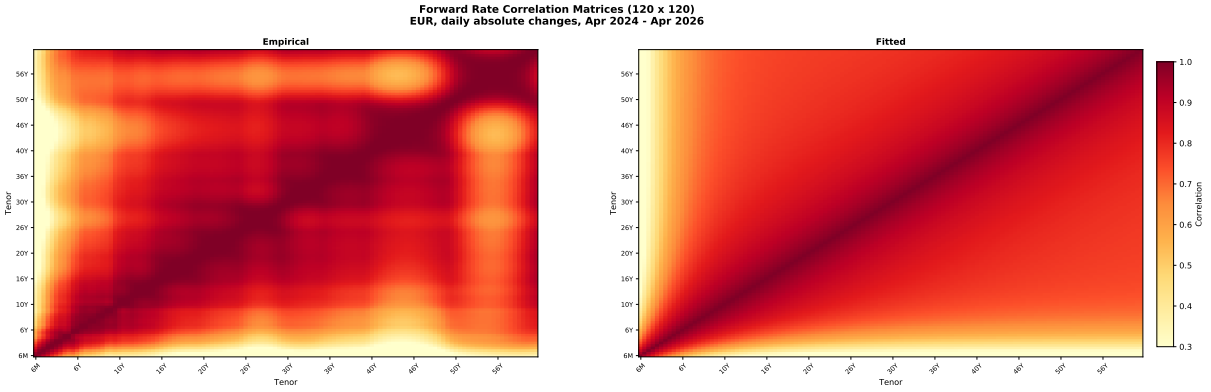


Figure 4.2: Forward rate correlation matrices (121×121) at the simulation tenors (6M to 60Y6M in 6-month steps). Left: empirical sample correlation of daily absolute changes in EUR forward rates, April 2024 to April 2026. Right: parametric correlation with the fitted parameters of Table 4.1.

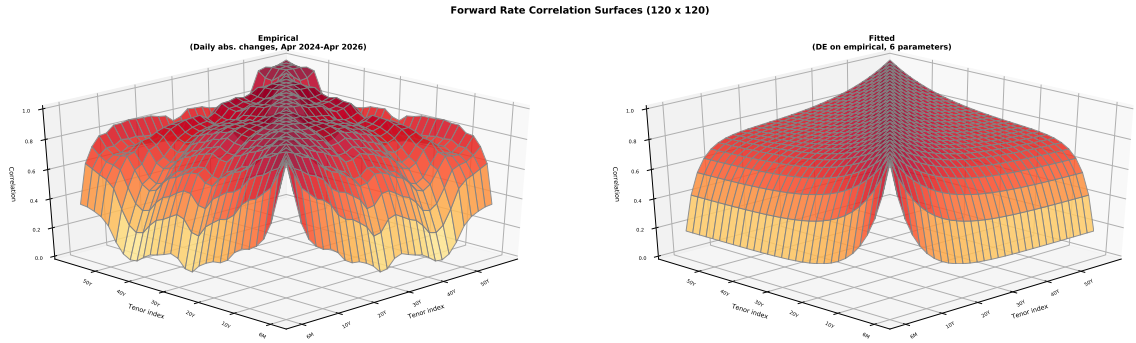


Figure 4.3: Three-dimensional correlation surfaces corresponding to the heatmaps of Figure 4.2. Left: empirical sample correlation. Right: parametric correlation with fitted parameters.

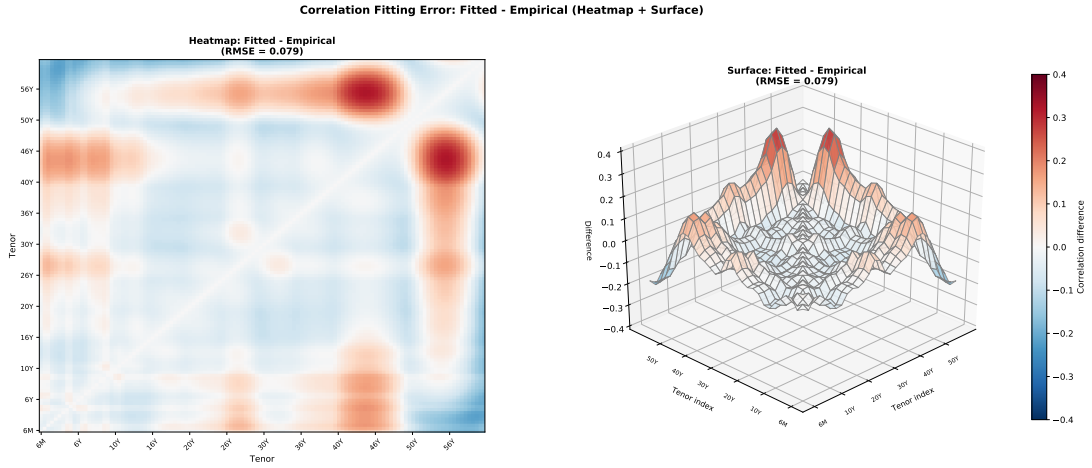


Figure 4.4: Correlation fitting error (parametric minus empirical) for the fitted parameters (RMSE = 0.079). The 3D view reveals the spatial distribution of fitting errors, with the largest residuals concentrated in the short-maturity region.

4.2 Objective Function

Given the volatility grid G and correlation structure ρ, the calibration minimises:

$$\begin{aligned}\mathcal{C}(G,\rho) = \frac{w_S}{N_S}\sum_{i=1}^{N_S}\left(\bar{\sigma}_{\text{swaption},i} - \hat{\sigma}_{\text{swaption},i}\right)^2 \\ + \frac{\lambda_t}{N_t N_x}\sum_{i,j}(\partial_t G_{i,j})^2 + \frac{\lambda_x}{N_t N_x}\sum_{i,j}(\partial_x G_{i,j})^2 \\ + \frac{\lambda_{tt}}{N_t N_x}\sum_{i,j}(\partial_t^2 G_{i,j})^2 + \frac{\lambda_{xx}}{N_t N_x}\sum_{i,j}(\partial_x^2 G_{i,j})^2,\end{aligned} \tag{4.6}$$

where w_S is a fitting weight and $\lambda_t, \lambda_x, \lambda_{tt}, \lambda_{xx}$ are regularisation weights.

4.3 Regularisation

The regularisation terms penalise:

- **First-order differences** (λ_t, λ_x): encourage smoothness and time-stationarity.
- **Second-order differences** (λ_{tt}, λ_{xx}): penalise curvature, promoting near-linear behaviour.

The weights are chosen by cross-validation or expert judgment to balance goodness of fit against smoothness.

Remark 4.3.1 (Economic motivation for first-order expiry regularisation)**.** The spot volatility $G_{i,j}$ at expiry t_i and tenor τ_j represents the instantaneous diffusion coefficient of the j-th forward rate at observation time t_i. In the absence of known future events, this quantity should evolve smoothly in t: abrupt changes between adjacent expiries would imply that the market anticipates a sudden volatility regime shift at a specific future date. The first-order expiry penalty

$$\mathcal{R}_{t,1} = \lambda_t \sum_{i=1}^{N_t-1}\sum_{j=1}^{N_x}(G_{i+1,j} - G_{i,j})^2 \tag{4.7}$$

enforces this prior by penalising the discrete time derivative of the volatility surface. This penalty is applied only along the expiry dimension; the tenor dimension is excluded because the volatility term structure exhibits a natural hump shape that a first-order penalty would suppress.

4.3.1 Selection of First-Order Expiry Weight

The first-order expiry weight λ_t controls the trade-off between calibration fit and smoothness of the volatility surface along the expiry dimension. We sweep the first-order expiry weight $\lambda_t \in \{0, 0.01, 0.05, 0.1, 0.5, 1, 2, 5\}$ with second-order weights fixed at $\lambda_{tt} = \lambda_{xx} = 0.5$ (see Section 4.3.2).

We record the optimiser residual $\mathrm{RMSE}_{\mathrm{A}}$ and the first-order total variation along expiry,

$$\mathrm{TV}_1^{(t)}(G) = \sum_{i=1}^{N_t-1} \sum_{j=1}^{N_x} (G_{i+1,j} - G_{i,j})^2. \tag{4.8}$$

All runs use 30,000 Monte Carlo paths with antithetic variates, 10 Brownian factors (rescaled PCA), Euler predictor–corrector discretisation under the spot measure, and a fixed random seed of 42. The grid comprises 12 expiries × 12 tenors (144 free parameters). The optimiser terminates on stationary function value after 5–6 evaluations in all cases.

As supported by Table 4.3 and Figure 4.5, $\lambda_t = 0.1$ achieves a good compromise between fit and smoothness.

λ_t	$\mathrm{RMSE}_{\mathrm{A}}$	$\mathrm{TV}_1^{(t)}$
0	0.312	2.296
0.01	0.318	2.129
0.05	0.338	1.909
0.1	0.362	1.640
0.5	0.482	1.345
2	0.708	0.933
5	0.901	0.604

Table 4.3: First-order expiry smoothing sweep: calibration diagnostics as a function of λ_t. Second-order weights fixed at $\lambda_{tt} = \lambda_{xx} = 0.5$. RMSE is computed over the 12 × 12 swaption grid; $\mathrm{TV}_1^{(t)}$: first-order total variation along expiry. All RMSE and grid values in bps.

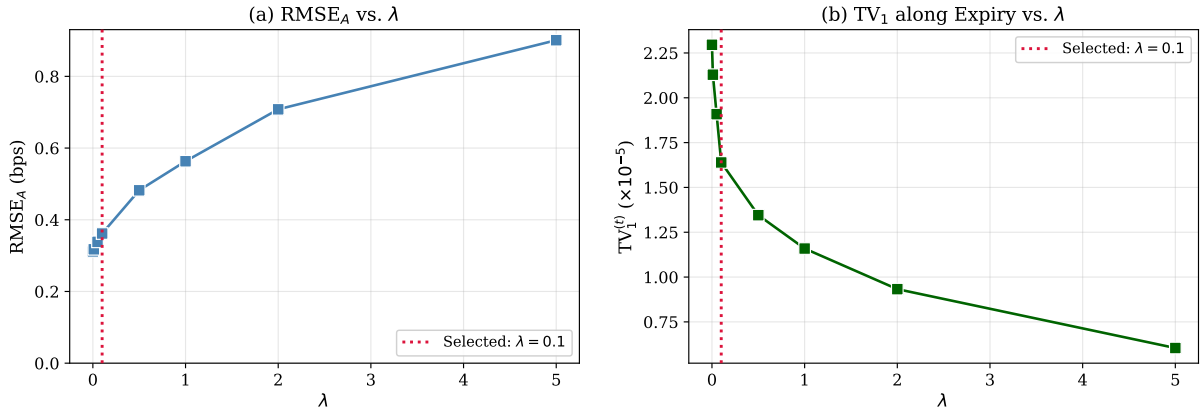


Figure 4.5: Calibration RMSE (approximate model vs. market) and first-order total variation along expiry as a function of the first-order expiry smoothing weight λ_t.

4.3.2 Selection of Second-Order Regularisation Weights

The regularisation weights λ_{tt} and λ_{xx} control the trade-off between goodness of fit and grid smoothness. The methodology is similar to the one of the previous section: for a range of regularisation strengths $\lambda = \lambda_{tt} = \lambda_{xx}$ (imposing symmetric penalisation across both dimensions), we record the optimiser residual $\mathrm{RMSE}_{\mathrm{A}}$ and the total second-order variation of the calibrated

grid,

$$\mathrm{TV}_2(G) = \sum_{i,j} \left[(G_{i+1,j} - 2G_{i,j} + G_{i-1,j})^2 + (G_{i,j+1} - 2G_{i,j} + G_{i,j-1})^2 \right]. \tag{4.9}$$

Figure 4.6 plots $\mathrm{RMSE_A}$ and $\mathrm{TV}_2(G)$ against λ for $\lambda \in \{0, 0.01, 0.05, 0.1, 0.5, 1.0, 2.0, 5.0\}$. The recommended weight $\lambda^* = 0.5$ corresponds to a compromise between fit quality and grid smoothness.

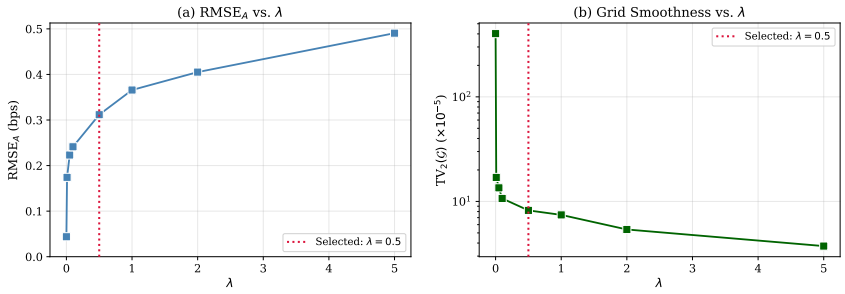


Figure 4.6: Plot of $\mathrm{RMSE_A}$ (bps) and $\mathrm{TV}_2(G)$ against λ.

4.4 Approximate Model-Implied Volatilities

Calculation of model-implied swaption volatilities $\bar{\sigma}_{\text{swaption}}$ is performed using the analytical approximation for swap rate dynamics developed in Section 14.4.2 of Andersen and Piterbarg [2]. The swap rate volatility $\sigma_S(t)$ of a swap that runs from T_j to T_k is approximated by a weighted sum of LIBOR volatilities $\lambda_n(t)$,

$$\sigma_S(t) = \sum_{n=j}^{k-1} w_n(0) \lambda_n(t)^\top, \tag{4.10}$$

where stochastic weights $w_n(t)$ are approximated by their time 0 values. Since we consider the normal case, the weights are defined by (see Eq.(14.31) of Andersen and Piterbarg [2])

$$w_n(t) = \frac{\partial S(t)}{\partial L_n(t)}. \tag{4.11}$$

The swap rate $S(t)$ is given by

$$S(t) = \frac{P(t, T_j) - P(t, T_k)}{A(t)}, \tag{4.12}$$

where the annuity is

$$A(t) \triangleq A_{j,k-j}(t) = \sum_{i=j}^{k-1} \eta_i P(t, T_{i+1}). \tag{4.13}$$

Here, $\eta_i \equiv \tau_i$ if the fixed leg payment schedule coincides with the floating leg schedule. However, if the periods of the fixed and floating legs differ, then $\eta_i = 0$ if T_{i+1} is not included in the fixed schedule. Otherwise, η_i is equal to the fixed leg period that pays at T_{i+1}.

Taking the partial derivative in Eq. (4.11) yields the following closed-form expression (see

the proof in Appendix A):

$$\frac{\partial S(t)}{\partial L_n(t)} = \frac{S(t)\tau_n}{1+\tau_n L_n(t)} \times \left[\frac{P(t,T_k)}{P(t,T_j)-P(t,T_k)} + \frac{1}{A(t)} \sum_{i=n}^{k-1} \eta_i P(t,T_{i+1}) \right]. \tag{4.14}$$

And we have

$$w_n(0) = \frac{S(0)\tau_n}{1+\tau_n L_n(0)} \times \left[\frac{P(0,T_k)}{P(0,T_j)-P(0,T_k)} + \frac{1}{A(0)} \sum_{i=n}^{k-1} \eta_i P(0,T_{i+1}) \right]. \tag{4.15}$$

The effective term swap rate volatility $\bar{\sigma}_{\text{swaption}}$ is defined by (see Section 7.6.2 of [2])

$$\bar{\sigma}_{\text{swaption}} = \left(\frac{1}{T_j} \int_0^{T_j} \|\sigma_S(t)\|^2 \mathrm{d}t \right)^{1/2} \approx \left(\frac{1}{T_j} \sum_{n=1}^{j} \tau_{n-1} \|\sigma_S(T_n)\|^2 \right)^{1/2}. \tag{4.16}$$

The squared norm of $\sigma_S(T_n)$ is given by

$$\begin{aligned} \|\sigma_S(T_n)\|^2 &= \left(\sum_{l=j}^{k-1} w_l(0)\lambda_l(T_n) \right)^\top \left(\sum_{m=j}^{k-1} w_m(0)\lambda_m(T_n) \right) \\ &= \sum_{l=j}^{k-1} \sum_{m=j}^{k-1} w_l(0) w_m(0) \lambda_l(T_n)^\top \lambda_m(T_n). \end{aligned} \tag{4.17}$$

The value $\bar{\sigma}_{\text{swaption}}$ is the model-implied volatility that enters the objective function.

4.5 Levenberg–Marquardt Algorithm

The calibration problem has the form $\min_\theta \|\boldsymbol{r}(\theta)\|^2$ where $\boldsymbol{r}(\theta) \in \mathbb{R}^M$ is the vector of weighted residuals. The Levenberg–Marquardt (LM) update is:

$$\delta = (J^\top J + \mu I)^{-1} J^\top \boldsymbol{r}(\theta), \tag{4.18}$$

where $J = \partial \boldsymbol{r}/\partial\theta$ is the $M \times P$ Jacobian and $\mu \geq 0$ is the damping parameter.

- When $\mu \to 0$: the step approaches the Gauss–Newton direction (fast convergence near optimum).
- When $\mu \to \infty$: the step approaches scaled gradient descent (more stable far from optimum).
- μ is adapted at each iteration based on the actual vs. predicted reduction in the residual.

Why LM over L-BFGS? The Jacobian provides $M \times P$ sensitivities per iteration (how each parameter affects each residual), whereas L-BFGS observes only P aggregated gradient components and builds a low-rank Hessian approximation from a limited history. For least-squares problems, LM converges faster and more robustly. This is verified empirically in Figure 4.7.

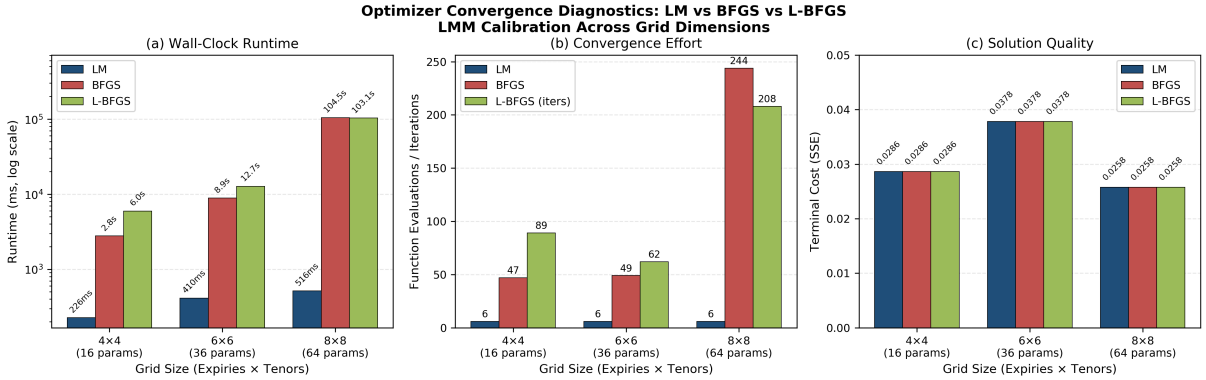


Figure 4.7: Optimiser convergence diagnostics for LMM parameter calibration across increasing grid dimensions. Panel (a) reports wall-clock runtime on a logarithmic scale; panel (b) reports the number of objective function evaluations (iterations for L-BFGS); panel (c) reports the terminal cost value (sum of squared errors). All three optimisers converge to effectively identical objective values, while the LM algorithm achieves convergence in one to two orders of magnitude less computation time.

Initial Guess The LM algorithm converges to a local minimum, and the calibration objective (4.6) is non-convex in the grid parameters G. The choice of initial guess therefore affects both the solution and the convergence speed.

All $N_t \times N_x$ volatility nodes are initialised to a common flat value:

$$G_{i,j}^{(0)} = \bar{\sigma} = 63.62\,\text{bps} \quad \text{for all } i = 1, \ldots, N_t, \quad j = 1, \ldots, N_x, \tag{4.19}$$

corresponding to the root mean square (RMS) of the market swaption volatility surface. The flat initialisation is deliberately agnostic about the shape of the volatility surface, ensuring that all structure in the calibrated grid is driven by the market data and the optimisation rather than by the starting point.

Remark 4.5.1 (Sensitivity to initialisation)**.** The sensitivity of the calibrated grid to the choice of initial guess is investigated in Section 5.5. Given the non-convexity of the objective function over $N_t \times N_x = 144$ parameters, the reported solution could in principle represent a local rather than global minimum. A systematic comparison of alternative initialisation strategies — including random perturbations of the flat guess, target-informed warm starts, and Latin Hypercube sampling — is presented in Section 5.5 and confirms that the regularised objective possesses a unique dominant basin of attraction.

Chapter 5

Results

The calibration is performed on market data as of 20 April 2026. The calibration basket consists of a 12×12 grid of at-the-money normal swaption implied volatilities, yielding 144 calibration instruments. The expiries and tenors are identical and given by:

$$\mathcal{T} = \{6\text{M},\ 1\text{Y},\ 2\text{Y},\ 3\text{Y},\ 4\text{Y},\ 5\text{Y},\ 7\text{Y},\ 10\text{Y},\ 12\text{Y},\ 15\text{Y},\ 20\text{Y},\ 30\text{Y}\}. \tag{5.1}$$

All volatilities are quoted in basis points of normal (Bachelier) implied volatility. In this section, we first present results for the unregularised calibration ($\lambda_t = \lambda_x = \lambda_{tt} = \lambda_{xx} = 0$) to swaptions only, in order to establish a baseline and to illustrate the necessity of regularisation. Regularised results are presented in Section 5.3.

All runtime measurements were performed on a single-threaded execution on an Intel Xeon Gold 6240R (2.40 GHz) with 64 GB RAM, running a 64-bit Windows environment.

5.1 Market Data

The calibration uses the following market data:

- **Swaption volatilities**: as specified above.
- **Forward rates**: the EUR forward curve retrieved for the same date.

Table 5.1 and Figure 5.1 report the market-observed ATM normal swaption implied volatilities used as calibration targets. The surface exhibits the characteristic features of the post-2020 rate environment: elevated short-expiry volatilities (approximately 70–71 bps for the 6M and 1Y expiries), a gradual decline with increasing expiry, and a mild downward slope along the tenor axis for short expiries that flattens for longer expiries.

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	70.65	70.65	73.84	71.19	68.97	66.98	63.83	60.30	59.56	58.93	57.65	56.08
1Y	70.16	70.16	71.22	69.08	67.48	66.06	63.47	60.89	60.36	59.93	58.96	58.10
2Y	68.23	68.23	67.54	66.73	66.13	65.50	63.86	62.62	62.12	61.57	60.86	60.90
3Y	67.08	67.08	66.91	66.23	65.78	65.65	64.43	63.77	63.16	62.37	61.56	61.95
4Y	67.24	67.24	66.91	66.06	65.71	65.66	64.81	64.26	63.67	62.84	62.13	62.05
5Y	67.25	67.25	66.85	66.25	65.84	65.81	65.16	64.45	63.81	62.89	62.17	61.85
7Y	67.47	67.47	66.93	66.19	65.78	65.52	64.92	64.51	63.79	62.72	62.04	61.83
10Y	66.46	66.46	66.84	66.09	65.30	64.95	64.31	63.68	62.97	61.90	61.33	60.83
12Y	66.02	66.02	66.16	65.43	64.83	64.45	63.70	62.89	62.14	61.01	60.24	59.77
15Y	65.08	65.08	65.22	64.34	63.65	63.25	62.39	61.50	60.74	59.62	58.70	58.29
20Y	63.91	63.91	64.24	63.31	62.40	61.77	60.44	59.45	58.70	57.59	56.68	56.22
30Y	61.58	61.58	61.97	61.18	60.34	59.06	57.33	55.71	54.84	53.54	53.14	52.57

Table 5.1: Market ATM normal swaption implied volatility surface (bps in swaption basis-point volatility space), 20 April 2026. The surface exhibits elevated short-expiry volatilities and a gradual decline with increasing expiry and tenor.

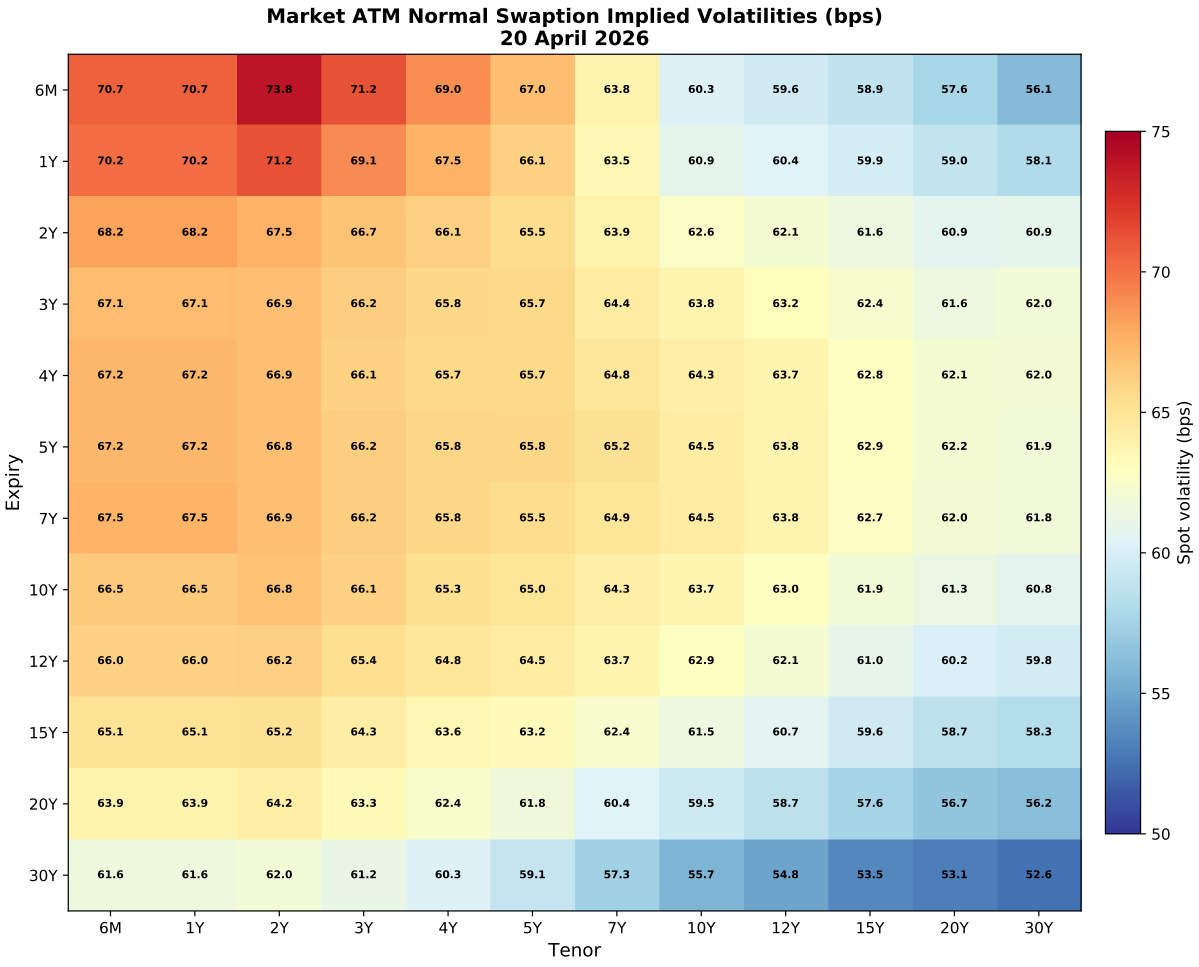


Figure 5.1: Heatmap corresponding to Table 5.1.

5.2 Unregularised Calibration (Baseline)

We first calibrate the volatility grid G to the swaption surface alone, with all regularisation weights set to zero ($\lambda_t = \lambda_x = \lambda_{tt} = \lambda_{xx} = 0$) and equal fitting weights across all instruments. This configuration serves two purposes: it establishes the best achievable fit to the swaption surface and it exposes the pathologies that arise in the absence of smoothness constraints.

5.2.1 Validation of the Swaption Volatility Approximation

The results in this subsection and in Sections 5.2.2–5.2.3 refer to the unregularised calibration with $\lambda_t = \lambda_x = \lambda_{tt} = \lambda_{xx} = 0$.

Before discussing calibration quality, we assess the impact of the frozen-weight approximation (4.16). Namely, we compare the approximate model-implied swaption volatilities obtained by formula (4.16) against those obtained by inverting the Bachelier formula after a full Monte Carlo pricing. Table 5.2 reports the differences.

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	−3.14	−1.66	−0.73	−0.53	−0.16	0.09	0.41	0.45	0.50	0.72	0.88	0.96
1Y	−1.22	0.75	0.14	0.50	0.84	1.02	1.38	1.33	1.32	1.32	1.29	1.28
2Y	−0.09	−0.02	0.33	0.81	1.10	1.22	1.25	1.33	1.39	1.44	1.48	1.35
3Y	−0.07	0.04	0.35	0.75	1.02	1.20	1.39	1.66	1.75	1.78	1.75	1.58
4Y	−0.03	0.07	0.28	0.66	0.99	1.24	1.64	1.88	1.91	1.92	1.83	1.44
5Y	−0.72	−0.82	0.22	0.48	0.71	1.03	1.49	1.80	1.81	1.79	1.65	1.23
7Y	1.23	0.35	0.17	0.33	0.61	0.78	1.39	1.65	1.76	1.66	1.57	1.21
10Y	0.03	0.20	0.66	0.88	0.95	1.01	1.18	1.47	1.49	1.46	1.36	0.92
12Y	0.31	0.97	0.79	0.74	0.71	0.83	1.15	1.29	1.30	1.23	1.11	0.63
15Y	−0.28	−0.32	−0.01	0.05	0.31	0.41	0.67	0.87	0.94	0.90	0.66	0.10
20Y	0.19	0.19	0.32	0.42	0.56	0.68	0.82	0.93	0.85	0.69	0.35	−0.39
30Y	−0.26	−0.27	−0.25	−0.20	−0.12	−0.02	0.17	0.33	0.34	0.24	0.02	−0.60

Table 5.2: Difference between Monte Carlo model-implied and approximate model-implied normal volatilities (bps). Positive values indicate that the Monte Carlo estimate exceeds the approximation.

The approximation error is within ±1.5 bps for the majority of the grid, with the largest deviations occurring at short expiries and short tenors (not in the intermediate region). In particular, the maximum absolute error is about 3.1 bps at the 6M expiry and 6M tenor, and errors remain elevated (around 1–2 bps) for 6M–1Y expiries across most tenors. Beyond roughly 2Y expiry, the approximation error is predominantly positive and typically around 0.7–1.9 bps, with the largest values concentrated at intermediate-to-long tenors (about 10Y–20Y), peaking near 1.9 bps around 4Y–5Y expiry. This pattern is consistent with known limitations of the frozen-weight approximation: as expiry increases, the swap-rate weights $w_n(t)$ drift further from their $t = 0$ values, and the convexity correction neglected by the linear approximation becomes more material.

Overall, the approximation appears sufficiently accurate for calibration purposes, though the $\sim$ 1.5–2 bps deviations should be borne in mind when interpreting calibration errors at the sub-basis-point level.

Remark 5.2.1 (Monte Carlo specification)**.** The Monte Carlo model-implied volatilities reported throughout this section are computed from 30,000 simulated paths under the spot measure $\mathbb{Q}^{\text{spot}}$, using an Euler scheme with predictor-corrector drift adjustment [19] and antithetic variates for variance reduction (effective sample size 60,000). The discretisation uses 60 uniform time steps ($\Delta t \approx 0.5\,\text{Y}$). The Brownian motion is projected onto the leading 10 eigenvectors of the parametric correlation matrix (Table 3.1), capturing 97.4% of the total variance.

5.2.2 Calibration Fit: Approximate Model vs. Market

Table 5.3 reports the difference between the approximate model-implied volatilities (the quantities actually minimised during calibration) and the market input volatilities. Since the optimiser targets these approximate volatilities, this table reflects the true calibration residuals.

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	0.00	−0.01	−0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1Y	0.00	−0.01	0.00	0.00	0.00	−0.01	0.01	−0.01	0.00	0.00	0.00	0.00
2Y	0.00	0.00	0.00	0.00	0.00	0.00	−0.01	0.02	−0.02	0.00	0.00	0.00
3Y	0.00	0.00	−0.01	0.00	−0.01	0.01	0.00	−0.02	0.03	−0.02	0.00	0.00
4Y	0.00	0.00	−0.01	−0.01	0.01	−0.02	0.03	−0.02	0.00	0.02	−0.01	0.00
5Y	0.00	0.00	0.00	0.00	−0.01	0.01	−0.03	0.04	−0.03	0.01	0.00	0.00
7Y	0.00	0.00	0.00	0.01	−0.04	0.03	0.00	−0.02	0.02	−0.02	0.01	0.00
10Y	0.00	0.01	−0.02	−0.01	0.06	−0.06	0.02	0.00	−0.02	0.03	−0.01	0.00
12Y	0.00	0.00	0.00	0.00	−0.02	0.03	−0.02	−0.01	0.03	−0.02	0.00	0.00
15Y	0.00	0.01	−0.05	0.05	0.04	−0.12	0.09	−0.06	−0.01	0.05	−0.01	−0.01
20Y	−0.01	0.10	−0.13	−0.01	0.05	−0.01	0.06	−0.11	−0.01	0.16	−0.12	0.01
30Y	−0.04	0.23	−0.19	−0.01	−0.11	0.07	0.03	0.03	−0.08	0.14	−0.08	0.02

Table 5.3: Calibration residuals: approximate model-implied minus market normal volatilities (bps).

The calibration residuals are within ± 0.25 bps. The RMSE across all 144 instruments is 0.04 bps and the maximum absolute error is 0.23 bps.

5.2.3 Calibrated Volatility Grid

Table 5.4 and Figure 5.2 report the calibrated spot volatility grid $G_{i,j}$.

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	69.40	67.94	79.49	64.79	68.31	63.89	62.22	57.69	70.87	64.25	60.60	55.27
1Y	69.74	62.44	68.71	69.35	65.58	66.81	63.45	66.23	66.57	73.54	60.43	72.72
2Y	60.23	58.83	66.53	66.03	70.03	70.41	65.84	73.35	63.76	69.29	68.51	75.56
3Y	72.04	56.92	66.07	60.64	64.28	78.12	63.75	73.13	67.11	55.73	69.71	73.29
4Y	70.06	58.85	65.73	57.27	56.65	76.85	67.17	65.28	65.78	67.44	56.67	59.93
5Y	81.48	51.22	75.34	57.26	49.51	78.72	54.11	77.00	52.49	62.86	56.90	57.64
7Y	86.69	50.19	77.42	50.22	50.18	67.43	57.83	75.02	64.52	44.84	66.01	62.53
10Y	81.98	28.53	82.55	58.95	65.20	46.37	53.75	70.33	68.55	48.37	49.47	65.33
12Y	95.11	40.36	74.19	85.95	33.20	66.94	45.16	71.38	66.59	32.11	55.94	52.54
15Y	78.31	32.90	103.30	30.59	83.12	36.14	60.21	61.68	74.10	36.41	44.65	61.94
20Y	91.07	18.15	80.65	62.69	48.19	81.70	39.53	63.90	69.43	49.85	30.41	62.07
30Y	88.91	14.32	74.01	85.95	89.57	48.45	27.27	85.45	58.01	18.13	31.64	36.70

Table 5.4: Calibrated spot volatility grid $G_{i,j}$ (bps), unregularised.

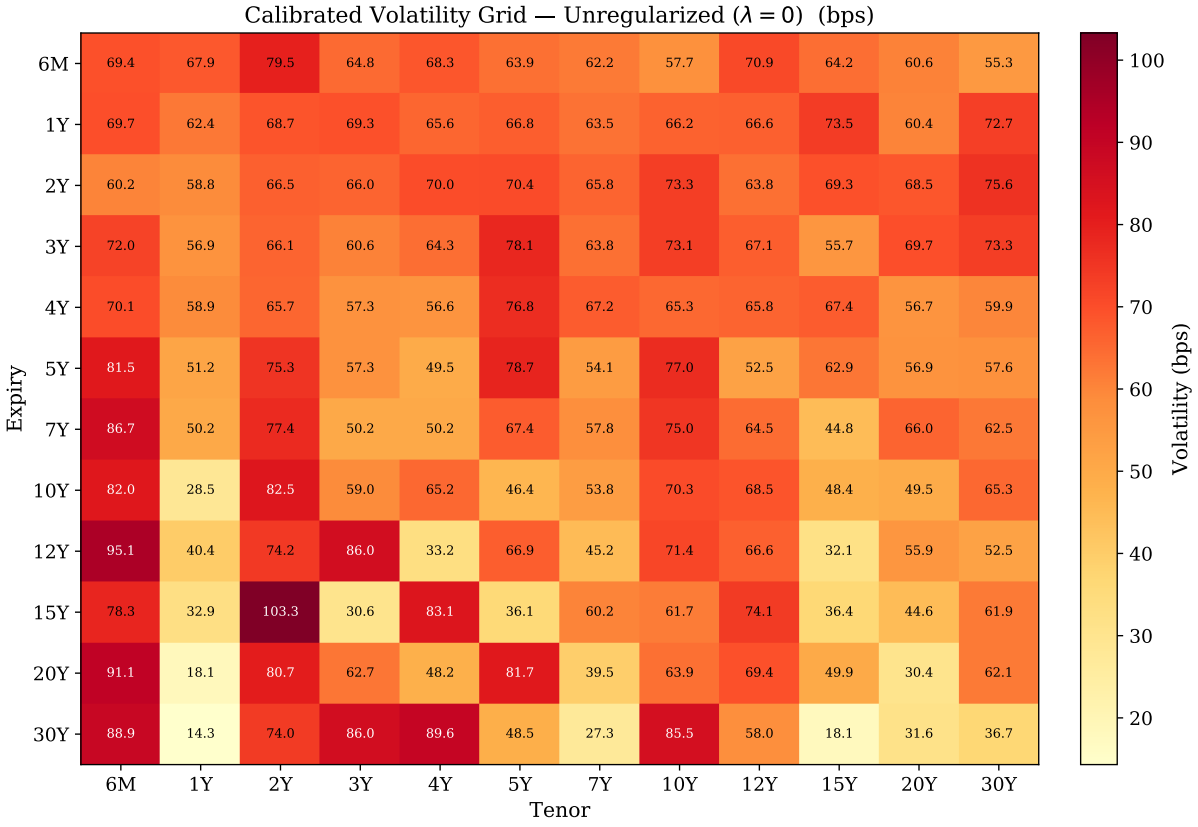


Figure 5.2: Heatmap of Table 5.4.

As visible in Figure 5.2, the calibrated grid exhibits severe pathologies that are characteristic of unregularised high-dimensional calibration.

The grid displays strong non-monotonicity and abrupt jumps across both expiry and tenor. For example, the 2Y tenor column exhibits a pronounced local spike at the 15Y expiry, reaching 103.30 bps, in the 1Y tenor column the volatility drops from 50.19 bps (7Y expiry) to 28.53 bps (10Y expiry) and then rises again to 40.36 bps (12Y expiry); similarly, in the 30Y tenor column

values vary widely, from 75.56 bps (2Y expiry) down to 36.70 bps (30Y expiry).

5.3 Regularised Calibration

As shown in Figure 5.3-Figure 5.5, the recommended regularisation configuration ($\lambda_{tt} = \lambda_{xx} = 0.5$, $\lambda_t = 0.1$) substantially improves the smoothness of the calibrated volatility surface. The extreme spike in the 2Y tenor column is completely removed, with values now ranging from 63 bps to 74 bps across expiries.

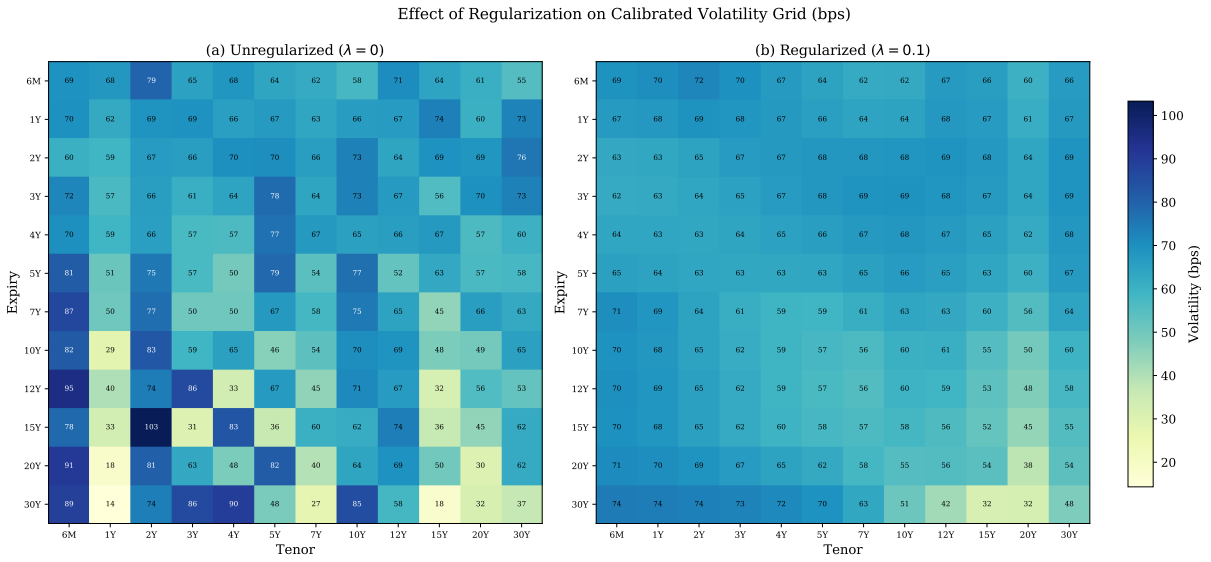


Figure 5.3: Side-by-side comparison of the calibrated volatility grids. Regularisation eliminates the spurious spikes and dips of the volatility surface.

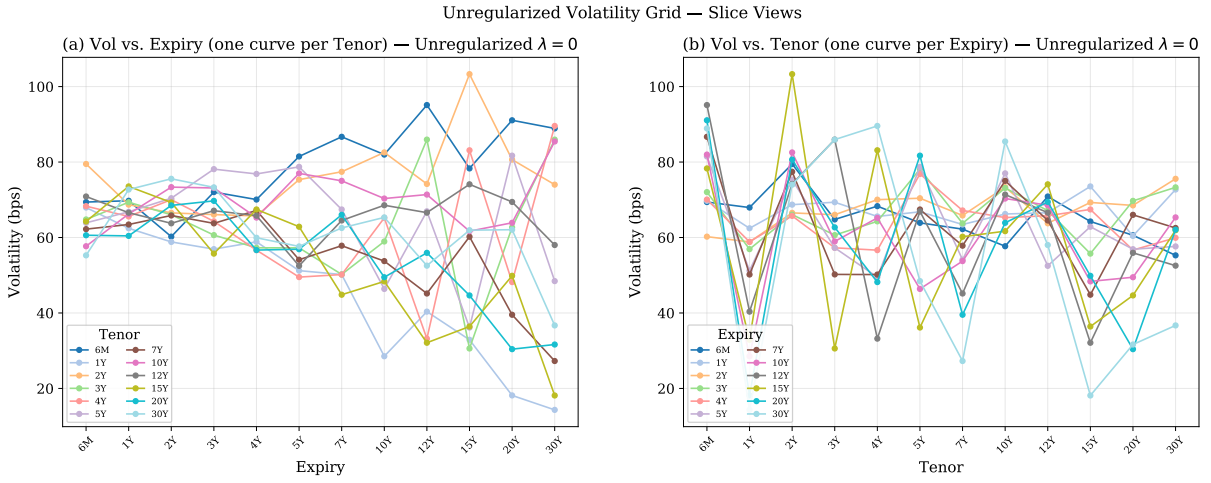


Figure 5.4: Tenor and expiry cross-sections of the calibrated volatility grid for the unregularised configuration.

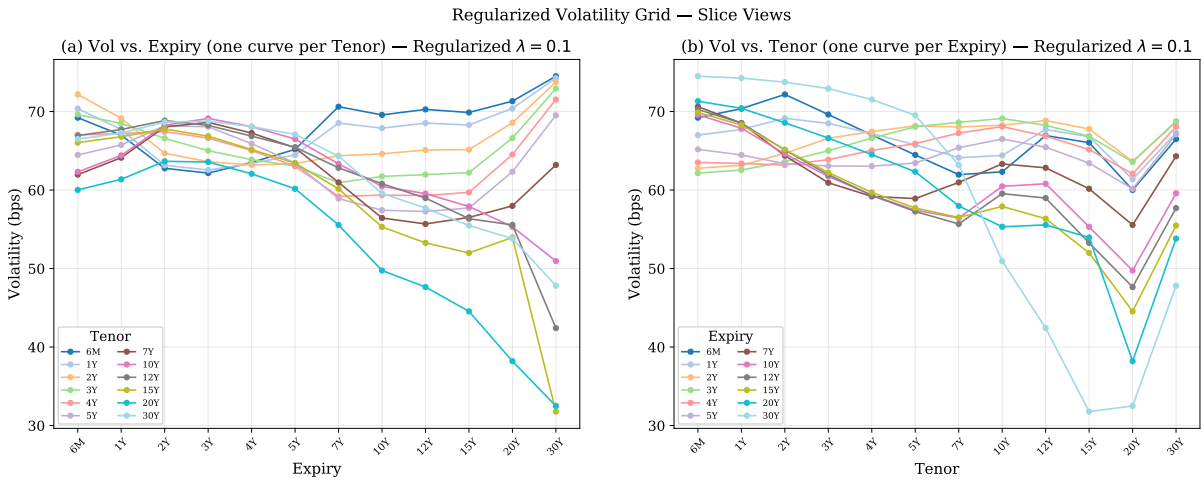


Figure 5.5: Tenor and expiry cross-sections of the calibrated volatility grid for the regularised configuration.

Figure 5.6 shows that errors of the regularised model stay within a reasonable range in comparison to the unregularised case with a difference in end-to-end RMSE less than 0.1 bps.

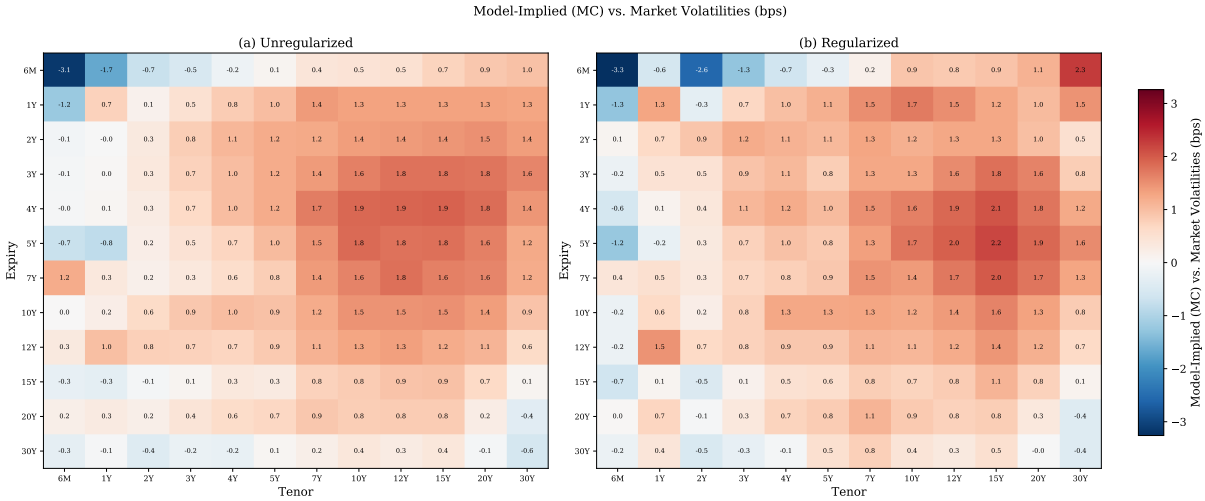


Figure 5.6: Comparison of model-implied volatility residuals (Monte Carlo model vs. market) for the unregularised and regularised configurations. RMSE are respectively 1.046 bps and 1.106 bps.

5.4 Computational Performance

Table 5.5 reports the computational performance of the Levenberg–Marquardt calibration for the unregularised and recommended regularisation configurations. The regularised calibration achieves a 7× speedup in total runtime (948 ms vs. 6607 ms) and a 8× reduction in function evaluations (6 vs. 48) compared to the unregularised case. This dramatic improvement is attributable to the convexifying effect of the regularisation, which stabilises convergence and prevents the optimiser from getting trapped in local minima associated with the oscillatory grid.

Configuration	$\lambda_{tt}, \lambda_{xx}$	λ_t	Function evals	Runtime (ms)
Unregularised	0	0	48	6607
Recommended	0.5	0.1	6	948

Table 5.5: Computational performance of the Levenberg–Marquardt calibration for various regularisation configurations. Grid size: 12×12 (144 parameters). All timings on Intel Xeon Gold 6240R @ 2.40 GHz, single-threaded.

Figure 5.7 shows how runtime scales with the number of parameters.

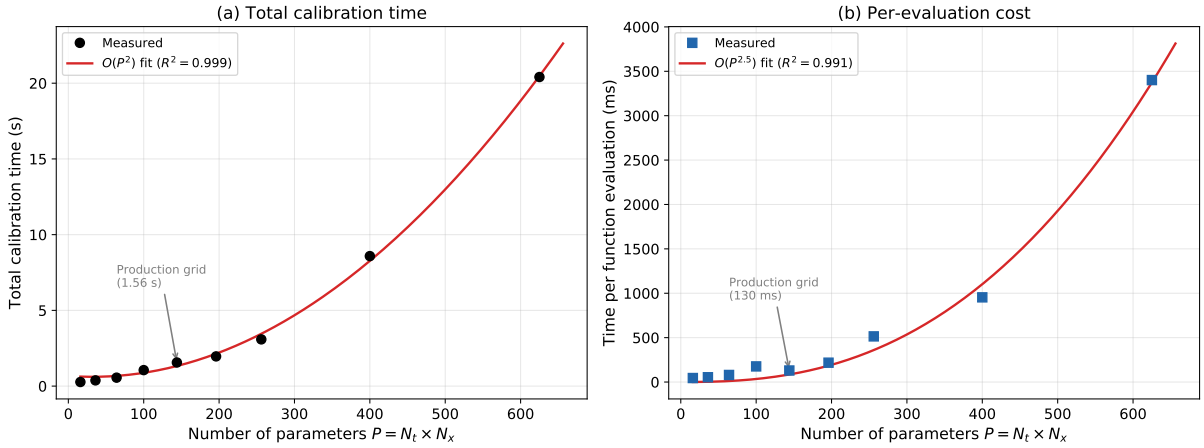


Figure 5.7: Computational scaling of the calibration routine with problem size $P = N_t \times N_x$. (a) Total calibration time scales as $O(P^2)$ ($R^2 = 0.997$). (b) Per-evaluation cost of the objective function scales as $O(P^{2.5})$ ($R^2 = 0.991$). Data points for $P < 64$ are excluded from fits to avoid the overhead-dominated regime. The production grid (12×12, $P = 144$) is annotated for reference.

The observed scaling exponents in Figure 5.7 admit a straightforward interpretation. Each function evaluation requires computing the Jacobian $J \in \mathbb{R}^{M \times P}$ via finite differences, entailing P perturbations of the parameter vector, each of which evaluates $M = P$ residuals through the Rebonato swaption approximation — yielding an $O(P^2)$ cost per evaluation. The total calibration time also scales as $O(P^2)$ rather than $O(P^{2.5})$ because the number of Levenberg–Marquardt iterations remains approximately constant (5–6 evaluations) across grid sizes once regularisation is active: the regularisation term convexifies the objective and stabilises convergence independently of P. The per-evaluation cost exhibits a slightly steeper $O(P^{2.5})$ scaling, which reflects the growing cost of the linear algebra in the LM update step $(J^\top J + \mu I)^{-1} J^\top r$, whose Cholesky factorization scales as $O(P^3)$ but is subdominant relative to the Jacobian computation at the grid sizes tested.

5.5 Sensitivity to Starting Point

The Levenberg–Marquardt algorithm is a local optimiser, and the non-convexity of the objective function over $N_t \times N_x = 144$ parameters raises the question of whether the calibration outcome depends on the initial guess. In this section, we assess the sensitivity of the calibration to

the choice of starting point by running the optimiser from multiple initialisation strategies and comparing the resulting convergence speed and accuracy.

The following strategies are tested:

- **Baseline (flat average):** the current production method. Every node is initialised to the same value, equal to the RMS average of the market swaption volatilities. This is the simplest possible guess and corresponds to a flat volatility surface.
- **Target-informed:** each grid node (t_i, x_j) is initialised to the market-implied normal volatility of the corresponding calibration swaption. The starting point therefore already captures the shape of the term structure, and the optimiser only needs to refine it.
- **Random perturbation:** the flat baseline guess is perturbed independently at each node by additive uniform noise of magnitude $\pm 50\%$ of the baseline value. This tests the robustness of convergence when the starting point is displaced from the flat guess in random directions.
- **Latin Hypercube Sampling (LHS):** the initial values are drawn via stratified random sampling [28] across the full feasible range $[0.2\bar{\sigma},\ 5\bar{\sigma}]$, where $\bar{\sigma}$ is the average target volatility. This produces starting points that may be very far from the true optimum, stress-testing the global convergence properties of the algorithm.
- **Target-informed with noise:** the target-informed initialisation is perturbed by $\pm 15\%$ uniform noise at each node, simulating the effect of imprecise prior information on the volatility surface.

For each stochastic strategy (random perturbation, LHS, target with noise), five independent trials are run. The results are summarised in Table 5.6.

Strategy	Trial	Final Cost	Max|Error| (vol)	RMS Error (vol)	Time (ms)
Baseline (flat avg)	—	0.1577	0.1216	0.0239	1152
Target-informed	—	0.1577	0.1216	0.0239	876
Random perturbation	0	0.1577	0.1216	0.0239	1320
Random perturbation	1	0.1577	0.1216	0.0239	1156
Random perturbation	2	0.1577	0.1216	0.0239	1333
Random perturbation	3	0.1577	0.1216	0.0239	1152
Random perturbation	4	0.1577	0.1216	0.0239	1282
Latin Hypercube	0	9.3941	1.2631	0.2056	3627
Latin Hypercube	1	0.1577	0.1216	0.0239	1184
Latin Hypercube	2	0.1577	0.1216	0.0239	1592
Latin Hypercube	3	0.1577	0.1216	0.0239	1689
Latin Hypercube	4	3.7052	0.3409	0.0825	14581
Target + noise	0	0.1577	0.1216	0.0239	1020
Target + noise	1	0.1577	0.1216	0.0239	1304
Target + noise	2	0.1577	0.1216	0.0239	1461
Target + noise	3	0.1577	0.1216	0.0239	1646
Target + noise	4	0.1577	0.1216	0.0239	2254

Table 5.6: Sensitivity of the Levenberg–Marquardt calibration to the choice of starting point. Final Cost is the sum of squared residuals (fitting and smoothing terms). Max|Error| and RMS Error are computed over the swaption fitting residuals only (in normal volatility units). All trials use the recommended regularisation ($\lambda_t = 0.1$, $\lambda_{tt} = \lambda_{xx} = 0.5$) on the 12×12 grid.

As shown in Table 5.6, the calibration is remarkably robust to the choice of starting point. All strategies that start within a reasonable neighbourhood of the solution — baseline, target-informed, random perturbation, and target with noise — converge to the same optimum (Final Cost ≈ 0.158, RMS ≈ 0.024) with identical accuracy. This confirms that the regularised objective possesses a single dominant basin of attraction, and the reported calibration represents a global (or near-global) minimum rather than a local artefact.

The target-informed strategy achieves a 24% reduction in calibration time (876 ms vs. 1152 ms) compared to the flat baseline, since the optimiser starts closer to the solution and requires fewer Levenberg–Marquardt iterations. This suggests that using the market volatility surface as the initial guess is a free performance improvement for production.

The only failures occur with Latin Hypercube trials 0 and 4, where the starting points are drawn far from the true solution (the feasible range spans $[0.2\bar{\sigma}, 5\bar{\sigma}]$). In these cases, the optimiser converges to a local minimum with substantially higher cost (9.39 and 3.71 respectively) and longer runtime (up to 14.6 s for trial 4). This confirms the expected behaviour: Levenberg–Marquardt's local convergence guarantees break down when initialised far from the global basin.

Chapter 6

Extension of the Calibration to Spread Options

6.1 Purpose and Set-up

In the previous sections we considered an initial version of the LMM with deterministic volatility calibrated on swaptions alone.

Though this version incorporates a relevant part of the market information, it does not allow the calibration of inter-rate correlation. Indeed, it is difficult to determine inter-rate correlation from swaptions alone, since swaption prices mainly constrain volatility levels. One possibility is therefore to include instruments highly sensitive to correlation, such as spread options.

In this chapter, we extend the calibration framework to include ATM spread options as additional calibration instruments. Including ATM spread options allows us to calibrate the instantaneous correlation to market-implied information, rather than relying on correlations estimated from historical time series. This distinction is important because historical correlations do not necessarily reflect the market's forward-looking view of the dependence between swap rates.

To this end, we modify the instantaneous correlation expression in Eq. (3.11) to include a calibrated time-dependent term.

6.2 Cost function and approximation of spread options

6.2.1 Cost function

The cost function introduced in Section 4.2 is extended to include the spread option residuals and smoothing of the additional time-dependent correlation term. The extended cost function

is given by:

$$\begin{aligned}\mathcal{C}(G,\rho) = &\frac{w_S}{N_S}\sum_{i=1}^{N_S}\left(\bar{\sigma}_{\text{swaption},i} - \hat{\sigma}_{\text{swaption},i}\right)^2 + \frac{w_{SO}}{N_{SO}}\sum_{i=1}^{N_{SO}}\left(\bar{\sigma}_{\text{spread},i} - \hat{\sigma}_{\text{spread},i}\right)^2 \\ &+ \frac{\lambda_t}{N_t N_x}\sum_{i,j}(\partial_t G_{i,j})^2 + \frac{\lambda_x}{N_t N_x}\sum_{i,j}(\partial_x G_{i,j})^2 \\ &+ \frac{\lambda_{tt}}{N_t N_x}\sum_{i,j}(\partial_t^2 G_{i,j})^2 + \frac{\lambda_{xx}}{N_t N_x}\sum_{i,j}(\partial_x^2 G_{i,j})^2 \\ &+ \frac{\lambda_{\rho,t}}{N_{SO}}\|\partial_t \rho\|^2 + \frac{\lambda_{\rho,tt}}{N_{SO}}\|\partial_t^2 \rho\|^2,\end{aligned} \tag{6.1}$$

The notation follows the conventions introduced in Eq. (4.6): $\hat{\sigma}$ denotes a market-implied volatility and $\bar{\sigma}$ denotes the corresponding model-implied volatility. In particular, $\hat{\sigma}_{\text{spread},i}$ is the market-implied normal volatility of spread option i, while $\bar{\sigma}_{\text{spread},i}$ is the approximate model-implied normal spread volatility entering the objective. The swaption residuals and the volatility-grid regularisation terms keep the same meaning as in Eq. (4.6); w_{SO} is the weight for the spread option residuals, N_{SO} is the number of spread options in the calibration basket, and $\lambda_{\rho,t}$ and $\lambda_{\rho,tt}$ are the regularisation parameters for the time-dependent correlation term. Section 6.2.2 gives the approximation used to determine $\bar{\sigma}_{\text{spread}}$, all other variables are determined as in Section 4.2.

6.2.2 Approximation of Model-Implied Normal Volatility of Spread Options

The payoff of a spread option at expiry T is given by:

$$\text{Payoff} = \max(S_1(T) - S_2(T) - K, 0), \tag{6.2}$$

where $S_1(T)$ and $S_2(T)$ are the two underlying swap rates and K is the strike price. Following Andersen and Piterbarg [2], we approximate the spread

$$\varepsilon(T) = S_1(T) - S_2(T) \tag{6.3}$$

as a Gaussian variable under the T-forward measure. Its variance is approximated by

$$\text{Var}^T[\varepsilon(T)] \approx \sum_{i=1}^{2}\int_0^T \|\sigma_{S_i}(t)\|^2\,\mathrm{d}t - 2\rho_{\text{term}}(0,T)\prod_{i=1}^{2}\left(\int_0^T \|\sigma_{S_i}(t)\|^2\,\mathrm{d}t\right)^{1/2}, \tag{6.4}$$

where $\sigma_{S_i}(t)$ denotes the effective instantaneous normal volatility vector of swap rate S_i, obtained from the swap-rate volatility approximation in Eq. (4.10). The term correlation over the interval $[T', T]$ is defined as

$$\begin{aligned}\rho_{\text{term}}(T',T) &= \text{Corr}^T\left(S_1(T) - S_1(T'), S_2(T) - S_2(T')\right) \\ &\approx \frac{\int_{T'}^T \sigma_{S_1}(t)^\top \sigma_{S_2}(t)\,\mathrm{d}t}{\left(\int_{T'}^T \|\sigma_{S_1}(t)\|^2\,\mathrm{d}t\right)^{1/2}\left(\int_{T'}^T \|\sigma_{S_2}(t)\|^2\,\mathrm{d}t\right)^{1/2}}.\end{aligned} \tag{6.5}$$

The corresponding approximate model-implied normal spread volatility is then obtained from

$$\bar{\sigma}_{\text{spread}}(T) \approx \left(\frac{1}{T} \operatorname{Var}^{T}[\varepsilon(T)] \right)^{1/2}, \tag{6.6}$$

and is fitted to the market normal volatilities through the spread-option residuals in Eq. (6.1).

Although the formula we use is slightly more sophisticated, it builds roughly on the same principles. We cannot disclose it due to confidentiality reasons.

6.3 Results

6.3.1 Fitting results

Additionally to the basket of swaptions described in Section 5.1, we include ATM spread options written on the 10Y-2Y and 30Y-2Y swap-rate spreads. For each spread, the calibration basket spans the 16 expiries $\{6\text{M}, 1\text{Y}, 2\text{Y}, 3\text{Y}, 4\text{Y}, 5\text{Y}, 6\text{Y}, 7\text{Y}, 8\text{Y}, 9\text{Y}, 10\text{Y}, 12\text{Y}, 15\text{Y}, 20\text{Y}, 25\text{Y}, 30\text{Y}\}$. The calibration is performed on the same market data as of 20 April 2026.

Tables 6.1 and 6.2 report the ATM spread-option normal volatilities across the 16 expiries. For each expiry we compare the market-implied normal volatility (the calibration target) against the calibrated LMM model-implied volatility — obtained by full Monte Carlo pricing followed by Bachelier inversion — and against the approximate volatility given by the closed-form expression used in the calibration objective.

We can see that, for the 10Y-2Y spread, the calibrated Monte-Carlo model reproduces the market volatilities to within 0.58 bps and the closed-form approximation tracks the Monte-Carlo model to within 0.42 bps across all expiries. For the wider 30Y-2Y spread both errors grow at the long end, where the model departs from the market by up to 1.75 bps and from the approximation by up to 1.69 bps (both at the 30Y expiry). These residuals are examined in detail below.

	Normal volatility (bps)			Difference (bps)		
Expiry	Market	Model	Approx.	Model−Mkt	Model−Approx	Approx−Mkt
6M	41.21	41.54	41.60	0.34	−0.06	0.40
1Y	41.64	41.06	41.22	−0.58	−0.16	−0.42
2Y	39.56	39.27	39.48	−0.30	−0.21	−0.08
3Y	37.67	37.83	37.76	0.16	0.07	0.08
4Y	36.33	36.29	36.29	−0.04	0.00	−0.04
5Y	35.20	35.11	35.15	−0.09	−0.05	−0.05
6Y	34.27	34.18	34.24	−0.09	−0.06	−0.03
7Y	33.46	33.49	33.49	0.03	0.00	0.03
8Y	32.69	32.55	32.74	−0.14	−0.19	0.05
9Y	32.05	31.98	32.18	−0.07	−0.20	0.13
10Y	31.63	31.39	31.55	−0.24	−0.16	−0.08
12Y	30.74	30.74	30.71	0.00	0.03	−0.03
15Y	29.78	29.88	29.83	0.10	0.06	0.04
20Y	28.90	28.97	28.91	0.07	0.06	0.01
25Y	28.39	28.20	28.40	−0.19	−0.20	0.01
30Y	28.10	27.65	28.07	−0.46	−0.42	−0.03

Table 6.1: ATM normal volatilities of the 10Y-2Y spread option (bps) across expiries. *Market* denotes the market-implied normal volatility used as calibration target; *Model* the calibrated LMM Monte Carlo model-implied volatility; *Approx* the closed-form approximation entering the calibration objective. The right-hand block reports the pairwise differences.

	Normal volatility (bps)			Difference (bps)		
Expiry	Market	Model	Approx.	Model−Mkt	Model−Approx	Approx−Mkt
6M	61.02	62.21	61.37	1.19	0.84	0.35
1Y	60.44	60.78	60.16	0.34	0.62	−0.28
2Y	57.12	57.03	56.97	−0.10	0.06	−0.15
3Y	54.24	54.69	54.28	0.45	0.41	0.03
4Y	52.14	52.36	52.12	0.22	0.25	−0.03
5Y	50.30	50.58	50.32	0.28	0.26	0.02
6Y	48.81	48.66	48.83	−0.15	−0.17	0.02
7Y	47.51	46.95	47.53	−0.55	−0.57	0.02
8Y	46.31	45.61	46.32	−0.70	−0.71	0.01
9Y	45.34	44.71	45.43	−0.63	−0.73	0.10
10Y	44.63	43.86	44.55	−0.76	−0.68	−0.08
12Y	43.21	42.33	43.20	−0.88	−0.87	−0.01
15Y	41.72	40.98	41.71	−0.74	−0.73	−0.01
20Y	40.22	39.57	40.28	−0.65	−0.71	0.06
25Y	40.33	39.02	40.34	−1.31	−1.32	0.01
30Y	40.67	38.92	40.61	−1.75	−1.69	−0.06

Table 6.2: ATM normal volatilities of the 30Y-2Y spread option (bps) across expiries. *Market* denotes the market-implied normal volatility used as calibration target; *Model* the calibrated LMM Monte Carlo model-implied volatility; *Approx* the closed-form approximation entering the calibration objective. The right-hand block reports the pairwise differences.

To check that the additional spread-option targets do not materially deteriorate the swaption fit, Table 6.3 reports the swaption volatility residuals obtained when calibrating with the 10Y-2Y spread options. Residuals are Monte Carlo model-implied minus market normal volatilities, in bps.

Swaption residuals remain of the same order as in the regularised calibration of Section 5.3. For the calibration including 10Y-2Y spread options, the swaption residual RMSE is 0.54 bps and the maximum absolute residual is 1.71 bps, comparable with the end-to-end regularised swaption residuals shown in Figure 5.6.

For the 10Y-2Y spread, the calibrated model reproduces the market spread volatilities closely: the model-implied differences (Model−Mkt) remain within ±0.6 bps across the entire expiry range (MC−market RMSE 0.24 bps over the 16 expiries), with the largest deviations at the 1Y (−0.58 bps) and 30Y (−0.46 bps) expiries and sub-0.3 bps errors elsewhere. The approximation used inside the objective is likewise accurate, with Model−Approx differences within ±0.42 bps, confirming that the closed-form spread-volatility formula is adequate for calibration purposes.

For the 30Y-2Y spread, the approximate volatilities remain very close to the market targets, with Approx−Mkt differences within ±0.35 bps and mostly close to zero beyond the 3Y expiry. The Monte Carlo model-implied volatilities are within roughly ±0.9 bps up to the 20Y expiry (MC−market RMSE 0.80 bps over the 16 expiries), while larger long-end deviations appear at 25Y (−1.31 bps) and 30Y (−1.75 bps).

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	−0.54	0.88	−1.71	−0.29	0.00	0.14	−0.08	0.41	0.33	0.19	0.50	1.53
1Y	−1.39	0.53	−0.77	−0.11	−0.06	−0.12	−0.01	0.44	0.39	0.20	0.20	0.32
2Y	0.09	1.33	0.77	0.79	0.54	0.46	0.77	0.82	0.72	0.47	0.27	−0.26
3Y	−0.60	0.71	0.34	0.38	0.33	−0.10	0.34	0.31	0.34	0.39	0.36	−0.51
4Y	−0.47	0.91	0.45	0.66	0.51	0.19	0.52	0.32	0.24	0.21	0.10	−0.26
5Y	−0.43	1.13	0.60	0.53	0.44	0.12	0.26	0.10	0.07	0.20	0.25	0.15
7Y	−0.70	0.66	0.12	0.25	0.25	0.29	0.21	−0.48	−0.40	−0.01	0.02	−0.23
10Y	−0.73	0.99	−0.08	−0.03	0.12	−0.12	−0.40	−0.74	−0.57	−0.22	−0.59	−0.69
12Y	−0.79	0.80	−0.20	−0.26	−0.29	−0.43	−0.41	−0.43	−0.29	−0.08	−0.41	−0.59
15Y	−1.10	0.46	−0.51	−0.24	−0.10	−0.15	−0.04	−0.31	−0.33	−0.26	−0.41	−0.73
20Y	−0.62	0.91	−0.11	0.04	0.20	0.14	0.26	−0.30	−0.43	−0.34	−0.52	−1.03
30Y	−0.77	0.58	−0.56	−0.60	−0.59	−0.19	−0.15	−0.61	−0.75	−0.43	−0.88	−1.39

Table 6.3: Swaption volatility residuals (Monte Carlo model-implied minus market normal volatility, bps) for the calibration including the 10Y-2Y spread-option targets.

Computational cost. Adding the spread-option targets substantially increases the calibration time relative to the swaption-only baseline. Calibrating the volatility grid together with the time-dependent correlation term takes 17.24 s for the 10Y-2Y basket and 27.73 s for the 30Y-2Y basket, against the 948 ms base case of the recommended regularised swaption-only calibration (144 swaptions, no spread, see Table 5.5). All timings were measured single-threaded on the Intel Xeon Gold 6240R (2.40 GHz) of Section 5.4.

6.3.2 Impact on correlation

Figures 6.1 and 6.2 compare, on the same 120×120 simulation grid, the fine-grid empirical fit of Section 5.3 against the two correlation matrices obtained by including the 10Y-2Y and 30Y-2Y CMS spread options in the objective.

Both spread-fitted matrices exhibit a markedly steep decrease of correlation as the tenor distance $|T_i - T_j|$ grows: the off-diagonal decay is faster than in the empirical fit, and the long-tenor plateau is pushed to lower correlation levels, with mildly negative values appearing between the very short and very long forward rates. The effect is more pronounced for the 10Y-2Y basket than for the 30Y-2Y one: averaging the correlation across all pairs separated by a 20-year tenor distance yields 0.76 under the empirical fit, 0.50 with the 30Y-2Y spread targets and 0.38 with the 10Y-2Y ones. Only the 10Y-2Y calibration produces (mildly) negative correlations at the extremes. The 3D surfaces of Figure 6.2 make this contrast visible: the empirical surface is comparatively flat with a broad high-correlation ridge, whereas the spread-fitted surfaces show a sharper ridge along the diagonal and steeper flanks away from it, most pronounced in the 10Y-2Y panel.

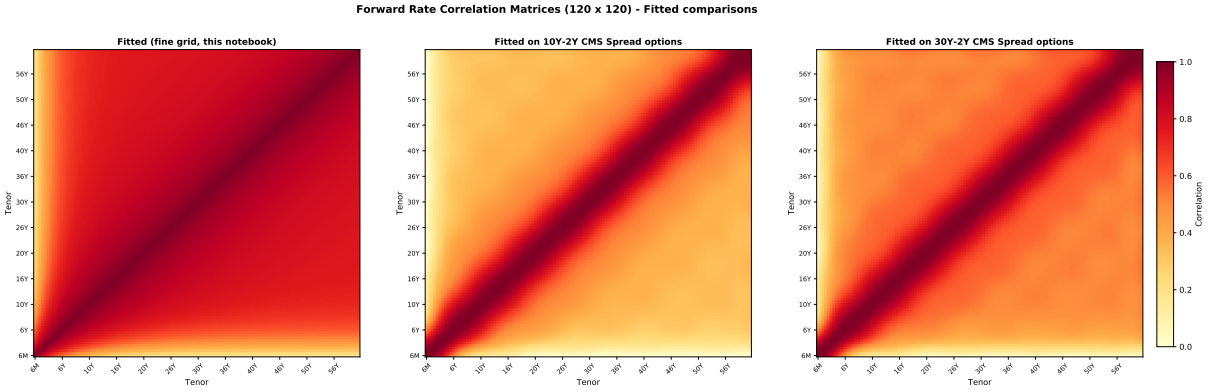


Figure 6.1: Forward rate correlation matrices (120×120) on the fine simulation grid (6M to 60Y in 6-month steps). Left: fitted parametric correlation from the empirical calibration of Section 5.3. Middle: correlation obtained by adding 10Y-2Y CMS spread options to the objective. Right: same with 30Y-2Y CMS spread options.

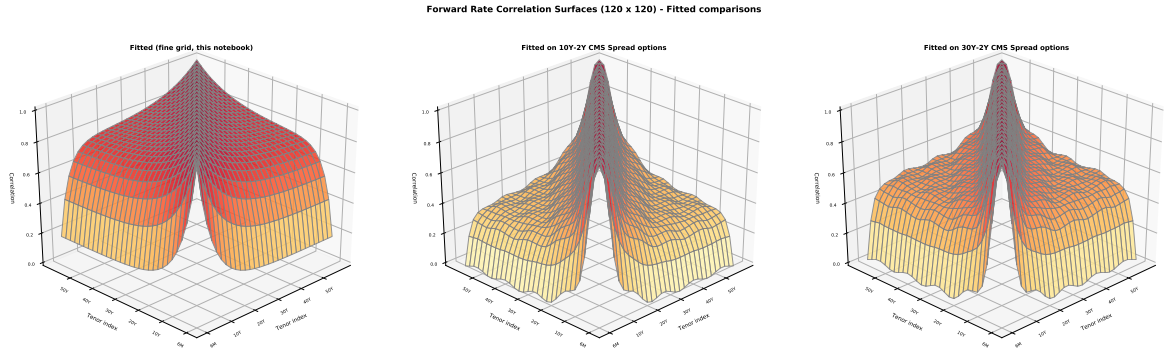


Figure 6.2: Three-dimensional correlation surfaces corresponding to the heatmaps of Figure 6.1. Including CMS spread-option targets produces a steeper decorrelation with tenor distance than the empirical fit, most pronounced for the 10Y-2Y basket.

Chapter 7

Extension to Displaced Log-Normal Specification

7.1 Purpose

So far we have considered only a normal specification for the volatility of the LIBOR rates. This specification produces implied volatilities that are independent of the strike. In reality, markets exhibit a volatility smile, with a strong dependence of the implied volatility on the strike.

A natural first step is to fit the skew of the smile, that is, its first-order slope near the ATM level; in practice one fits the smile at a small number of points around ATM. One way to introduce such a skew into the LIBOR-rate dynamics is to specify them as displaced log-normal processes. This way, one can gain an additional degree of freedom in the calibration, which allows us to fit the market-implied skew.

In this chapter, we extend the calibration framework to a displaced log-normal specification for the forward rates. This displaced log-normal model allows for more flexible interest-rate dynamics and lets us calibrate to the swaption smile through a displacement surface fitted to the market-implied skew.

7.2 Definitions and dynamics

We introduce a displaced log-normal volatility specification by adding a local volatility component to the diffusion coefficient of each forward rate. Under the T_{n+1}-forward measure $\mathbb{Q}^{n+1}$, the forward rate dynamics are written as

$$\mathrm{d}L_n(t) = \Big(b_n(t)L_n(t) + (1 - b_n(t))L_n(0)\Big)\lambda_n(t)^\top \mathrm{d}W^{n+1}(t),$$

and we define

$$\varphi_n(t, x) = b_n(t)x + (1 - b_n(t))L_n(0).$$

The function $b_n(t)$ controls the interpolation between normal-style dynamics ($b_n(t) = 0$) and log-normal dynamics ($b_n(t) = 1$). It is represented on an expiry-tenor grid and recovered between nodes by the same piecewise-linear interpolation used for $\|\lambda_n(t)\|$.

7.3 Calibration

7.3.1 Cost function

Let G denote the instantaneous volatility grid, b the displacement surface, and ρ the calibrated correlation term structure. The displaced log-normal cost function is

$$\begin{aligned}\mathcal{C}_{\mathrm{DLN}}(G,b,\rho) = &\frac{w_S}{N_S}\sum_{i=1}^{N_S}\left(\bar{\sigma}^{\mathrm{DLN}}_{\mathrm{swaption},i} - \hat{\sigma}^{\mathrm{DLN}}_{\mathrm{swaption},i}\right)^2 + \frac{w_{SO}}{N_{SO}}\sum_{i=1}^{N_{SO}}\left(\bar{\sigma}_{\mathrm{spread},i} - \hat{\sigma}_{\mathrm{spread},i}\right)^2 \\ &+ \frac{w_{b_S}}{N_{b_S}}\sum_{i=1}^{N_{b_S}}\left(\bar{b}_{\mathrm{S},i} - \hat{b}_{\mathrm{S},i}\right)^2 \\ &+ \frac{\lambda_t}{N_tN_x}\sum_{i,j}(\partial_t G_{i,j})^2 + \frac{\lambda_x}{N_tN_x}\sum_{i,j}(\partial_x G_{i,j})^2 \\ &+ \frac{\lambda_{tt}}{N_tN_x}\sum_{i,j}(\partial_t^2 G_{i,j})^2 + \frac{\lambda_{xx}}{N_tN_x}\sum_{i,j}(\partial_x^2 G_{i,j})^2 \\ &+ \frac{\lambda_{b,1}}{N_tN_x}\sum_{i,j}\|\nabla b_{i,j}\|^2 + \frac{\lambda_{b,2}}{N_tN_x}\sum_{i,j}\|\nabla^2 b_{i,j}\|^2 \\ &+ \frac{\lambda_{\rho,t}}{N_{SO}}\|\partial_t\rho\|^2 + \frac{\lambda_{\rho,tt}}{N_{SO}}\|\partial_t^2\rho\|^2.\end{aligned} \tag{7.1}$$

The notation follows Eq. (4.6) and Eq. (6.1). The term $\hat{\sigma}^{\mathrm{DLN}}_{\mathrm{swaption},i}$ is the market-implied scalar displaced log-normal volatility (we discuss how this quantity is determined in Section 7.3.2), while $\bar{\sigma}^{\mathrm{DLN}}_{\mathrm{swaption},i}$ is the corresponding model-implied displaced log-normal swaption volatility. The spread-option volatility terms $\hat{\sigma}_{\mathrm{spread},i}$ and $\bar{\sigma}_{\mathrm{spread},i}$ remain normal spread volatilities, quoted in the same Bachelier convention as in Chapter 6 ($\hat{\sigma}_{\mathrm{spread},i}$ being given by the market). Here $\bar{b}_{\mathrm{S},i}$ is the model-implied swap-rate displacement (as obtained in Section 4.4) and $\hat{b}_{\mathrm{S},i}$ is the corresponding market-implied displacement (as obtained in Section 7.3.2). The two b-regularisation terms are first- and second-order finite-difference smoothing penalties on the expiry-tenor displacement grid. If spread options are not included, the spread-volatility residual block and the ρ-regularisation terms are omitted.

To calibrate the displaced log-normal specification, we need enhanced approximations for $\bar{\sigma}^{\mathrm{DLN}}_{\mathrm{swaption}}$ and $\bar{\sigma}_{\mathrm{spread}}$ beyond the normal formulas used in Sections 4.4 and 6.2.2. These enhancements are discussed in Sections 7.3.3 and 7.3.4, respectively.

7.3.2 Market-implied swaption skew

Let us consider a payer swaption with maturity T_j, with the underlying being a fixed-for-floating swap paying fixed coupons of c against LIBOR at times $T_{j+1},\ldots,T_k$, where $j<k$. The annuity for the swap is defined as:

$$A(t) = A_{j,k-j}(t) = \sum_{n=j}^{k-1}\tau_n P(t,T_{n+1}), \quad \tau_n = T_{n+1}-T_n, \tag{7.2}$$

and the par forward swap rate is defined as:

$$S(t) = S_{j,k}(t) = \frac{P(t,T_j) - P(t,T_k)}{A(t)}.$$

We suppose that $S(t)$ follows the following dynamics with constant displacement under the annuity measure $\mathbb{Q}^A$:

$$\mathrm{d}S(t) = (\hat{b}_S S(t) + (1-\hat{b}_S)S(0))\hat{\sigma}_S^{\mathrm{DLN}}\mathrm{d}W^A(t).$$

Under this assumption, the price of a payer swaption is given by:

$$\bar{V}^{\mathrm{DLN}}_{\mathrm{swaption}} = A(0)\, c_B\big(0,\, S(0)/\hat{b}_S;\, T,\, c - S(0) + S(0)/\hat{b}_S;\, \hat{b}_S\hat{\sigma}_S^{\mathrm{DLN}}\big),$$

where $c_B(t,S;T,K;\sigma)$ is the Black call option formula with time to expiry $T-t$, spot S, strike K and volatility σ.

We then invert the price with the Bachelier formula to obtain a swaption normal volatility $\bar{\sigma}^{\mathrm{normal}}(\hat{\sigma}_S, \hat{b}_S; K)$ at strike K.

To determine the market-implied skew, we solve the following optimisation problem for each swaption, fitting jointly across the quoted strikes $K \in \mathcal{K}$:

$$\hat{b}_S = \arg\min_{b_S} \min_{\sigma_S} \sum_{K\in\mathcal{K}} \big(\bar{\sigma}^{\mathrm{normal}}(\sigma_S, b_S; K) - \hat{\sigma}^{\mathrm{normal}}_{\mathrm{swaption}}(K)\big)^2, \tag{7.3}$$

where $\mathcal{K}$ denotes the set of quoted strike offsets (here ATM $-$ 25, ATM and ATM $+$ 25 bp): a single strike would leave the pair (σ_S, b_S) underdetermined, whereas fitting several strikes simultaneously identifies the displacement $\hat{b}_S$.

7.3.3 Enhancement of swap rate approximation

The previous approximation we used for the swap rate dynamics did not account for the displacement and was derived as an approximation in the normal case. The goal of this subsection is therefore to derive an enhanced approximation for the swap-rate dynamics that can yield both an effective volatility and an effective displacement (skew) for the swap rate. We mostly follow Andersen and Piterbarg [2].

We consider the same swaption as in Section 7.3.2 as well as the same notations. Let us denote by $\mathbf{L}(t)$ the vector of all LIBOR rates. By the Itô formula (see Appendix B for the details), the dynamics of the swap rate $S(t)$ under the annuity measure $\mathbb{Q}^A$ are given by:

$$\mathrm{d}S(t) = \lambda_S(t, \mathbf{L}(t))^\top \mathrm{d}W^A(t),$$

where W^A is a multi-dimensional Brownian motion under $\mathbb{Q}^A$ and where

$$\lambda_S(t, \mathbf{L}(t)) = \sum_{n=j}^{k-1} \frac{\partial S(t)}{\partial L_n(t)} \varphi_n(t, L_n(t))\lambda_n(t).$$

We can write this as a one-dimensional form:

$$\mathrm{d}S(t) = \|\lambda_S(t, \mathbf{L}(t))\|\,\mathrm{d}Y^A(t),$$

where Y^A is a one-dimensional Brownian motion under $\mathbb{Q}^A$.

We use Markovian projection to approximate the dynamics of $S(t)$ under $\mathbb{Q}^A$ by a process whose diffusion coefficient depends on the current swap rate alone — obtained from the conditional expectation of the instantaneous variance given $S(t)$ (see Section 15.2 and Appendix A of [2] for the derivation):

$$\mathrm{d}S(t) \approx p_S(t)\Big(b_S(t)S(t) + (1 - b_S(t))S(0)\Big)\mathrm{d}Y^A(t),$$

with

$$\begin{aligned} p_S(t) &= \frac{1}{S(0)}\,\|\lambda_S(t, \mathbb{E}^A[\mathbf{L}(t)])\|, \\ b_S(t) &= \frac{1}{S(0)\,p_S(t)} \sum_{n=j}^{k-1} \frac{\partial\|\lambda_S(t, \mathbb{E}^A[\mathbf{L}(t)])\|}{\partial L_n(t)}\,\frac{\int_0^t \lambda_n(u)^\top \lambda_S(u, \mathbf{L}(0))\,\mathrm{d}u}{\int_0^t \|\lambda_S(u, \mathbf{L}(0))\|^2\,\mathrm{d}u}. \end{aligned}$$

By a convexity adjustment (a change of measure to the annuity measure followed by a first-order Taylor expansion; see Appendix B for the details), we obtain:

$$\mathbb{E}^A[L_n(t)] \approx L_n(0)(1 + c_n(t)), \tag{7.4}$$

with

$$c_n(t) = \frac{1}{L_n(0)Q_n(0)} \sum_{i=j}^{k-1} \frac{\partial Q_n(0)}{\partial L_i(0)} \int_0^t \lambda_i(s)^\top \lambda_n(s)\,\mathrm{d}s, \qquad Q_n(t) = \frac{A(t)}{P(t, T_{n+1})}. \tag{7.5}$$

To approximate the swap rate skew, let us define:

$$v_n(t) = \frac{\partial S(t)}{\partial L_n(t)}, \qquad v_{n,n'}(t) = \frac{\partial^2 S(t)}{\partial L_n(t)\partial L_{n'}(t)}, \tag{7.6}$$

$$r_{S,i}(t) = \lambda_i(t)^\top \lambda_S(t, \mathbf{L}(0)), \qquad r_S(t) = \|\lambda_S(t, \mathbf{L}(0))\|^2, \tag{7.7}$$

$$\xi_i(t) = \frac{r_{S,i}(t)\int_0^t r_{S,i}(u)\,\mathrm{d}u}{r_S(t)\int_0^t r_S(u)\,\mathrm{d}u}. \tag{7.8}$$

Matching the slope of the projected local volatility to the displaced log-normal form, via a second-order expansion of $\|\lambda_S\|$ in the forwards (see Appendix B for the details), we obtain the following approximation for the swap-rate skew:

$$b_S(t) = \sum_{i,n=j}^{k-1} \frac{r_{S,n}(t)}{r_{S,i}(t)} v_{i,n}(0)\xi_i(t) + \sum_{i=j}^{k-1} b_i(t)v_i(0)\xi_i(t). \tag{7.9}$$

We can then obtain the following model-implied volatility and skew:

$$\bar{\sigma}_S^{\mathrm{DLN}} = \left(\frac{1}{T}\int_0^T p_S(t)^2\,\mathrm{d}t\right)^{1/2}, \tag{7.10}$$

$$\bar{b}_S = S(0) \int_0^T b_S(t) w_T(t)\,\mathrm{d}t, \tag{7.11}$$

where w_T and ν are defined as:

$$w_T(t) = \frac{\nu(t)^2 p_S(t)^2}{\int_0^T \nu(u)^2 p_S(u)^2\,\mathrm{d}u}, \qquad \nu(t)^2 = \int_0^t p_S(s)^2\,\mathrm{d}s.$$

7.3.4 Spread option approximations from the literature

The goal of this subsection is to derive an enhanced approximation for the spread option price and the model-implied spread normal volatility.

In the normal specification of Section 6.2.2, each swap rate is approximately Gaussian, so the spread $\varepsilon = S_1 - S_2$ is Gaussian too and its ATM normal volatility follows from its variance alone. Under the displaced log-normal specification each swap rate is instead displaced-lognormal, and therefore skewed, so the spread is no longer Gaussian and its distribution can no longer be summarised by a single variance. Thus a richer approximation is required.

Approximating the spread option volatility under the displaced log-normal LMM proved substantially harder. We first present the method described in Andersen and Piterbarg [2].

Let us consider a CMS spread option with maturity T and strike K between swap rate S_1 and S_2. For simplicity, we suppose that the payment date T_p is equal to the maturity T. The undiscounted value of the option under the LIBOR Market Model can be approximated by:

$$V_{CMS}(0) \approx \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \left((\Psi_1^T)^{-1}(\Phi(z_1)) - (\Psi_2^T)^{-1}(\Phi(z_2)) - K\right)^+ \phi(z_1, z_2; \rho_{\text{term}}(0,T))\,\mathrm{d}z_1\,\mathrm{d}z_2,$$

where Ψ_i^T is the T-forward measure CDF of the swap rate S_i, Φ is the standard Gaussian CDF, and ϕ is a two-dimensional Gaussian density defined, for all $z_1, z_2 \in \mathbb{R}$ and correlation R, by

$$\phi(z_1, z_2; R) = \frac{1}{2\pi\sqrt{1-R^2}} \exp\left(-\frac{z_1^2 - 2Rz_1z_2 + z_2^2}{2(1-R^2)}\right),$$

and ρ_{term} is defined in (6.5).

To obtain Ψ^T, one can use the following formula:

$$\forall s \in \mathbb{R}, \quad \Psi^T(s) = \frac{A(0)}{P(0,T)} \int_{-\infty}^{s} \alpha(u)\psi^A(u)\,\mathrm{d}u,$$

where A is the annuity defined in (7.2), ψ^A is the probability density function of the swap rate under its own annuity measure — available in closed form because the swap rate is approximated by a displaced log-normal process — and α is an annuity mapping function defined as:

$$\forall s \in \mathbb{R}, \quad \alpha(s) = E^A\left[\frac{1}{A(T)}\middle| S(T) = s\right].$$

Let $\mathbf{L}(T) = (L_0(T), \ldots, L_{N-1}(T))$ denote the vector of all LIBOR rates at time T. We can

approximate $\alpha(s)$ by:

$$\alpha(s) = E^A\left[\frac{1}{A(T)}\middle| S(T) = s\right] = E^A[\rho(\mathbf{L}(T)) \mid S(T) = s]$$
$$\approx \rho(E^A[\mathbf{L}(T) \mid S(T) = s]),$$

where

$$\rho(x) = \left(\sum_{n=0}^{N-1} \tau_n \prod_{i=0}^{n} (1 + \tau_i x_i)^{-1}\right)^{-1}.$$

Still following [2], we can approximate, for all $n \in \{0, \ldots, N-1\}$,

$$E^A[L_n(T) \mid S(T) = s] \approx L_n(0)\left(1 + \beta_n \frac{s - S(0)}{S(0)}\right),$$

where $\beta_n = \frac{S(0)\int_0^T \lambda_n(t)^\top \lambda_S(t)\,\mathrm{d}t}{L_n(0)\int_0^T \|\sum_{i=0}^{N-1} w_i \lambda_i(t)\|^2\,\mathrm{d}t}$ and $w_i = \frac{\partial S(0)}{\partial L_i(0)}$ is the swap-rate sensitivity to the LIBOR forward rate, given in closed form in Appendix A. Thus

$$\alpha(s) \approx \left(\sum_{n=0}^{N-1} \tau_n \prod_{i=0}^{n} (1 + \tau_i l_i(s))^{-1}\right)^{-1},$$

where $l_n(s) = L_n(0)\left(1 + \beta_n \frac{s - S(0)}{S(0)}\right)$.

It is useful to correct α to take arbitrage into account. By no-arbitrage, we have

$$E^A[\alpha(S(T))] = E^A\left[\frac{P(T,T)}{A(T)}\right] = \frac{P(0,T)}{A(0)}.$$

But due to numerical approximations, this equality can be violated. We then correct α by using instead

$$\tilde{\alpha}(\cdot) = \frac{P(0,T)}{A(0)} \frac{\alpha(\cdot)}{E^A[\alpha(S(T))]}.$$

Along with the approximation just derived, [2] also propose a simpler approximation that does not take the displaced log-normal dynamics into account.

Figures 7.1 and 7.2 show the performance of these two approximations — the displaced log-normal annuity mapping derived above and the simpler variant that ignores the displacement — against the Monte-Carlo reference for the 10Y-2Y CMS spread, reported respectively in terms of the ATM implied normal volatility and the convexity-adjusted forward. The accuracy of both tends to deteriorate at longer expiries.

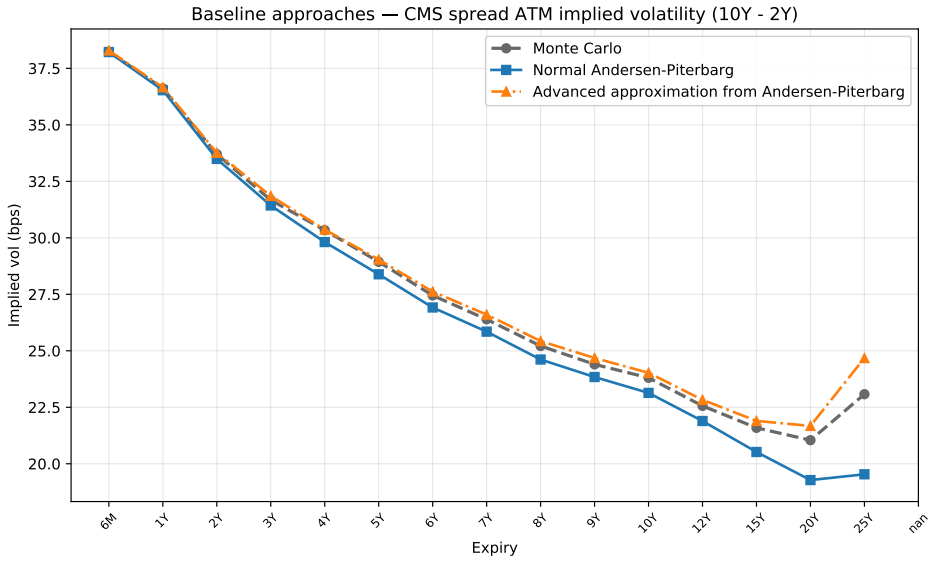


Figure 7.1: 10Y-2Y CMS spread: ATM implied normal volatility (bps) across expiries for the two Andersen–Piterbarg approximations — with and without the displaced log-normal adjustment — against the Monte-Carlo reference (black).

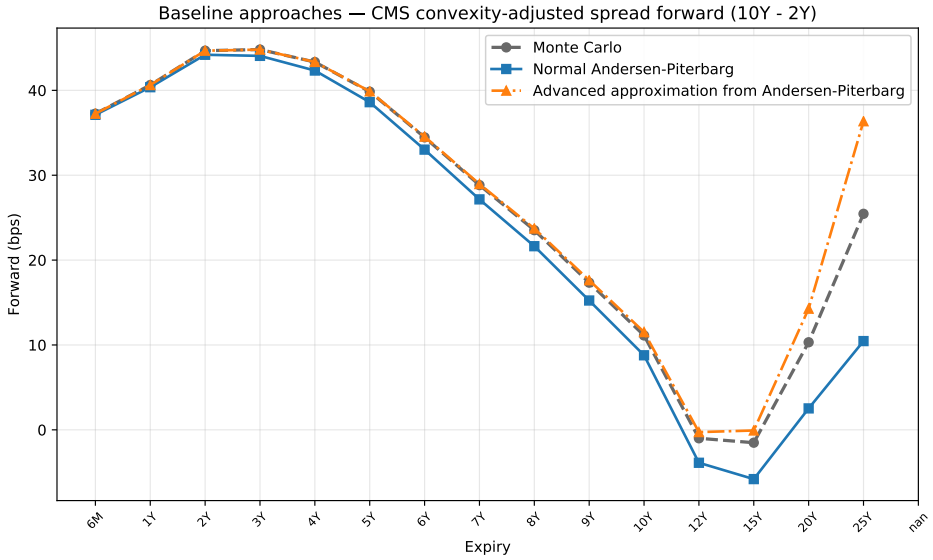


Figure 7.2: 10Y-2Y CMS spread: convexity-adjusted forward (bps) across expiries for the two Andersen–Piterbarg approximations — with and without the displaced log-normal adjustment — against the Monte-Carlo reference (black).

7.3.5 Internally developed approximation

As we were not satisfied with the performance of the approximation mentioned above we investigated several other possibilities. For confidentiality reasons, the exact formulas are not disclosed here. Different possibilities exist when it comes to improving the approximation of the CMS spread normal volatility:

- Improve the approximation of $E^A[L_n(T) \mid S(T) = s]$;
- Improve the approximation of $\alpha(s) := E^A\left[\frac{1}{A(T)}\middle| S(T) = s\right]$;

- Follow a different approach altogether, e.g. based on Kirk's formula or on moment matching.

Annuity-mapping approximations We first chose to focus on the improvement of the annuity mapping function α. This led us to test the following approaches:

- **Approach 1:** a second-order Taylor expansion of the Andersen and Piterbarg annuity mapping function.
- **Approach 2:** an improved approximation of $E^A[L_n(T) \mid S(T) = s]$, which is then used to compute $\alpha(s)$.
- **Approach 3:** a variant of approach 2 combining both the improved approximation of $E^A[L_n(T) \mid S(T) = s]$ and a second-order Taylor expansion of the Andersen and Piterbarg annuity mapping function.
- **Approach 4:** an annuity-mapping function based on the covariance of the swap rate and the LIBOR rates (that we call here "Hessian-based variance/covariance" approximation).

We assess the quality of the approximations of the annuity-mapping function $\alpha(s)$ against a Monte-Carlo reference. Figures 7.3 and 7.4 plot $\alpha(s)$ as a function of the terminal swap rate s (in bp) for the two legs of a 25Y-expiry CMS spread payoff, comparing five approximations: the leading-order Andersen and Piterbarg mapping (0th Taylor approximation), its second-order Taylor correction, two improved conditional-expectation variants, and the Hessian-based variance/covariance method. The Monte-Carlo curve (black) is the reference.

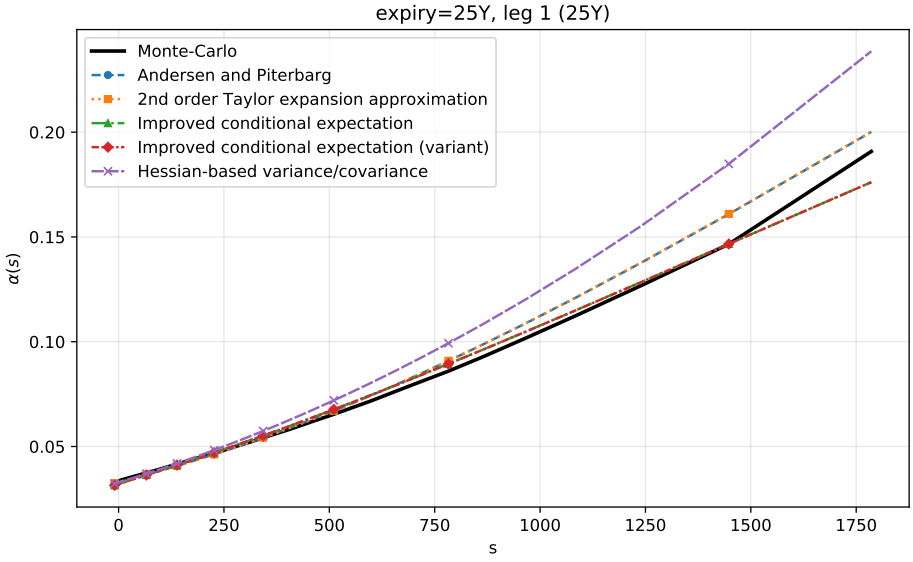


Figure 7.3: Annuity-mapping function $\alpha(s)$ for leg 1 (25Y tenor) of the 25Y-expiry payoff: approximations vs. Monte-Carlo reference (black).

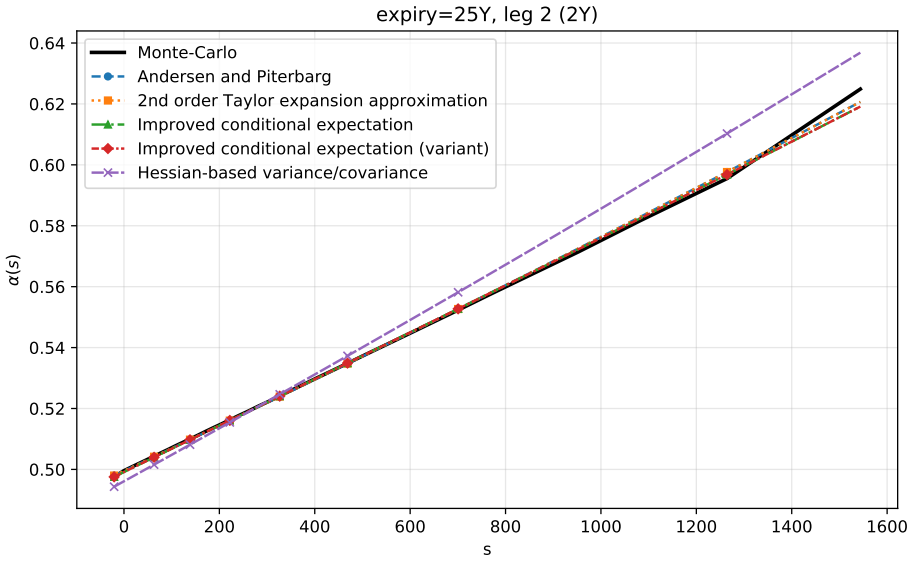


Figure 7.4: Annuity-mapping function $\alpha(s)$ for leg 2 (2Y tenor) of the 25Y-expiry payoff: approximations vs. Monte-Carlo reference (black).

Table 7.1 quantifies the error between the true annuity function obtained via Monte Carlo and the approximation on leg 1. The leading-order Andersen–Piterbarg mapping and its second-order Jensen correction are numerically indistinguishable (maximum pairwise deviation $\sim 1.5 \times 10^{-4}$): the Jensen term brings no discernible improvement at this maturity, and the two curves are superimposed. Both track the bulk of the reference but slightly overshoot in the right tail. The enhanced conditional-expectation variants likewise coincide and achieve the lowest mean error — about 30% below Andersen–Piterbarg on average — though their worst-case tail error is comparable.

Leg 2 (2Y tenor) exhibits the same ordering with tighter spreads across all methods, reflecting the smaller volatility contribution of the short leg.

Approximation	Max $\lvert\Delta\rvert$ vs. MC	Mean $\lvert\Delta\rvert$ vs. MC
0th order approximation (Andersen and Piterbarg)	1.43×10^{-2}	2.43×10^{-3}
2nd order approximation	1.44×10^{-2}	2.45×10^{-3}
Enhanced conditional-expectation	1.47×10^{-2}	1.74×10^{-3}
Enhanced conditional-expectation + 2nd order approximation	1.47×10^{-2}	1.74×10^{-3}
Hessian-based variance/covariance	4.78×10^{-2}	7.26×10^{-3}

Table 7.1: Absolute deviation of the annuity-mapping approximations from the Monte-Carlo reference on leg 1 (25Y tenor, 25Y expiry).

Other approaches (including selected approach) As summarised previously, errors obtained on volatilities with the previous approaches remain significant, even after our improvements. As we were not satisfied with the results obtained with the annuity-mapping approximations, we moved to an entirely different type of approach:

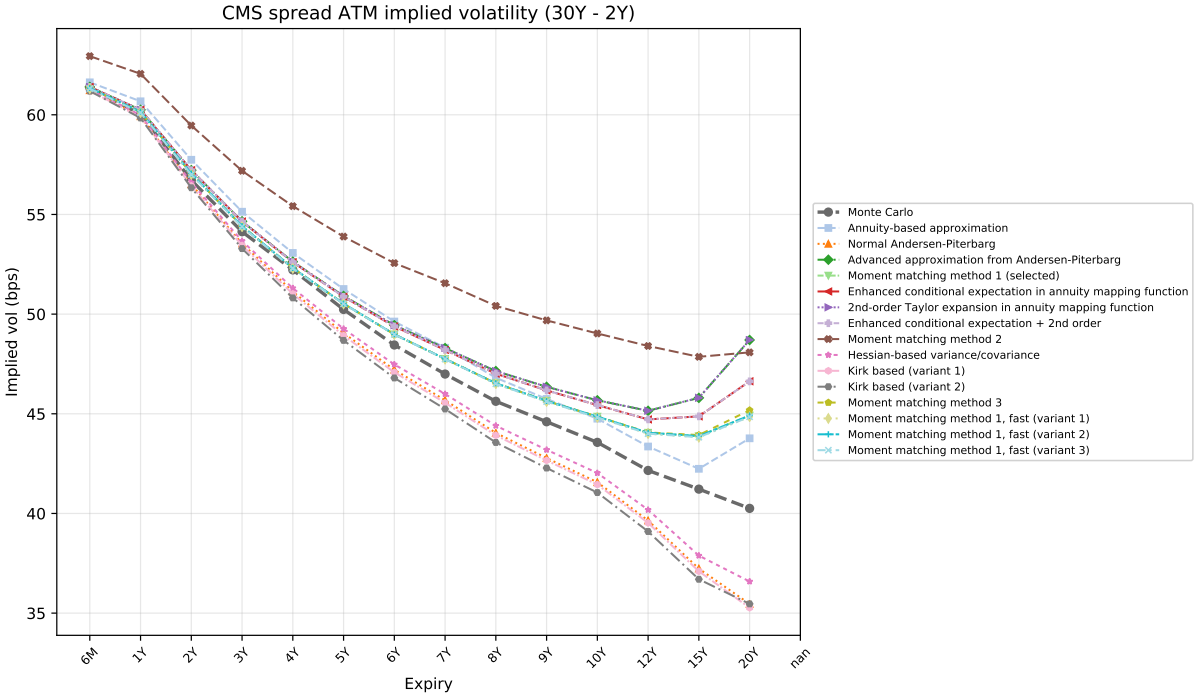
- **Approach 5:** an approximation based on Kirk's formula.

- **Approach 6:** an analytical expression for the convexity adjustment based on moment matching.
- **Approach 7 (chosen):** an analytical expression for the convexity adjustment based on moment matching, coupled with an enhanced way to compute the spread volatility.

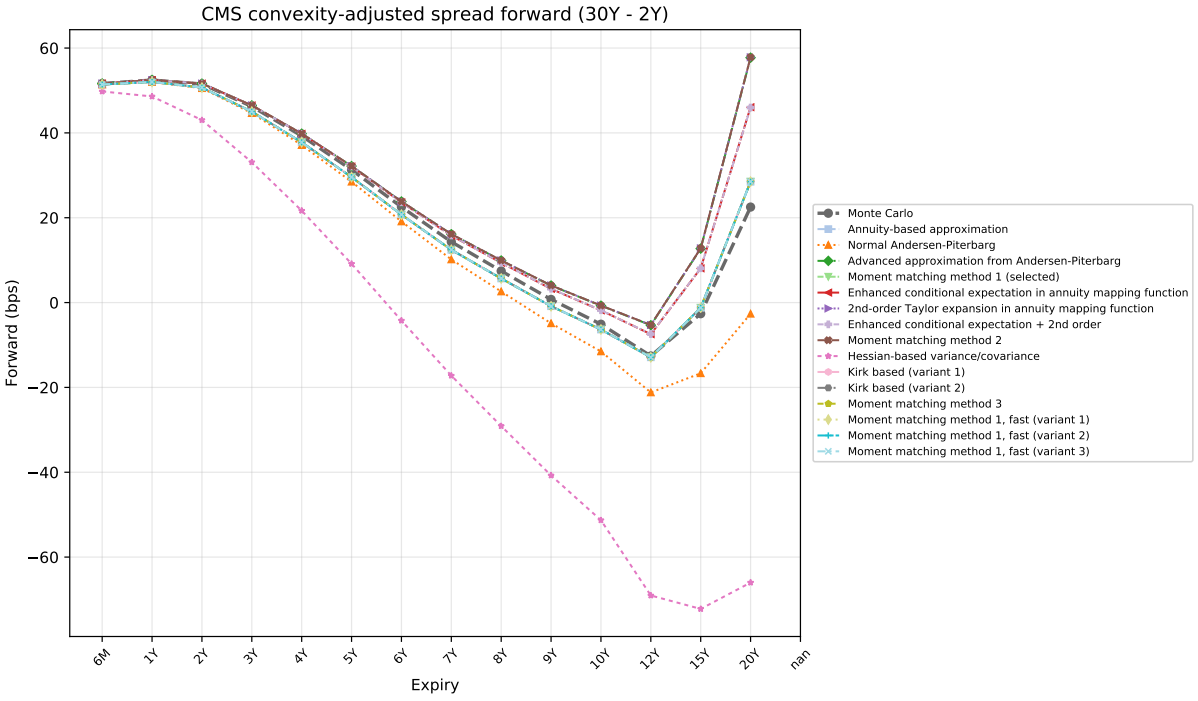
While only the main methods are described above, we tested fifteen approximations in total: the two Andersen and Piterbarg methods [2], the approaches listed above, and further variations of them. The annuity-mapping methods are also carried over into the comparison below for completeness.

Figures 7.5 and 7.6 compare, across expiries, the Monte-Carlo benchmark (solid black) against the fifteen analytical approximations for the two CMS spreads. For each spread we report both the ATM implied normal volatility and the convexity-adjusted forward. Most approximations cluster tightly around the Monte-Carlo reference at short and intermediate expiries. The agreement deteriorates towards the long end, where the dispersion between methods grows and the longest points become unreliable; this is why the 30Y expiry — and, for the wider 30Y-2Y spread, also the 25Y expiry — has been trimmed. Comparing the two baskets, the approximations agree more closely for the narrower 10Y-2Y spread than for the wider 30Y-2Y spread, where the larger volatility of the long leg amplifies the differences between methods, on both the implied-volatility and the convexity-adjusted-forward plots.

As Approach 7 (named "Moment matching method 1" in the graphs below) yields the most satisfying results, we select it as the production method for calibration.

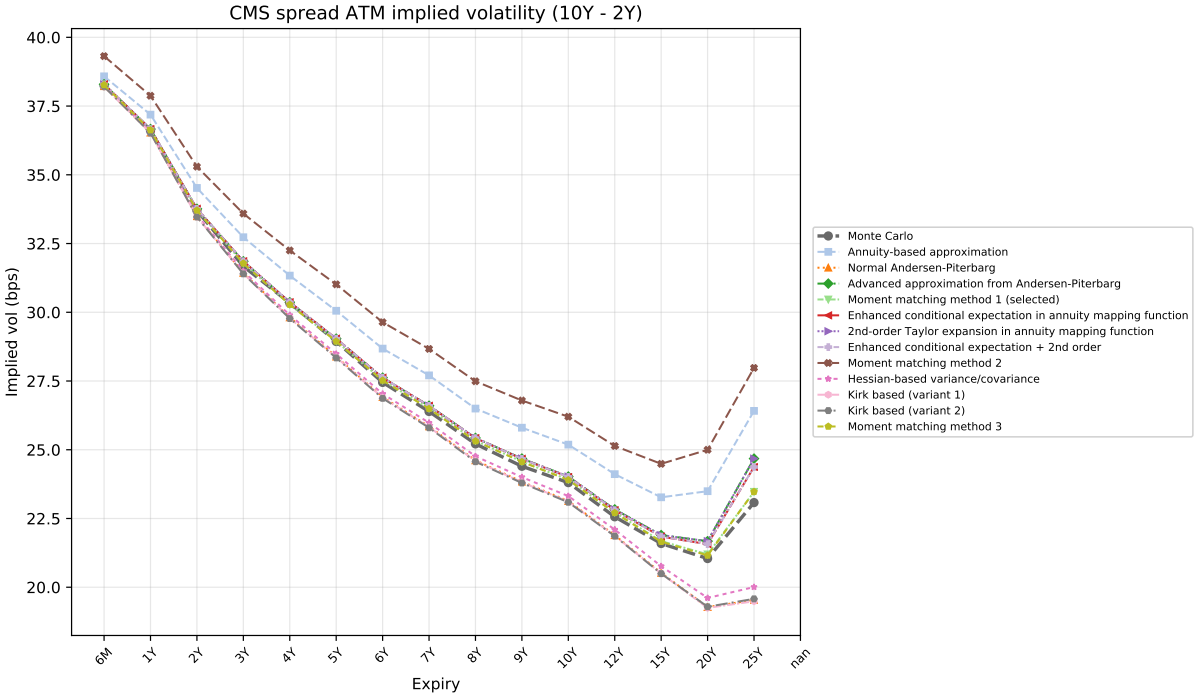


(a) ATM implied normal volatility (bps).

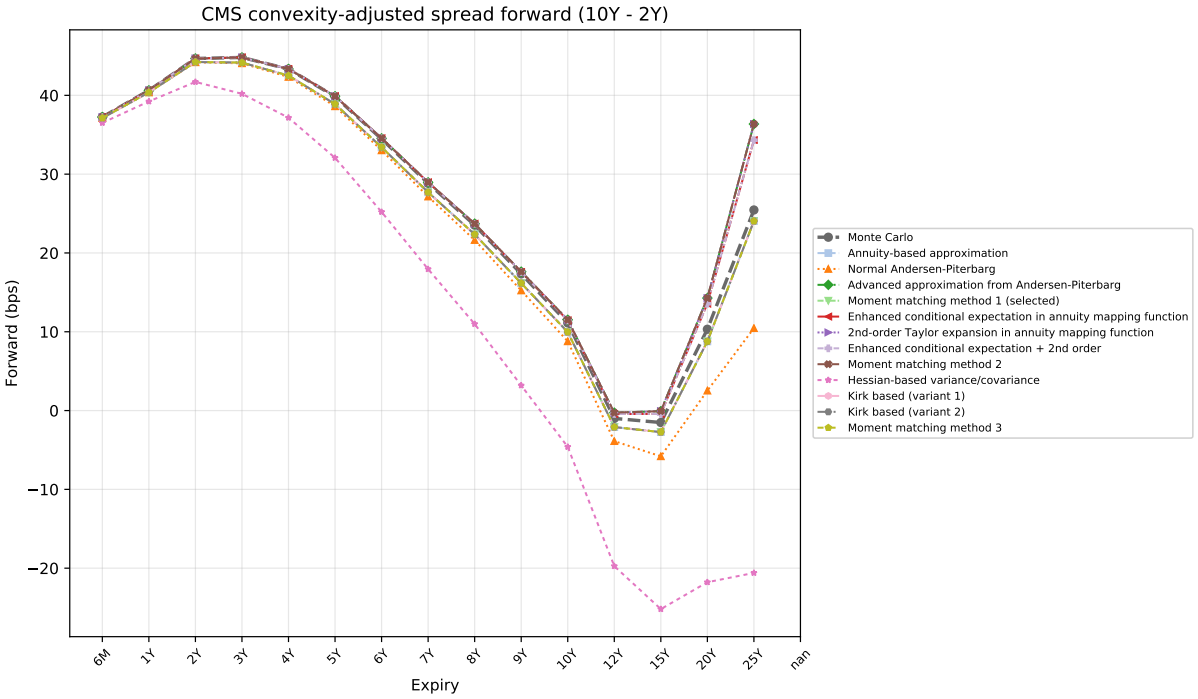


(b) Convexity-adjusted forward (bps).

Figure 7.5: 30Y-2Y CMS spread: Monte-Carlo reference vs. 15 analytical approximations across expiries (6M–20Y; the 25Y and 30Y points are removed as unreliable). Top: ATM implied normal volatility; bottom: convexity-adjusted forward.



(a) ATM implied normal volatility (bps).



(b) Convexity-adjusted forward (bps).

Figure 7.6: 10Y-2Y CMS spread: Monte-Carlo reference vs. 15 analytical approximations across expiries (6M–20Y; the 30Y point is removed as unreliable). Top: ATM implied normal volatility; bottom: convexity-adjusted forward.

7.4 Results

The displaced log-normal specification is calibrated jointly to the swaption smile — sampled at three strike offsets (ATM − 25, ATM, ATM + 25 bp) — and to the CMS spread options, on the market data of 20 April 2026. To assess accuracy we report the residuals of the full Monte-Carlo model both against the market quotes and against the closed-form approximation used inside the calibration objective.

Figures 7.7, 7.8 and 7.9 show the market ATM normal-volatility surfaces of the 30Y-2Y calibration basket at the three strike offsets, which constitute the swaption-smile targets of the joint calibration.

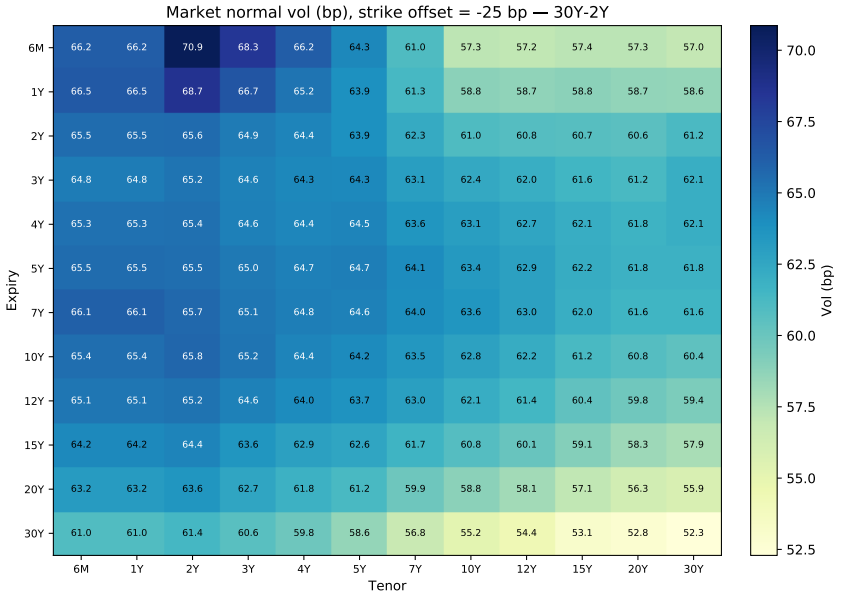


Figure 7.7: Market normal swaption volatility surface (bps) for the 30Y-2Y calibration basket at the ATM − 25 bp strike offset, 20 April 2026.

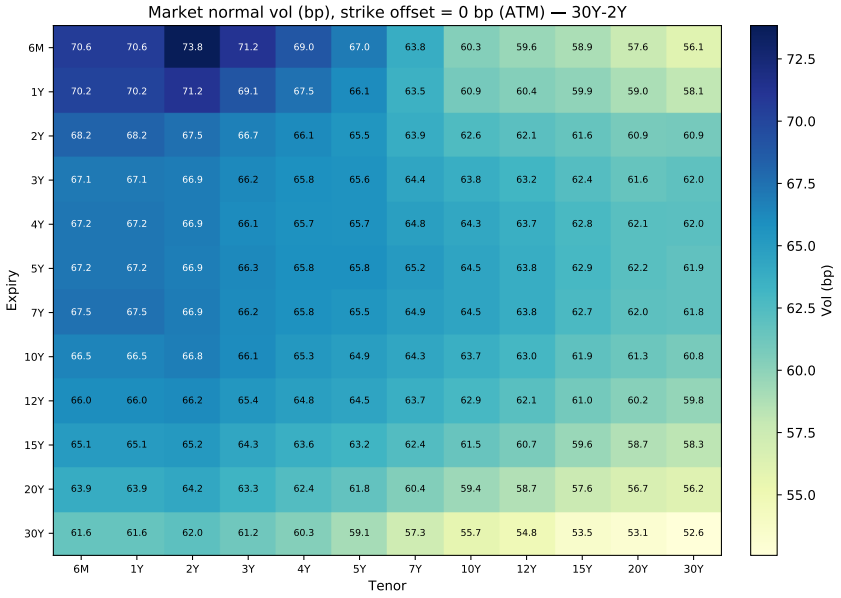


Figure 7.8: Market normal swaption volatility surface (bps) for the 30Y-2Y calibration basket at the ATM strike, 20 April 2026.

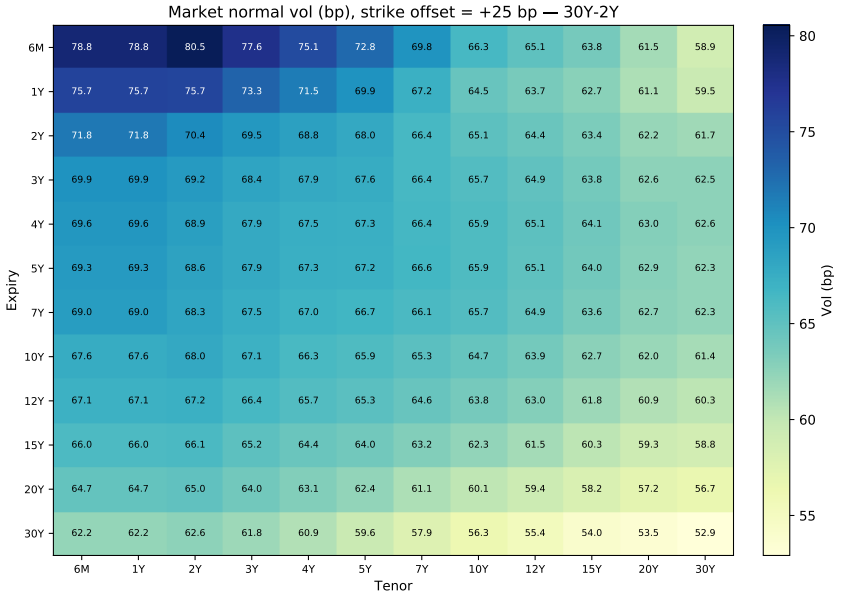


Figure 7.9: Market normal swaption volatility surface (bps) for the 30Y-2Y calibration basket at the ATM + 25 bp strike offset, 20 April 2026.

Figure 7.10 shows the associated market-implied displacement (skew) surface $\hat{b}_S$, obtained from the per-swaption optimisation (7.3) of Section 7.3.2; it is the skew target $\hat{b}_S$ of the displaced log-normal cost function (7.1).

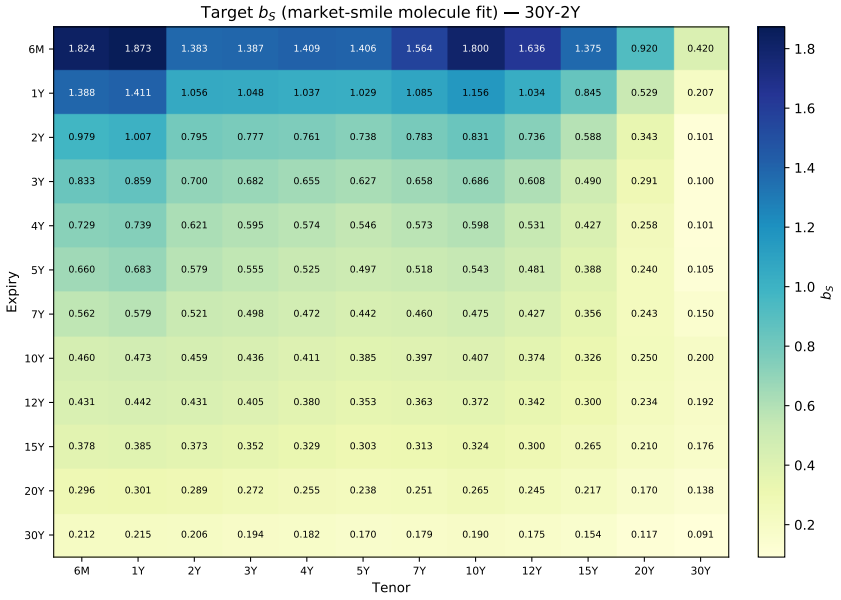


Figure 7.10: Market-implied displacement $\hat{b}_S$ (skew) surface for the 30Y-2Y calibration basket, obtained from the per-swaption skew optimisation (7.3), 20 April 2026.

Table 7.2 summarises the fit on all instruments. Tables 7.3–7.8 give the full expiry×tenor swaption residual grids (MC − market and MC − approximation for each strike), and Tables 7.9 and 7.10 report the CMS spread residuals per expiry together with the Monte-Carlo standard error.

Across the swaption smile the Monte-Carlo model matches the market to well under a basis point on average (RMSE between 0.52 and 0.76 bps per strike), and the closed-form approximation tracks the Monte-Carlo model even more tightly (RMSE ≈ 0.3 bps at every strike); the larger maximum residuals (3–3.4 bps) occur only at the shortest expiry of the off-ATM strikes. For the CMS spreads, the narrower 10Y-2Y basket is reproduced within 0.6 bps, whereas the wider 30Y-2Y basket shows a systematic long-expiry bias (RMSE 1.4 bps, up to 2.2 bps), consistent with the harder-to-price wide spread. These residuals must be read against the Monte-Carlo standard error, obtained as half the model-implied one-standard-error band and reported per expiry in Tables 7.9 and 7.10: it stays below 0.27 bps at every expiry. The 10Y-2Y residuals are therefore only a few standard errors and largely within simulation noise, whereas the 30Y-2Y long-expiry bias is roughly an order of magnitude larger than the noise and hence a genuine approximation bias rather than a sampling artefact. The approximation and the Monte-Carlo model remain in close agreement throughout, confirming that the closed form is adequate for calibration.

	MC − Market (bp)		MC − Approx. (bp)	
Instrument	RMSE	Max	RMSE	Max
Swaptions, ATM−25 bp	0.66	3.07	0.32	1.54
Swaptions, ATM	0.52	1.48	0.32	1.46
Swaptions, ATM+25 bp	0.76	3.44	0.32	1.38
CMS spread 30Y − 2Y	1.40	2.18	1.41	2.19
CMS spread 10Y − 2Y	0.30	0.61	0.26	0.38

Table 7.2: Displaced-lognormal calibration accuracy: RMSE and maximum absolute residual (bps) of the Monte-Carlo model relative to the market and to the closed-form approximation, for the swaption smile (three strike offsets) and the two CMS spreads.

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	−1.93	−0.10	−3.07	−1.10	−0.78	−0.60	−0.06	+0.50	−0.18	−0.53	−1.32	−1.34
1Y	−1.13	+0.66	−0.84	−0.20	−0.47	−0.82	−0.79	−0.84	−1.11	−1.17	−1.43	−1.09
2Y	−0.23	+1.19	+0.17	+0.12	−0.21	−0.35	−0.02	−0.23	−0.28	−0.40	−0.71	−0.79
3Y	−0.43	+0.86	+0.03	+0.05	+0.06	−0.31	−0.03	−0.38	−0.34	−0.35	−0.57	−0.95
4Y	−0.30	+1.06	+0.19	+0.41	+0.31	−0.10	−0.02	−0.29	−0.40	−0.49	−0.86	−0.83
5Y	−0.52	+0.98	+0.24	+0.27	+0.17	−0.26	−0.21	−0.19	−0.30	−0.33	−0.72	−0.52
7Y	−0.19	+1.09	+0.23	+0.09	−0.06	−0.06	+0.24	−0.13	−0.13	−0.10	−0.51	−0.68
10Y	−0.46	+1.08	−0.28	−0.24	+0.16	+0.07	+0.12	+0.10	+0.07	+0.01	−0.54	−0.69
12Y	−0.62	+1.09	+0.28	+0.13	+0.01	−0.10	+0.16	+0.12	+0.04	+0.01	−0.35	−0.76
15Y	−0.19	+1.28	+0.18	+0.33	+0.41	+0.28	+0.37	+0.08	−0.08	−0.17	−0.36	−1.01
20Y	−0.33	+1.08	−0.05	+0.01	+0.13	+0.04	+0.26	−0.10	−0.27	−0.36	−0.67	−1.38
30Y	+0.32	+1.67	+0.55	+0.47	+0.42	+0.74	+0.71	+0.13	−0.19	−0.13	−0.61	−1.26

Table 7.3: Swaption residuals, Monte-Carlo model − market normal volatility (bps), strike offset ATM−25 bp (RMSE = 0.66 bps, max = 3.07 bps).

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	+0.41	+1.30	−1.26	+0.32	+0.45	+0.49	+0.73	+0.67	+0.27	+0.31	+0.31	+1.02
1Y	+0.06	+1.44	+0.47	+1.06	+0.81	+0.52	+0.64	+0.28	+0.07	+0.09	+0.13	+0.71
2Y	0.00	+1.34	+0.86	+0.75	+0.40	+0.27	+0.42	−0.05	+0.02	+0.09	+0.16	+0.29
3Y	−0.37	+0.85	+0.36	+0.36	+0.37	+0.02	+0.14	−0.40	−0.27	−0.09	+0.03	−0.17
4Y	−0.35	+0.97	+0.36	+0.59	+0.53	+0.16	+0.12	−0.27	−0.28	−0.19	−0.28	−0.14
5Y	−0.59	+0.89	+0.37	+0.42	+0.37	−0.01	−0.09	−0.22	−0.23	−0.09	−0.23	0.00
7Y	−0.33	+0.97	+0.22	+0.12	+0.02	+0.08	+0.32	−0.18	−0.10	+0.02	−0.22	−0.40
10Y	−0.50	+1.07	−0.30	−0.22	+0.20	+0.14	+0.11	+0.03	+0.04	+0.06	−0.42	−0.64
12Y	−0.67	+1.05	+0.23	+0.12	+0.03	−0.04	+0.16	+0.06	+0.02	+0.02	−0.29	−0.74
15Y	−0.35	+1.16	+0.07	+0.26	+0.38	+0.30	+0.35	−0.01	−0.15	−0.22	−0.37	−1.07
20Y	−0.41	+1.04	−0.11	−0.02	+0.12	+0.05	+0.19	−0.28	−0.44	−0.51	−0.76	−1.48
30Y	−0.13	+1.26	+0.15	+0.10	+0.10	+0.44	+0.40	−0.19	−0.47	−0.36	−0.75	−1.40

Table 7.4: Swaption residuals, Monte-Carlo model − market normal volatility (bps), strike offset ATM (RMSE = 0.52 bps, max = 1.48 bps).

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	−1.41	−1.37	−3.44	−2.07	−1.95	−1.81	−1.76	−2.25	−2.50	−2.18	−1.58	−0.37
1Y	−0.80	+0.25	−0.17	+0.37	+0.20	+0.06	+0.32	−0.30	−0.49	−0.40	−0.08	+0.58
2Y	−0.72	+0.56	+0.57	+0.41	+0.05	−0.05	−0.06	−0.82	−0.63	−0.39	−0.03	+0.34
3Y	−0.93	+0.24	+0.08	+0.06	+0.09	−0.23	−0.27	−1.02	−0.79	−0.46	−0.03	−0.03
4Y	−0.83	+0.44	+0.10	+0.34	+0.30	−0.04	−0.21	−0.72	−0.60	−0.37	−0.20	+0.07
5Y	−1.03	+0.45	+0.16	+0.24	+0.21	−0.13	−0.32	−0.59	−0.52	−0.26	−0.16	+0.15
7Y	−0.68	+0.62	0.00	−0.07	−0.13	−0.03	+0.13	−0.47	−0.33	−0.10	−0.20	−0.37
10Y	−0.66	+0.92	−0.46	−0.34	+0.11	+0.06	−0.05	−0.21	−0.15	−0.09	−0.50	−0.77
12Y	−0.82	+0.92	+0.08	−0.01	−0.07	−0.11	+0.02	−0.14	−0.16	−0.11	−0.38	−0.88
15Y	−0.59	+0.95	−0.14	+0.10	+0.25	+0.22	+0.22	−0.21	−0.33	−0.38	−0.50	−1.24
20Y	−0.57	+0.91	−0.23	−0.13	+0.03	−0.01	+0.05	−0.54	−0.71	−0.74	−0.96	−1.66
30Y	−0.62	+0.80	−0.30	−0.31	−0.29	+0.10	+0.03	−0.58	−0.81	−0.65	−0.97	−1.59

Table 7.5: Swaption residuals, Monte-Carlo model − market normal volatility (bps), strike offset ATM+25 bp (RMSE = 0.76 bps, max = 3.44 bps).

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	−0.11	−0.12	−0.15	−0.14	−0.11	−0.02	+0.05	+0.03	+0.02	−0.01	−0.06	−0.18
1Y	−0.02	−0.01	−0.02	−0.04	+0.04	+0.01	+0.01	−0.11	−0.16	−0.20	−0.24	−0.32
2Y	+0.28	+0.30	+0.28	+0.42	+0.37	+0.41	+0.31	+0.13	+0.03	−0.06	−0.16	−0.32
3Y	+0.13	+0.14	+0.36	+0.28	+0.34	+0.27	+0.11	−0.04	−0.12	−0.17	−0.25	−0.45
4Y	+0.23	+0.24	+0.24	+0.21	+0.18	+0.13	0.00	−0.16	−0.22	−0.28	−0.36	−0.63
5Y	+0.16	+0.18	+0.18	+0.15	+0.09	+0.03	−0.08	−0.18	−0.20	−0.23	−0.32	−0.62
7Y	+0.24	+0.23	+0.19	+0.15	+0.12	+0.17	+0.08	+0.02	−0.02	−0.06	−0.20	−0.59
10Y	−0.03	−0.01	0.00	+0.02	+0.04	+0.07	+0.09	+0.08	+0.05	−0.05	−0.23	−0.72
12Y	+0.02	+0.06	+0.08	+0.10	+0.13	+0.14	+0.15	+0.12	+0.04	−0.09	−0.32	−0.90
15Y	+0.25	+0.27	+0.27	+0.26	+0.25	+0.22	+0.16	−0.01	−0.11	−0.25	−0.52	−1.17
20Y	+0.19	+0.19	+0.14	+0.09	+0.04	−0.01	−0.10	−0.22	−0.30	−0.42	−0.66	−1.34
30Y	+0.31	+0.32	+0.31	+0.29	+0.25	+0.22	+0.11	−0.08	−0.22	−0.42	−0.75	−1.54

Table 7.6: Swaption residuals, Monte-Carlo model − closed-form approximation (bps), strike offset ATM−25 bp (RMSE = 0.32 bps, max = 1.54 bps).

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	−0.22	−0.24	−0.24	−0.22	−0.18	−0.11	−0.07	−0.11	−0.13	−0.16	−0.19	−0.27
1Y	−0.10	−0.13	−0.14	−0.11	+0.02	+0.02	+0.03	−0.07	−0.13	−0.23	−0.31	−0.37
2Y	+0.16	+0.17	+0.17	+0.31	+0.27	+0.30	+0.18	+0.04	−0.07	−0.17	−0.25	−0.40
3Y	+0.04	+0.05	+0.27	+0.18	+0.22	+0.14	0.00	−0.15	−0.25	−0.31	−0.36	−0.53
4Y	+0.17	+0.17	+0.16	+0.11	+0.09	+0.05	−0.04	−0.18	−0.24	−0.30	−0.37	−0.64
5Y	+0.17	+0.18	+0.15	+0.12	+0.06	+0.02	−0.10	−0.22	−0.25	−0.26	−0.33	−0.68
7Y	+0.22	+0.21	+0.17	+0.12	+0.10	+0.15	+0.08	0.00	−0.04	−0.09	−0.23	−0.64
10Y	−0.02	0.00	+0.01	+0.01	+0.01	+0.04	+0.06	+0.07	+0.03	−0.05	−0.26	−0.74
12Y	+0.02	+0.06	+0.09	+0.11	+0.13	+0.14	+0.15	+0.13	+0.04	−0.10	−0.35	−0.89
15Y	+0.23	+0.26	+0.25	+0.24	+0.22	+0.20	+0.14	0.00	−0.11	−0.28	−0.56	−1.17
20Y	+0.20	+0.20	+0.15	+0.09	+0.05	−0.01	−0.11	−0.24	−0.33	−0.47	−0.71	−1.32
30Y	+0.32	+0.33	+0.31	+0.28	+0.24	+0.20	+0.10	−0.09	−0.22	−0.43	−0.74	−1.46

Table 7.7: Swaption residuals, Monte-Carlo model − closed-form approximation (bps), strike offset ATM (RMSE = 0.32 bps, max = 1.46 bps).

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	−0.32	−0.35	−0.38	−0.38	−0.34	−0.26	−0.21	−0.26	−0.28	−0.30	−0.27	−0.29
1Y	−0.12	−0.15	−0.19	−0.18	−0.04	−0.01	+0.07	−0.02	−0.10	−0.22	−0.31	−0.42
2Y	+0.06	+0.08	+0.07	+0.22	+0.17	+0.20	+0.08	−0.03	−0.12	−0.24	−0.35	−0.44
3Y	−0.03	−0.02	+0.20	+0.10	+0.13	+0.04	−0.09	−0.23	−0.34	−0.42	−0.43	−0.53
4Y	+0.12	+0.11	+0.09	+0.03	+0.01	−0.03	−0.10	−0.20	−0.24	−0.30	−0.36	−0.59
5Y	+0.16	+0.17	+0.14	+0.10	+0.04	0.00	−0.12	−0.24	−0.28	−0.31	−0.37	−0.69
7Y	+0.20	+0.19	+0.14	+0.10	+0.08	+0.13	+0.06	−0.01	−0.05	−0.11	−0.27	−0.67
10Y	0.00	+0.01	+0.01	+0.01	0.00	+0.01	+0.03	+0.06	+0.02	−0.08	−0.30	−0.75
12Y	+0.04	+0.08	+0.10	+0.12	+0.14	+0.15	+0.15	+0.13	+0.05	−0.11	−0.39	−0.89
15Y	+0.22	+0.24	+0.23	+0.22	+0.20	+0.18	+0.13	−0.01	−0.12	−0.31	−0.60	−1.15
20Y	+0.19	+0.20	+0.15	+0.10	+0.05	0.00	−0.10	−0.25	−0.36	−0.52	−0.77	−1.28
30Y	+0.32	+0.33	+0.30	+0.27	+0.23	+0.18	+0.08	−0.10	−0.23	−0.44	−0.73	−1.38

Table 7.8: Swaption residuals, Monte-Carlo model − closed-form approximation (bps), strike offset ATM+25 bp (RMSE = 0.32 bps, max = 1.38 bps).

Expiry	MC − Market (bp)	MC − Approx. (bp)	MC std. err. (bp)
6M	−0.07	−0.21	0.26
1Y	−0.47	−0.41	0.26
2Y	−0.59	−0.59	0.25
3Y	−0.76	−0.80	0.23
4Y	−0.62	−0.64	0.22
5Y	−0.73	−0.79	0.21
6Y	−0.96	−0.92	0.21
7Y	−1.11	−1.18	0.20
8Y	−1.39	−1.36	0.19
9Y	−1.53	−1.60	0.19
10Y	−1.67	−1.67	0.18
12Y	−1.93	−1.95	0.17
15Y	−2.18	−2.19	0.16
20Y	−2.13	−2.15	0.16
25Y	−2.01	−2.02	0.17
30Y	−1.74	−1.74	0.19

Table 7.9: ATM 30Y − 2Y CMS spread-option residuals (bps): Monte-Carlo model minus market and minus the closed-form approximation, together with the Monte-Carlo standard error (half the reported one-standard-error band). RMSE (max) = 1.40 (2.18) vs. market and = 1.41 (2.19) vs. approximation; the MC standard error stays ≤ 0.26 bps across expiries.

Expiry	MC − Market (bp)	MC − Approx. (bp)	MC std. err. (bp)
6M	+0.45	−0.11	0.19
1Y	−0.61	−0.14	0.19
2Y	−0.32	−0.25	0.18
3Y	−0.16	−0.30	0.17
4Y	−0.30	−0.34	0.17
5Y	−0.17	−0.22	0.16
6Y	−0.15	−0.08	0.16
7Y	+0.02	−0.11	0.15
8Y	−0.23	−0.27	0.15
9Y	−0.21	−0.34	0.15
10Y	−0.25	−0.23	0.14
12Y	−0.10	−0.08	0.14
15Y	−0.18	−0.19	0.13
20Y	−0.34	−0.33	0.13
25Y	−0.35	−0.38	0.13
30Y	−0.43	−0.38	0.14

Table 7.10: ATM 10Y − 2Y CMS spread-option residuals (bps): Monte-Carlo model minus market and minus the closed-form approximation, together with the Monte-Carlo standard error (half the reported one-standard-error band). RMSE (max) = 0.30 (0.61) vs. market and = 0.26 (0.38) vs. approximation; the MC standard error stays ≤ 0.19 bps across expiries.

Computational cost. Fitting the displacement surface jointly with the volatility grid and the time-dependent correlation raises the calibration time further: the full displaced log-normal calibration of the 30Y-2Y basket takes 96.14 s, against the 27.73 s of the correlation-only spread calibration reported above and the 948 ms swaption-only baseline (Table 5.5). The increase reflects the additional displacement degrees of freedom and the two extra smile strikes (ATM $\pm$ 25 bp) evaluated at each iteration. All timings were measured single-threaded on the Intel Xeon Gold 6240R (2.40 GHz) of Section 5.4.

Chapter 8

Conclusion

We presented a complete framework for global calibration of the LIBOR Market Model to the EUR swaption surface, and extended it in two directions: a correlation calibration to CMS spread options and a displaced log-normal extension that introduces skew. The main contributions and findings are summarised below.

Calibration methodology (Normal volatility case). The instantaneous volatility is parametrised on a 12×12 grid (144 free parameters) with bilinear interpolation between nodes. A combined first- and second-order regularisation ($\lambda_t = 0.1$, $\lambda_{tt} = \lambda_{xx} = 0.5$) enforces smoothness while preserving calibration accuracy. The Levenberg–Marquardt algorithm, operating on the Rebonato frozen-weight swaption volatility approximation, converges in 6 function evaluations (948 ms) for the recommended configuration — a 7× speedup over the unregularised case.

Calibration accuracy (Normal volatility case). The optimiser achieves an end-to-end RMSE of 1.106 bps against the market swaption surface.

Correlation estimation (Normal volatility case). The six-parameter Andersen–Piterbarg correlation structure was estimated from two years of daily EUR forward rate changes (121 rates, 7,260 correlation pairs). The fitted parameters achieve an RMSE of 0.079 against the empirical correlation matrix. Factor reduction via PCA retains 10 factors capturing 97.4% of the total variance.

Market-implied correlation. Beyond the historical estimate, the correlation surface was calibrated to CMS spread options by introducing a time-dependent correlation term and extending the calibration objective with the spread-option residuals, using an LMM spread-volatility approximation validated against Monte-Carlo. The spread information markedly steepens the term decorrelation: averaged over all forward-rate pairs separated by a 20-year tenor distance, the correlation falls from 0.76 under the historical fit to 0.50 with the 30Y-2Y spread targets and 0.38 with the 10Y-2Y ones, the effect being more pronounced for the narrower basket.

Skew calibration (displaced log-normal). To capture the volatility smile, the model was extended with a time-dependent per-rate displacement b_n. Closed-form moment matching maps

the shifted dynamics onto a blended swap rate and feeds the same spread-volatility machinery, so that the swaption smile (three strike offsets) and the CMS spreads are calibrated jointly without degrading the ATM fit. The Monte-Carlo model reproduces the swaption smile within 0.52–0.76 bps RMSE per strike and the 10Y-2Y CMS spread to within about 0.6 bps — a few times the Monte-Carlo standard error — while the wider 30Y-2Y spread retains a long-expiry error of up to 2.2 bps.

Limitations. The framework operates under several simplifying assumptions:

- The single-curve framework does not account for OIS–LIBOR basis spreads;
- The displaced log-normal extension reproduces the skew but not the full smile curvature; capturing the wings would require a stochastic-volatility overlay.

Current and future work.

- **Stochastic-volatility overlay:** adding smile dynamics on top of the displaced log-normal skew (CEV or stochastic volatility) to capture the smile wings.
- **Extension to SOFR rates:** adaptation of the framework to the SOFR markets, building on the framework established in [25] and [30].

Appendix A

Derivative of a swap rate with respect to a LIBOR forward rate

This appendix proves the closed-form expression for the stochastic weight $w_n(t) = \partial S(t)/\partial L_n(t)$ used in the swap-rate volatility approximation (4.10) and, at $t = 0$, in the spread-option conditional-expectation coefficient β_n of Section 7.3.4.

Setting. Let $P(t, T_m)$ be the time-t price of the zero-coupon bond maturing at T_m, and consider the forward swap rate and its annuity

$$S(t) = \frac{P(t, T_j) - P(t, T_k)}{A(t)}, \qquad A(t) = A_{j,k-j}(t) = \sum_{i=j}^{k-1} \eta_i \, P(t, T_{i+1}), \tag{A.1}$$

with accrual weights η_i as in (7.2). Fix $n \in \{j, \dots, k-1\}$ and recall the tenor relation

$$1 + \tau_n L_n(t) = \frac{P(t, T_n)}{P(t, T_{n+1})}. \tag{A.2}$$

Claim.

$$\frac{\partial S(t)}{\partial L_n(t)} = \frac{S(t)\,\tau_n}{1 + \tau_n L_n(t)} \left[\frac{P(t, T_k)}{P(t, T_j) - P(t, T_k)} + \frac{1}{A(t)} \sum_{i=n}^{k-1} \eta_i \, P(t, T_{i+1}) \right]. \tag{A.3}$$

Proof. For readability write $P_m := P(t, T_m)$, $L_n := L_n(t)$ and $\alpha_n := \dfrac{\tau_n}{1 + \tau_n L_n}$.

Bond sensitivities. For $m \geq n + 1$, telescoping the tenor relation gives

$$\frac{P_n}{P_m} = \prod_{r=n}^{m-1} \frac{P_r}{P_{r+1}} = \prod_{r=n}^{m-1} (1 + \tau_r L_r), \qquad \text{hence} \qquad P_m = P_n \prod_{r=n}^{m-1} (1 + \tau_r L_r)^{-1}. \tag{A.4}$$

Differentiating with respect to L_n, holding P_n and L_r $(r \neq n)$ fixed,

$$\frac{\partial P_m}{\partial L_n} = P_n \prod_{r=n+1}^{m-1} (1 + \tau_r L_r)^{-1} \frac{\partial}{\partial L_n} (1 + \tau_n L_n)^{-1} = -\frac{\tau_n}{1 + \tau_n L_n} P_m = -\alpha_n P_m, \qquad m \geq n + 1. \tag{A.5}$$

For $m \le n$, P_m does not depend on L_n, so $\partial P_m / \partial L_n = 0$. Since $n \ge j$, this yields

$$\frac{\partial P_j}{\partial L_n} = 0, \qquad \frac{\partial P_k}{\partial L_n} = -\alpha_n P_k. \tag{A.6}$$

Numerator and annuity. With $N(t) := P_j - P_k$,

$$\frac{\partial N(t)}{\partial L_n} = 0 - \big(-\alpha_n P_k\big) = \alpha_n P_k. \tag{A.7}$$

In $A(t) = \sum_{i=j}^{k-1} \eta_i P_{i+1}$, only the terms with $i+1 \ge n+1$, i.e. $i \ge n$, depend on L_n, so

$$\frac{\partial A(t)}{\partial L_n} = \sum_{i=n}^{k-1} \eta_i \frac{\partial P_{i+1}}{\partial L_n} = -\alpha_n \sum_{i=n}^{k-1} \eta_i P_{i+1}. \tag{A.8}$$

Quotient rule. Since $S = N/A$,

$$\frac{\partial S(t)}{\partial L_n} = \frac{1}{A}\frac{\partial N}{\partial L_n} - \frac{N}{A^2}\frac{\partial A}{\partial L_n} = \frac{\alpha_n P_k}{A} + \frac{N}{A^2}\,\alpha_n \sum_{i=n}^{k-1} \eta_i P_{i+1}. \tag{A.9}$$

Factoring out $S\alpha_n = (N/A)\alpha_n$ and recalling $N = P_j - P_k$ and $\alpha_n = \tau_n/(1+\tau_n L_n)$,

$$\frac{\partial S(t)}{\partial L_n} = S(t)\,\frac{\tau_n}{1+\tau_n L_n}\left[\frac{P_k}{P_j - P_k} + \frac{1}{A}\sum_{i=n}^{k-1} \eta_i P_{i+1}\right], \tag{A.10}$$

which is exactly (A.3). ∎

Appendix B

Markovian projection of the swap rate onto a displaced log-normal process

This appendix derives the swap-rate approximation of Section 7.3.3: the projection of the multi-factor swap-rate dynamics onto a one-factor displaced log-normal process, and the resulting model-implied volatility $\bar{\sigma}_S^{\mathrm{DLN}}$ and skew $\bar{b}_S$. We work under the annuity measure $\mathbb{Q}^A$ and follow the Markovian-projection programme of Andersen and Piterbarg [2].

Step 1: Exact one-factor dynamics. The par swap rate $S(t) = S(t, \mathbf{L}(t))$ is a smooth function of the live forwards $\mathbf{L}(t) = (L_j(t), \dots, L_{k-1}(t))$ and a martingale under $\mathbb{Q}^A$. Applying Itô's formula and discarding the drift—which vanishes by the martingale property—leaves only the diffusion term,

$$\mathrm{d}S(t) = \lambda_S(t, \mathbf{L}(t))^\top \, \mathrm{d}W^A(t), \qquad \lambda_S(t, \mathbf{L}) = \sum_{n=j}^{k-1} \frac{\partial S}{\partial L_n}(t, \mathbf{L}) \, \varphi_n(t, L_n) \, \lambda_n(t), \tag{B.1}$$

where $\varphi_n(t, x) = b_n(t)x + (1 - b_n(t))L_n(0)$ is the displaced log-normal forward volatility and the sensitivity $\partial S / \partial L_n$ is given in closed form in Appendix A. Projecting the vector increment onto the unit direction of λ_S, i.e. setting $\mathrm{d}Y^A(t) = \lambda_S(t, \mathbf{L}(t))^\top \mathrm{d}W^A(t) / \|\lambda_S(t, \mathbf{L}(t))\|$, defines a scalar $\mathbb{Q}^A$-Brownian motion by Lévy's characterisation (it is a continuous local martingale with quadratic variation t), and (B.1) becomes the *exact* one-factor SDE

$$\mathrm{d}S(t) = \|\lambda_S(t, \mathbf{L}(t))\| \, \mathrm{d}Y^A(t). \tag{B.2}$$

Please note that, while exact, equation (B.2) depends on the whole vector $\mathbf{L}(t)$, not on $S(t)$ alone. We thus develop an approximation dependent on $S(t)$ to approximate it by a tractable quantity.

Step 2: Markovian projection (Gyöngy's theorem). By the mimicking theorem of Gyöngy [11], the Itô process (B.2) shares, at every t, its one-dimensional marginal law with

the local-volatility diffusion

$$\mathrm{d}\hat{S}(t) = \eta(t, \hat{S}(t))\,\mathrm{d}\hat{Y}^A(t), \qquad \eta(t,x)^2 = \mathbb{E}^A[\,\|\lambda_S(t, \mathbf{L}(t))\|^2 \,|\, S(t) = x\,]. \tag{B.3}$$

Because a swaption payoff depends on $S(T)$ only through its marginal distribution, the Markovian surrogate $\hat{S}$ prices it exactly; the problem reduces to approximating the conditional expectation in (B.3).

Step 3: Convexity-adjusted forward. Equation (7.4) is a first-order convexity expansion. The forward rate L_n is a martingale under its own forward measure $\mathbb{Q}^{n+1}$ (numéraire $P(\cdot, T_{n+1})$) but not under the annuity measure $\mathbb{Q}^A$; the change of numéraire gives

$$\mathbb{E}^A[L_n(t)] = \mathbb{E}^{n+1}\left[L_n(t)\,\frac{Q_n(t)}{Q_n(0)}\right], \qquad Q_n(t) = \frac{A(t)}{P(t, T_{n+1})}, \tag{B.4}$$

since the Radon–Nikodym density $\mathrm{d}\mathbb{Q}^A/\mathrm{d}\mathbb{Q}^{n+1}$ restricted to $\mathcal{F}_t$ equals $Q_n(t)/Q_n(0)$. Regarding $Q_n(t) = Q_n(\mathbf{L}(t))$ as a function of the live forwards and expanding to first order about $\mathbf{L}(0)$,

$$Q_n(\mathbf{L}(t)) \approx Q_n(0) + \sum_{i=j}^{k-1} \frac{\partial Q_n(0)}{\partial L_i(0)}\,(L_i(t) - L_i(0)), \tag{B.5}$$

and using the martingale property $\mathbb{E}^{n+1}[L_n(t)] = L_n(0)$ gives

$$\mathbb{E}^A[L_n(t)] \approx L_n(0) + \frac{1}{Q_n(0)} \sum_{i=j}^{k-1} \frac{\partial Q_n(0)}{\partial L_i(0)}\,\mathbb{E}^{n+1}[L_n(t)\,(L_i(t) - L_i(0))]. \tag{B.6}$$

Then, by approximating the quantity by a linear regression, we have:

$$\mathbb{E}^{n+1}[L_n(t)\,(L_i(t) - L_i(0))] \approx \mathrm{Cov}^{n+1}(L_n(t), L_i(t)) \approx \int_0^t \lambda_n(s)^\top \lambda_i(s)\,\mathrm{d}s. \tag{B.7}$$

Factoring out $L_n(0)$ then gives $\mathbb{E}^A[L_n(t)] \approx L_n(0)(1 + c_n(t))$ with the convexity coefficient (7.5).

Step 4: Freezing and Gaussian regression. Two approximations make η explicit. First, we freeze the state dependence of the volatility by evaluating it at the conditional mean of the forwards,

$$\eta(t,x)^2 \approx \|\lambda_S(t, \ell(t,x))\|^2, \qquad \ell_n(t,x) := \mathbb{E}^A[L_n(t) \mid S(t) = x]. \tag{B.8}$$

Second, if the small skew of the forwards is neglected, the pair $(\mathbf{L}(t), S(t))$ is approximately jointly Gaussian, and this conditional mean is then an affine function of the swap rate — the least-squares regression of the forward $L_n(t)$ on $S(t)$:

$$\ell_n(t,x) \approx \mathbb{E}^A[L_n(t)] + \xi_n(t)\,(x - S(0)), \qquad \xi_n(t) = \frac{\mathrm{Cov}^A(L_n(t), S(t))}{\mathrm{Var}^A(S(t))}. \tag{B.9}$$

Freezing the diffusion coefficients at their time-zero values $\mathbf{L}(0)$ inside the covariance and variance integrals yields

$$\xi_n(t) \approx \frac{\int_0^t \lambda_n(u)^\top \lambda_S(u, \mathbf{L}(0))\,\mathrm{d}u}{\int_0^t \|\lambda_S(u, \mathbf{L}(0))\|^2\,\mathrm{d}u}, \tag{B.10}$$

while the unconditional mean is the convexity-adjusted forward $\mathbb{E}^A[L_n(t)] = L_n(0)(1 + c_n(t))$ of (7.4). Retaining the instantaneous covariance $r_{S,n}(t) = \lambda_n(t)^\top \lambda_S(t, \mathbf{L}(0))$ alongside its time integral upgrades (B.10) to the second-order weight $\xi_i(t)$ of (7.8).

Step 5: Displaced log-normal fit. We approximate the projected local volatility by the displaced log-normal form $\eta(t,x) \approx p_S(t)\big(b_S(t)x + (1 - b_S(t))S(0)\big)$, matching value and first derivative at the forward $x = S(0)$, where $\ell_n(t, S(0)) = \mathbb{E}^A[L_n(t)]$. The level match $\eta(t, S(0)) = p_S(t)S(0)$ gives

$$p_S(t) = \frac{1}{S(0)}\,\|\lambda_S(t, \mathbb{E}^A[\mathbf{L}(t)])\|, \tag{B.11}$$

in agreement with Section 7.3.3. For the slope, the chain rule $\partial_x \ell_n(t,x) = \xi_n(t)$ from (B.9) gives

$$\frac{\partial\,\eta(t,x)}{\partial x}\Big|_{x=S(0)} = \sum_{n=j}^{k-1} \frac{\partial\,\|\lambda_S(t, \mathbb{E}^A[\mathbf{L}(t)])\|}{\partial L_n}\,\xi_n(t), \tag{B.12}$$

and equating this to the slope of the displaced log-normal form determines the blend $b_S(t)$. Carrying the expansion of $\|\lambda_S\|$ to second order in the forwards—through the first and second sensitivities $v_n = \partial S/\partial L_n$, $v_{n,n'} = \partial^2 S/\partial L_n \partial L_{n'}$ of (7.6) and the covariances $r_{S,i}$ of (7.7)—yields the explicit skew (7.9).

Step 6: Time-homogenisation. The projected coefficients $p_S(t), b_S(t)$ are still time-dependent; a second, temporal averaging in the spirit of Piterbarg replaces them by the constant parameters that preserve the terminal variance and the variance-weighted skew. With accumulated variance $\nu(t)^2 = \int_0^t p_S(s)^2\,\mathrm{d}s$, the model-implied normal volatility and skew fed to the calibration objective (7.1) are

$$\bar\sigma_S^{\mathrm{DLN}} = \left(\frac{1}{T}\int_0^T p_S(t)^2\,\mathrm{d}t\right)^{1/2}, \qquad \bar b_S = S(0)\int_0^T b_S(t)\,w_T(t)\,\mathrm{d}t, \quad w_T(t) = \frac{\nu(t)^2 p_S(t)^2}{\int_0^T \nu(u)^2 p_S(u)^2\,\mathrm{d}u}, \tag{B.13}$$

which are Eqs. (7.10)–(7.11). Inverting the displaced-Black swaption price at the pair $(\bar\sigma_S^{\mathrm{DLN}}, \bar b_S)$ into a Bachelier volatility, as in Section 7.3.2, produces the model-implied swaption normal volatility $\bar\sigma^{\mathrm{DLN}}_{\mathrm{swaption}}$ that enters the objective (7.1). ■

Bibliography

[1] Leif B. G. Andersen. A simple approach to the pricing of Bermudan swaptions in the multi-factor LIBOR market model. *SSRN Working Paper*, 1999.

[2] Leif B. G. Andersen and Vladimir V. Piterbarg. *Interest Rate Modeling*, volume 2. Atlantic Financial Press, 2010.

[3] Marco Bianchetti. Two curves, one price. *Risk Magazine*, pages 74–80, August 2010.

[4] Fischer Black and Piotr Karasinski. Bond and option pricing when short rates are lognormal. *Financial Analysts Journal*, 47(4):52–59, 1991.

[5] Alan Brace, Dariusz Gatarek, and Marek Musiela. The market model of interest rate dynamics. *Mathematical Finance*, 7(2):127–155, 1997.

[6] Damiano Brigo, Antoine Jacquier, Nicola Pede, and Bo Yuan. Deep learning interpretability for rough volatility, 2024.

[7] Damiano Brigo and Fabio Mercurio. *Interest Rate Models—Theory and Practice: With Smile, Inflation and Credit*. Springer, 2nd edition, 2006.

[8] John C. Cox, Jonathan E. Ingersoll, and Stephen A. Ross. A theory of the term structure of interest rates. *Econometrica*, 53(2):385–407, 1985.

[9] Carl Eckart and Gale Young. The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3):211–218, 1936.

[10] Paul Glasserman. *Monte Carlo Methods in Financial Engineering*. Stochastic Modelling and Applied Probability. Springer, New York, NY, 1 edition, 2003.

[11] István Gyöngy. Mimicking the one-dimensional marginal distributions of processes having an Itô differential. *Probability Theory and Related Fields*, 71(4):501–516, 1986.

[12] Patrick S. Hagan, Deep Kumar, Andrew S. Lesniewski, and Diana E. Woodward. Managing smile risk. *Wilmott Magazine*, pages 84–108, 2002.

[13] David Heath, Robert Jarrow, and Andrew Morton. Bond pricing and the term structure of interest rates: A new methodology for contingent claims valuation. *Econometrica*, 60(1):77–105, 1992.

[14] Marc Henrard. *Interest Rate Modelling in the Multi-Curve Framework: Foundations, Evolution and Implementation*. Palgrave Macmillan, 2014.

[15] Pierre Henry-Labordère. Unifying the BGM and SABR models: A short ride in hyperbolic geometry. *SSRN Working Paper*, 2007.

[16] Steven L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2):327–343, 1993.

[17] Blanka Horvath, Aitor Muguruza, and Mehdi Tomas. Deep learning volatility: A deep neural network perspective on pricing and calibration in (rough) volatility models. *Quantitative Finance*, 21(1):11–27, 2021.

[18] John Hull and Alan White. *Pricing Interest-Rate-Derivative Securities*, volume 3. 1990.

[19] Christopher J. Hunter, Peter Jäckel, and Mark S. Joshi. Getting the drift: Drift approximation in the LMM. *Risk Magazine*, July 2001.

[20] Farshid Jamshidian. LIBOR and swap market models and measures. *Finance and Stochastics*, 1(4):293–330, 1997.

[21] E. Kirk. Correlation in the energy markets. *Managing Energy Price Risk*, pages 71–78, 1995.

[22] Robert Litterman and José Scheinkman. Common factors affecting bond returns. *The Journal of Fixed Income*, 1(1):54–61, 1991.

[23] Dong C. Liu and Jorge Nocedal. On the limited memory BFGS method for large scale optimization. *Mathematical Programming*, 45(1–3):503–528, 1989.

[24] Francis A. Longstaff and Eduardo S. Schwartz. Valuing American options by simulation: A simple least-squares approach. *The Review of Financial Studies*, 14(1):113–147, 2001.

[25] Andrei Lyashenko and Fabio Mercurio. Looking forward to backward-looking rates: A modeling framework for term rates replacing LIBOR. *SSRN Working Paper*, 2019.

[26] William Margrabe. The value of an option to exchange one asset for another. *The Journal of Finance*, 33(1):177–186, 1978.

[27] Donald W. Marquardt. An algorithm for least-squares estimation of nonlinear parameters. *Journal of the Society for Industrial and Applied Mathematics*, 11(2):431–441, 1963.

[28] Michael D. McKay, Richard J. Beckman, and William J. Conover. A comparison of three methods for selecting values of input variables in the analysis of output from a computer code. *Technometrics*, 21(2):239–245, 1979.

[29] Fabio Mercurio. Interest rates and the credit crunch: New formulas and market models. *Bloomberg Portfolio Research Paper*, 2009.

[30] Fabio Mercurio. A simple multi-curve model for pricing SOFR futures and other derivatives. *SSRN Working Paper*, 2018.

[31] Fabio Mercurio and Massimo Morini. Joining the SABR and Libor models together. *Risk Magazine*, March 2009.

[32] Jorge J. Moré. The Levenberg-Marquardt algorithm: Implementation and theory. *Numerical Analysis. Lecture Notes in Mathematics*, 630:105–116, 1978.

[33] Nicola Moreni and Andrea Pallavicini. Parsimonious HJM modelling for multiple yield curve dynamics. *Quantitative Finance*, 14(2):199–210, 2014.

[34] Marek Musiela and Marek Rutkowski. Continuous-time term structure models: Forward measure approach. *Finance and Stochastics*, 1(4):261–291, 1997.

[35] Jorge Nocedal and Stephen J. Wright. *Numerical Optimization*. Springer, 2nd edition, 2006.

[36] Morten Bjerregaard Pedersen. Calibrating Libor market models. 1998. Working paper, SSRN, posted 7 September 1998.

[37] Vladimir V. Piterbarg. A practitioner's guide to pricing and hedging callable Libor exotics in forward Libor models. *SSRN Working Paper*, 2003.

[38] Riccardo Rebonato. On the simultaneous calibration of multi-factor lognormal interest rate models to Black volatilities and to the correlation matrix. *Journal of Computational Finance*, 2(4):5–27, 1999.

[39] Riccardo Rebonato. Modern pricing of interest-rate derivatives: The LIBOR market model and beyond. 2002.

[40] Riccardo Rebonato. A time-homogeneous, SABR-consistent extension of the LMM. *Risk Magazine*, November 2007.

[41] Riccardo Rebonato and Peter Jäckel. The most general methodology to create a valid correlation matrix for risk management and option pricing purposes. 1999. Quantitative Research Centre of the NatWest Group, Working Paper.

[42] Riccardo Rebonato and Richard White. Linking caplets and swaptions prices in the LMM: A semiempirical approach. *Journal of Computational Finance*, 7(2), 2004.

[43] Peter Ritchken and L. Sankarasubramanian. Volatility structures of forward rates and the dynamics of the term structure. *Mathematical Finance*, 5(1):55–72, 1995.

[44] John Schoenmakers and Brian Coffey. Systematic generation of parametric correlation structures for the LIBOR market model. *International Journal of Theoretical and Applied Finance*, 6(5):507–519, 2003.

[45] Jakob Sidenius. LIBOR market models in practice. *Journal of Computational Finance*, 3(3):5–26, 2000.

[46] Rainer Storn and Kenneth Price. Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. *Journal of Global Optimization*, 11(4):341–359, 1997.

[47] Oldřich A. Vasicek. An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5(2):177–188, 1977.